



Conversión CA/CC

Rectificación Controlada
Monofásica



Rectificación controlada vs no controlada

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Rectificación controlada vs no controlada

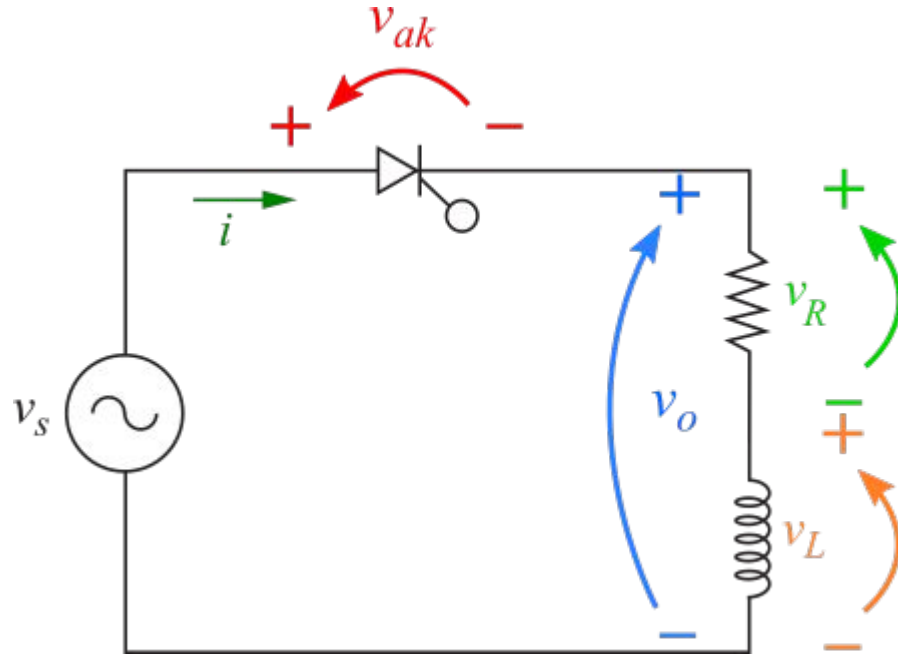
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- Circuitualmente, pasar de rectificación no controlada a controlada consiste en cambiar los diodos por **tiristores**.

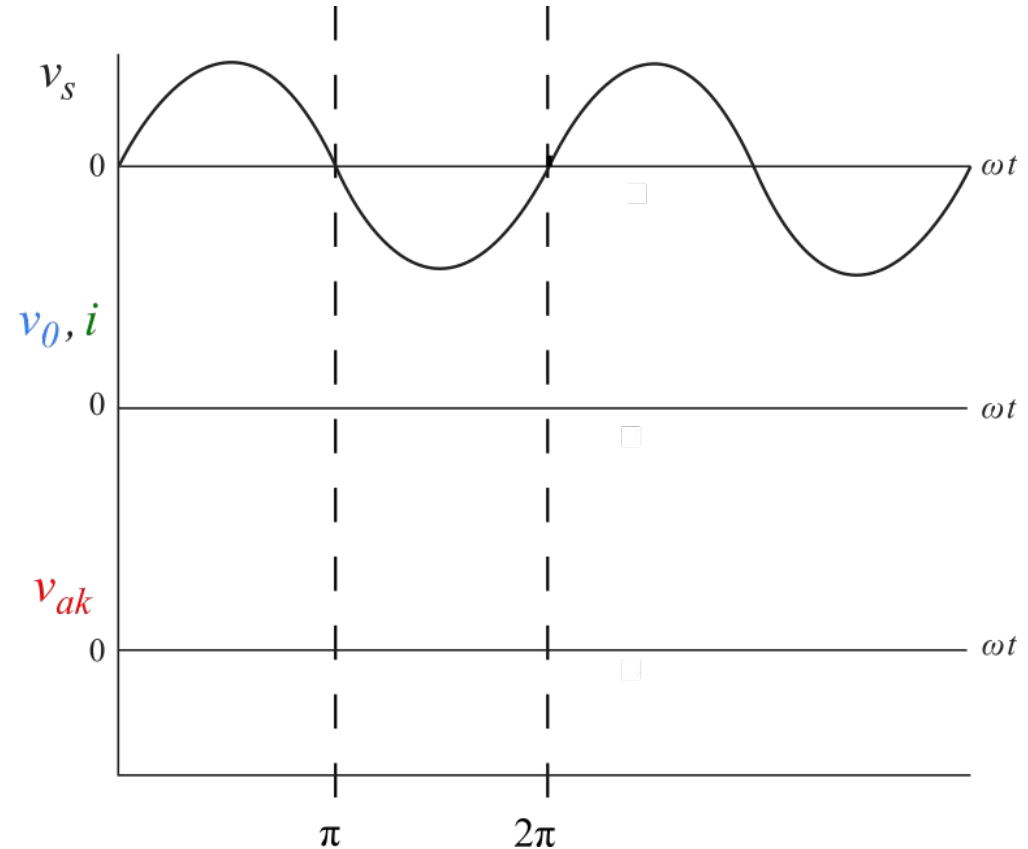
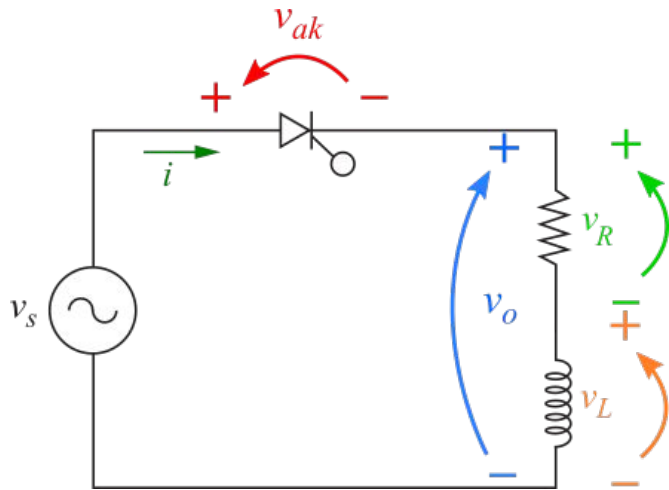
Rectificador controlado de media onda

- Carga R-L



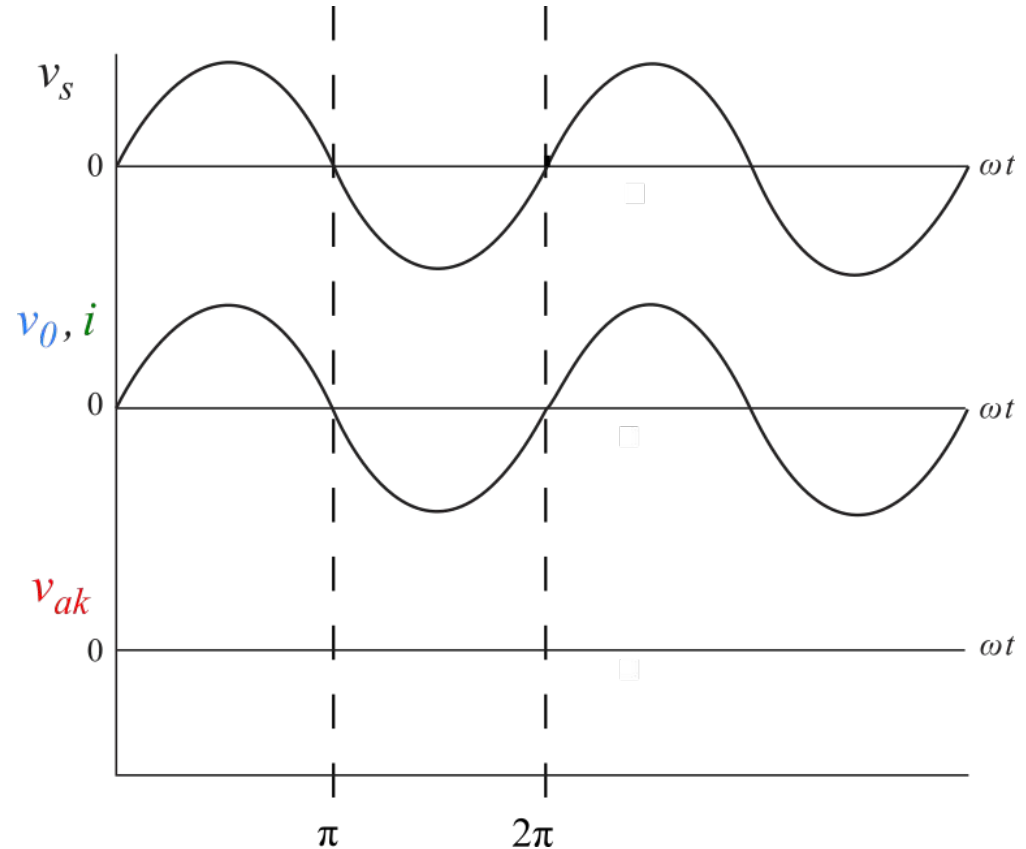
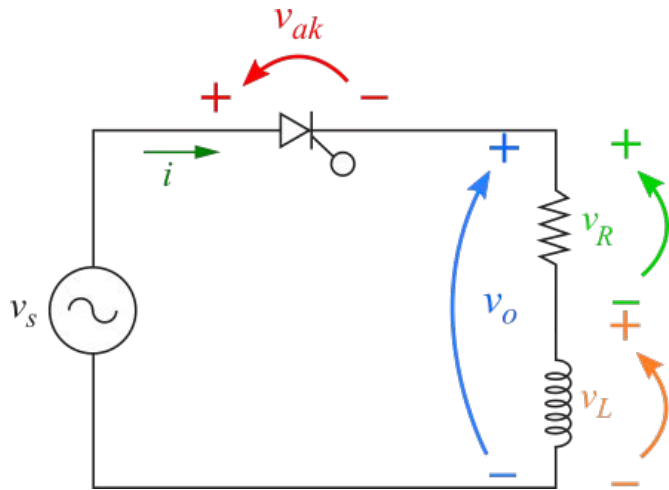
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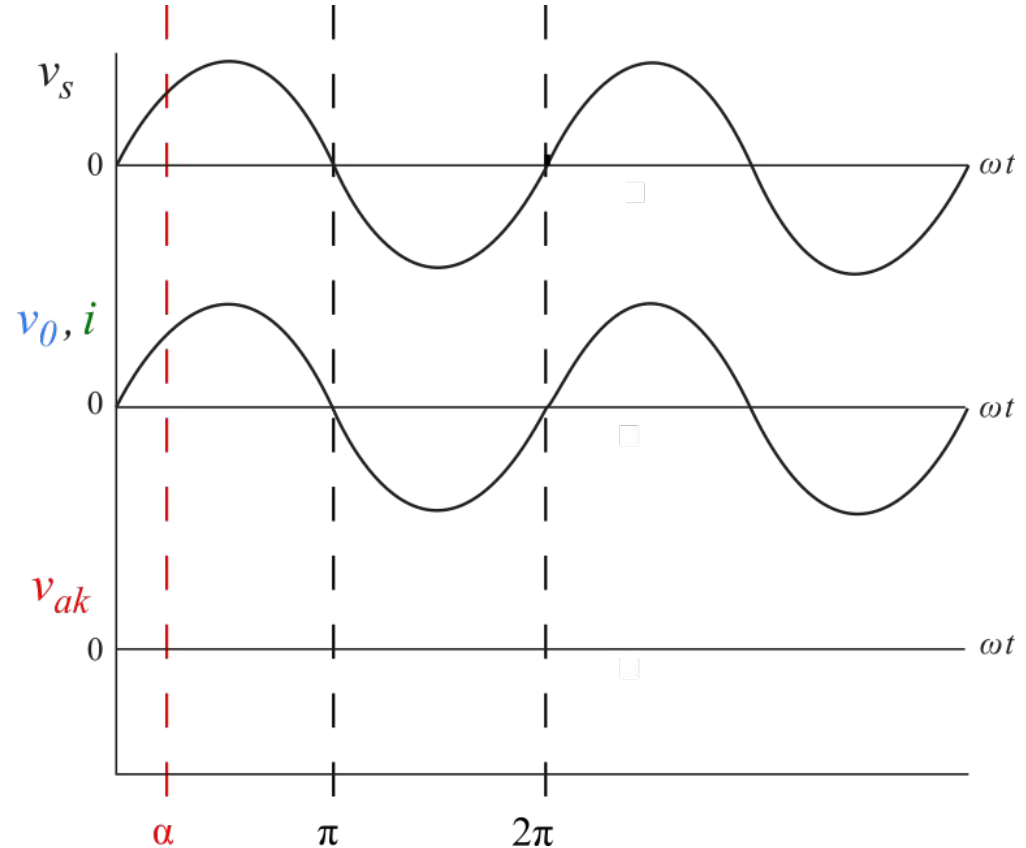
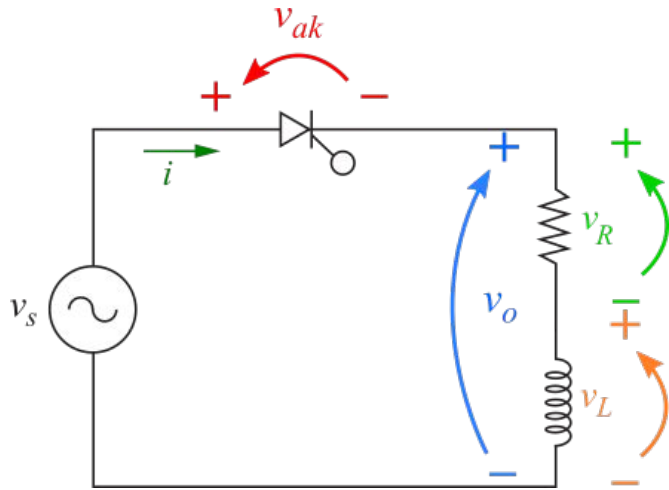
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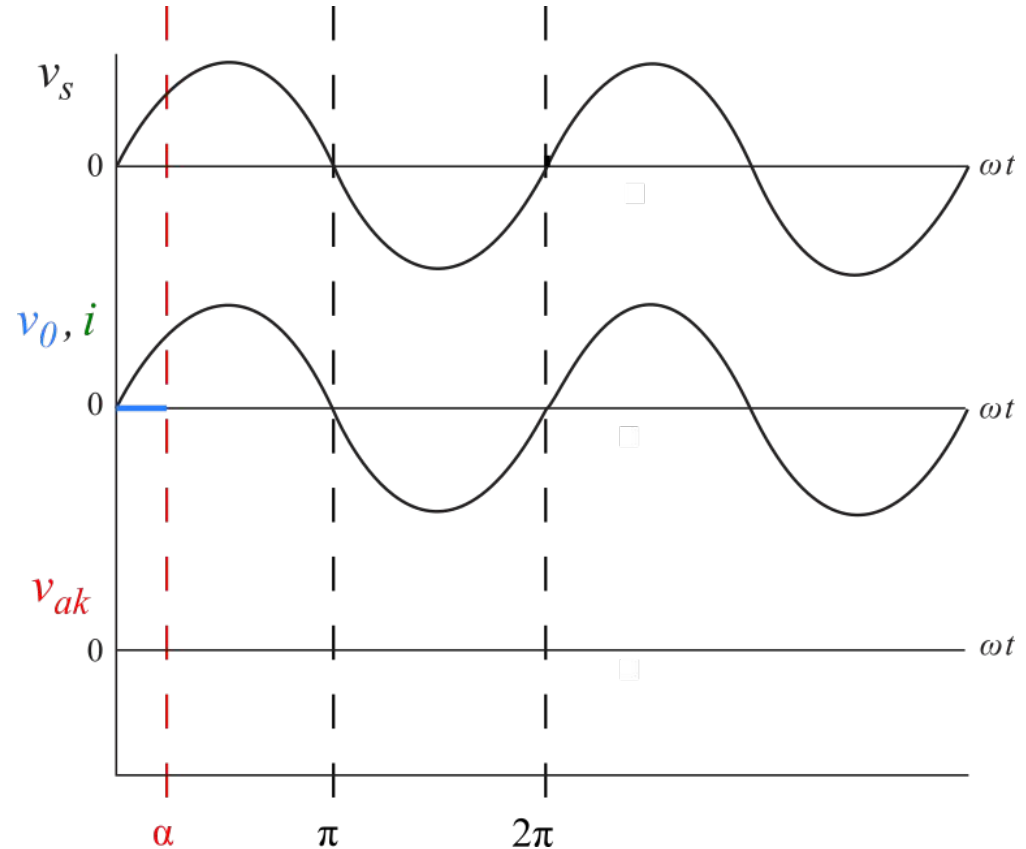
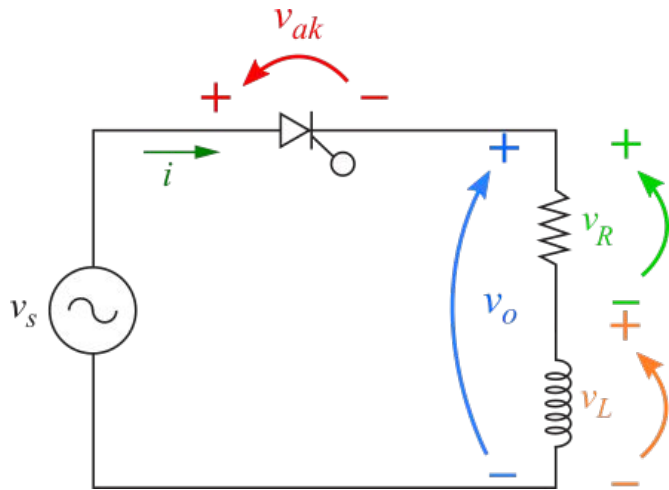
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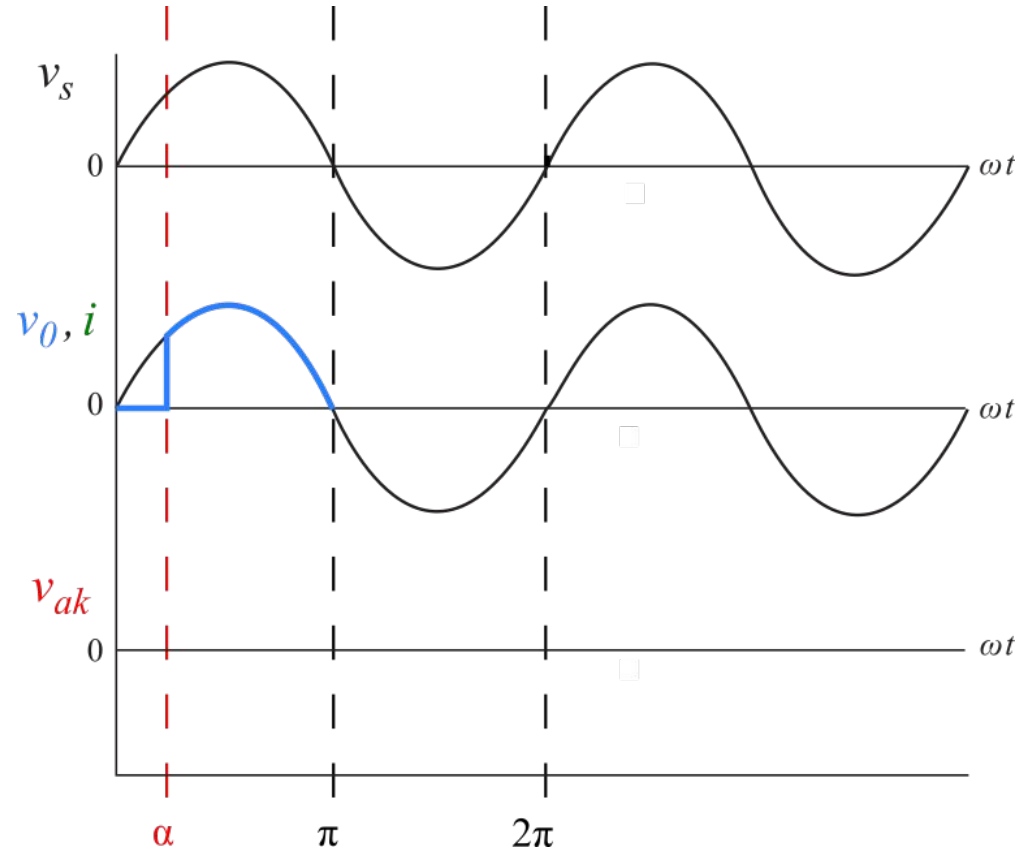
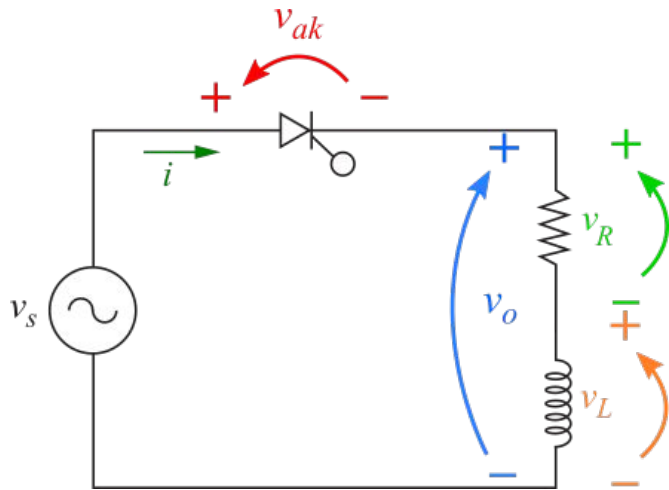
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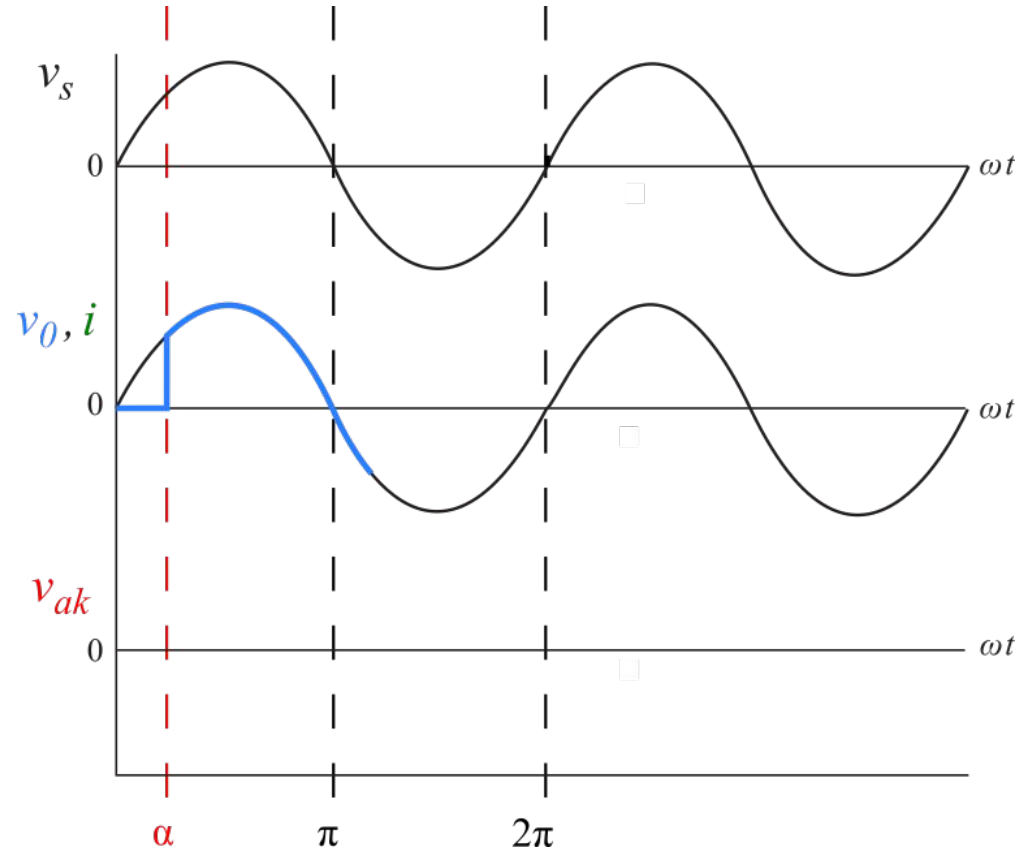
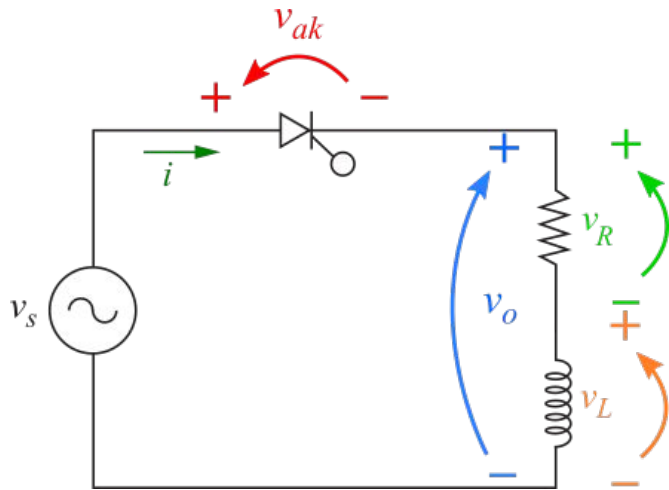
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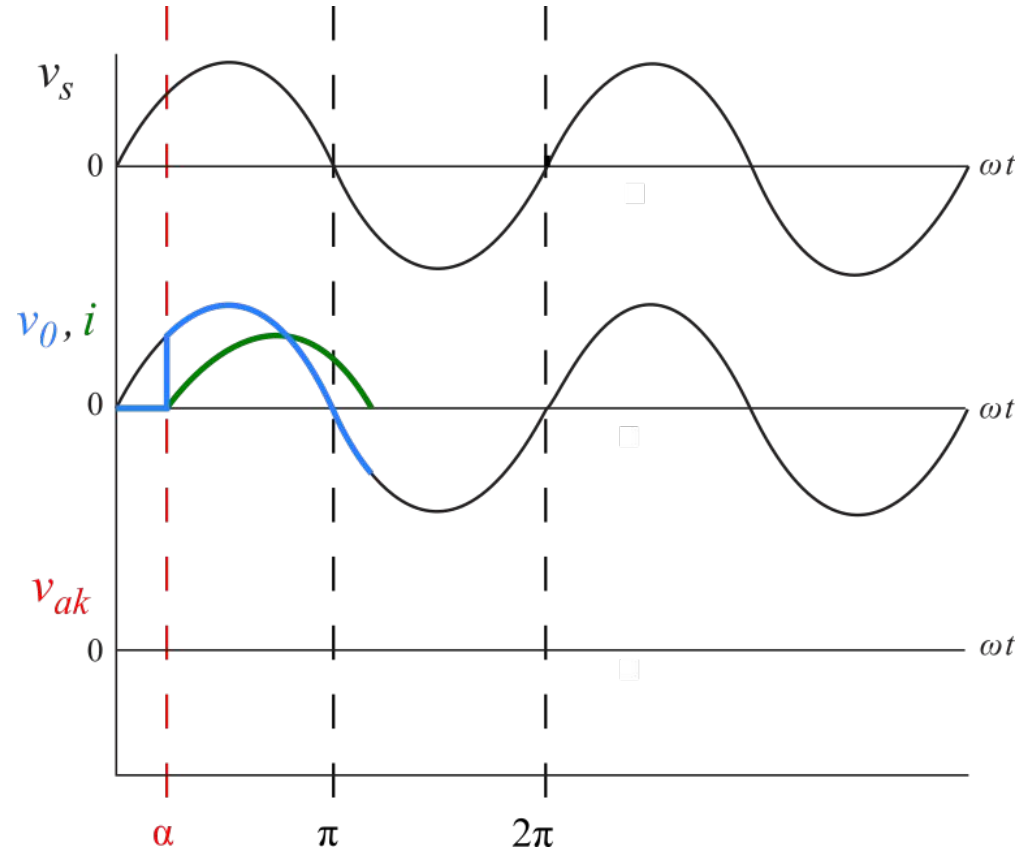
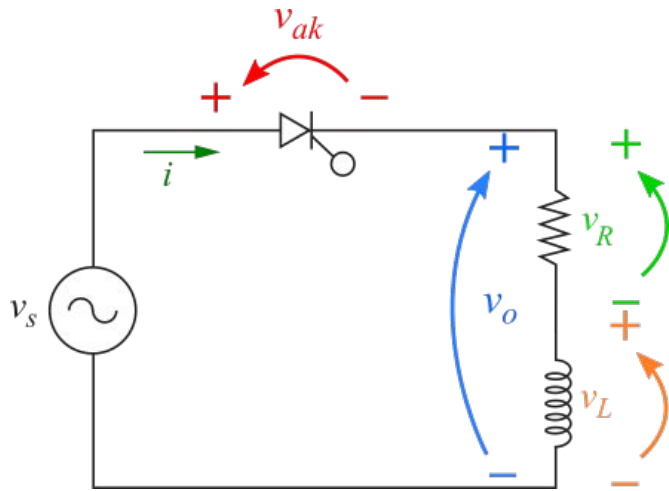
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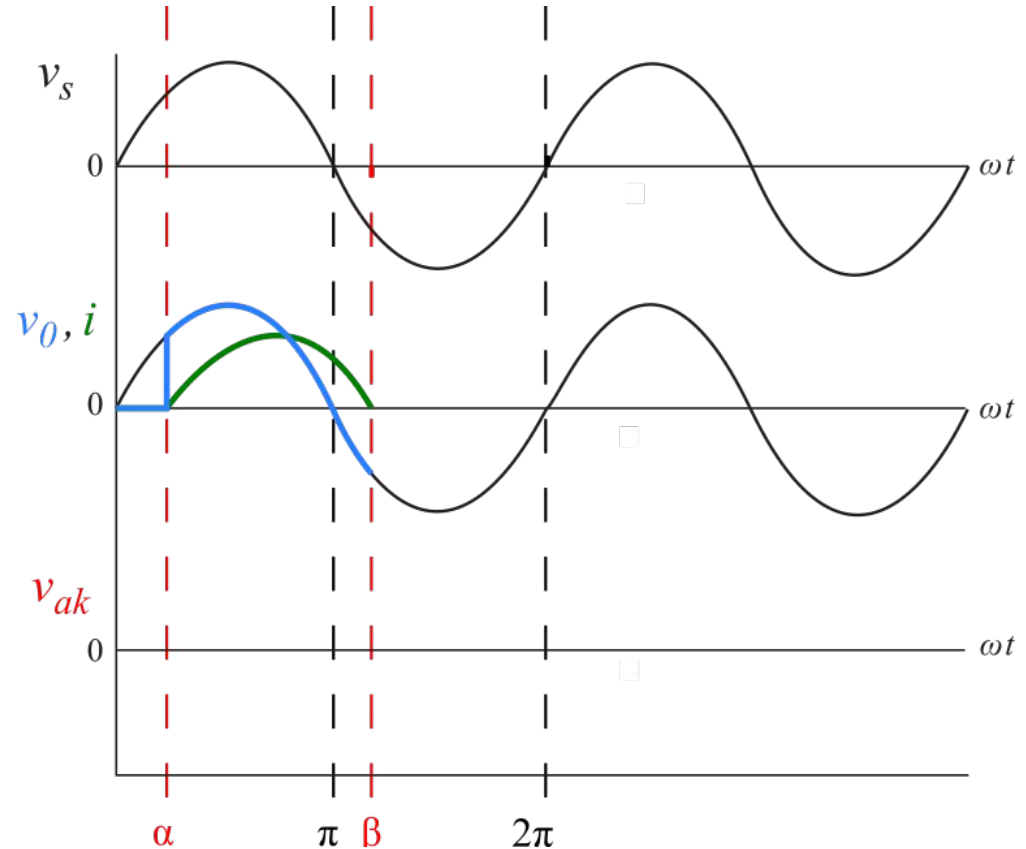
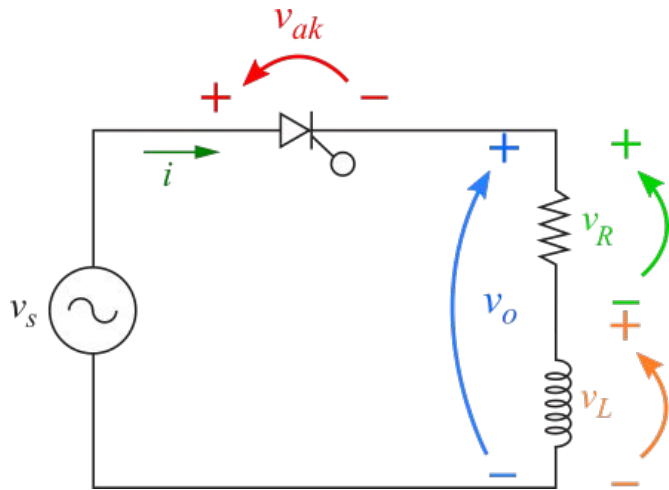
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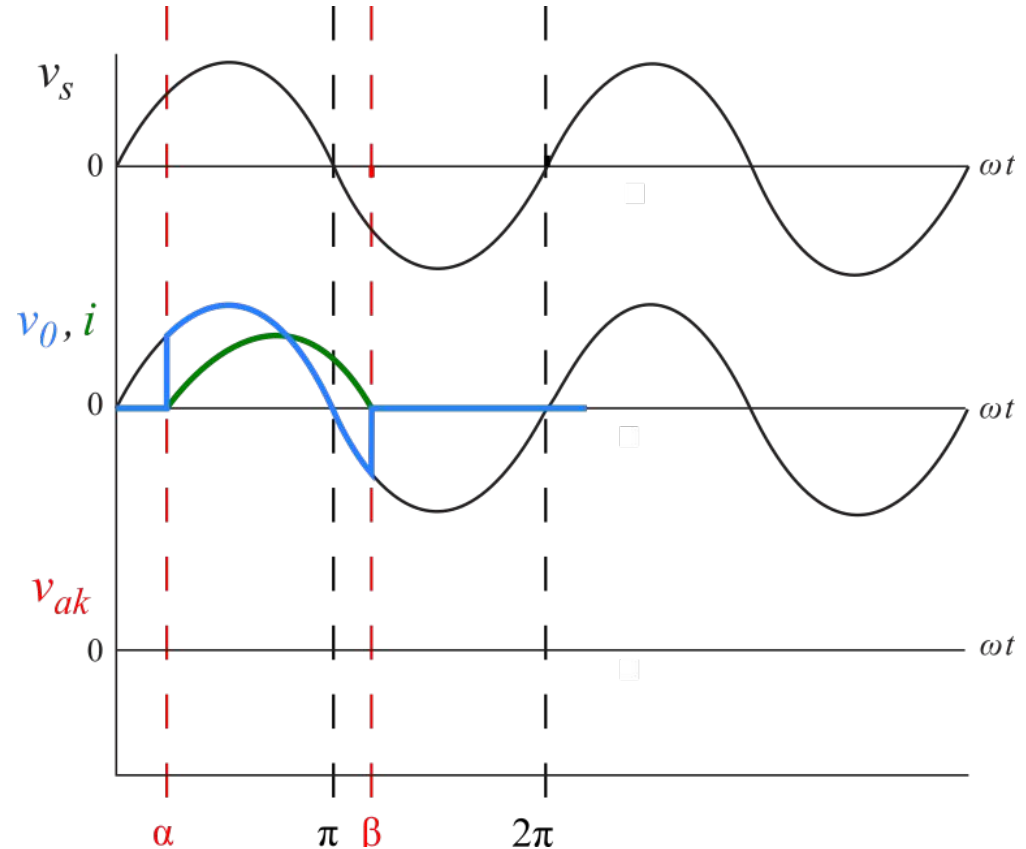
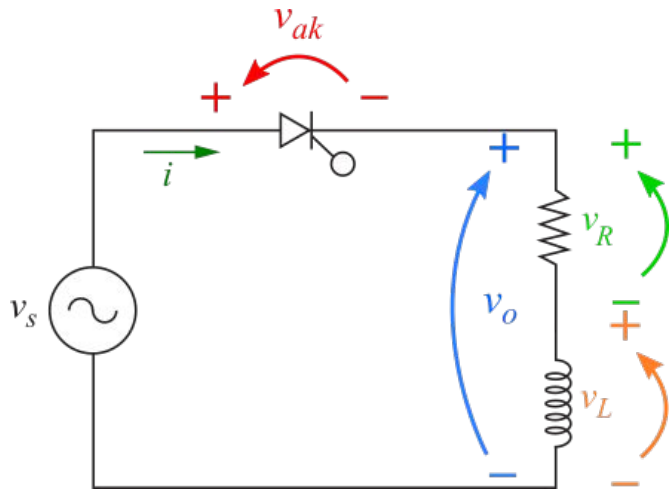
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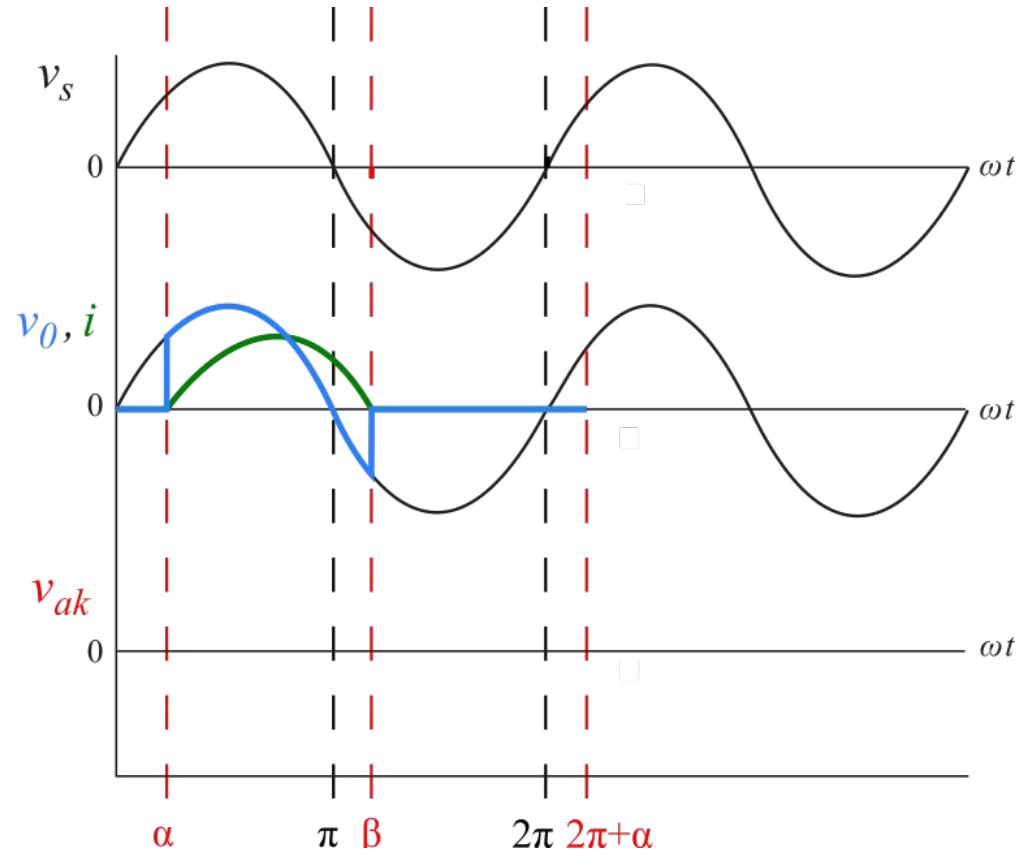
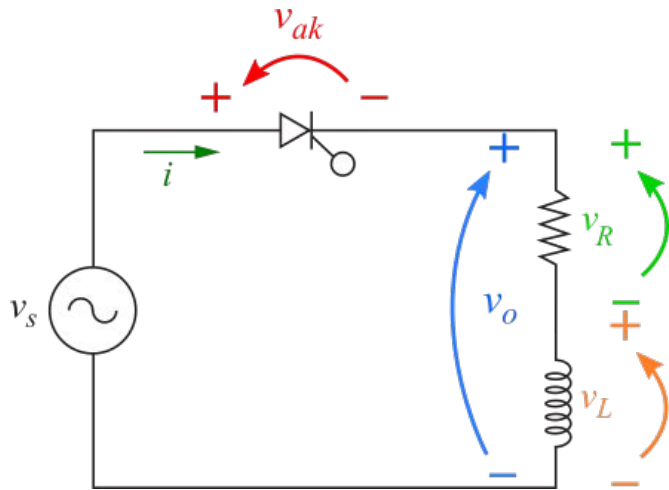
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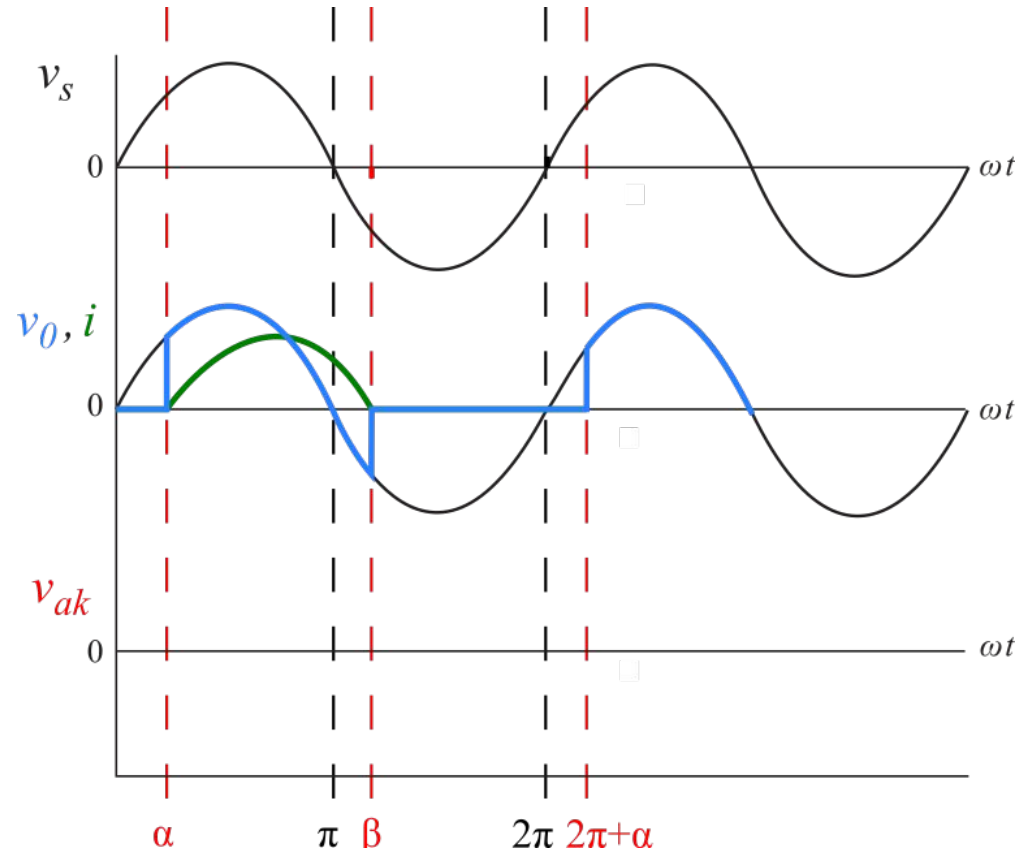
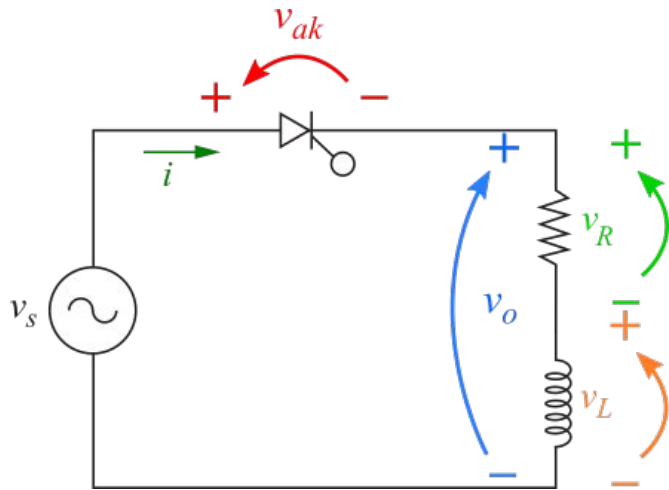
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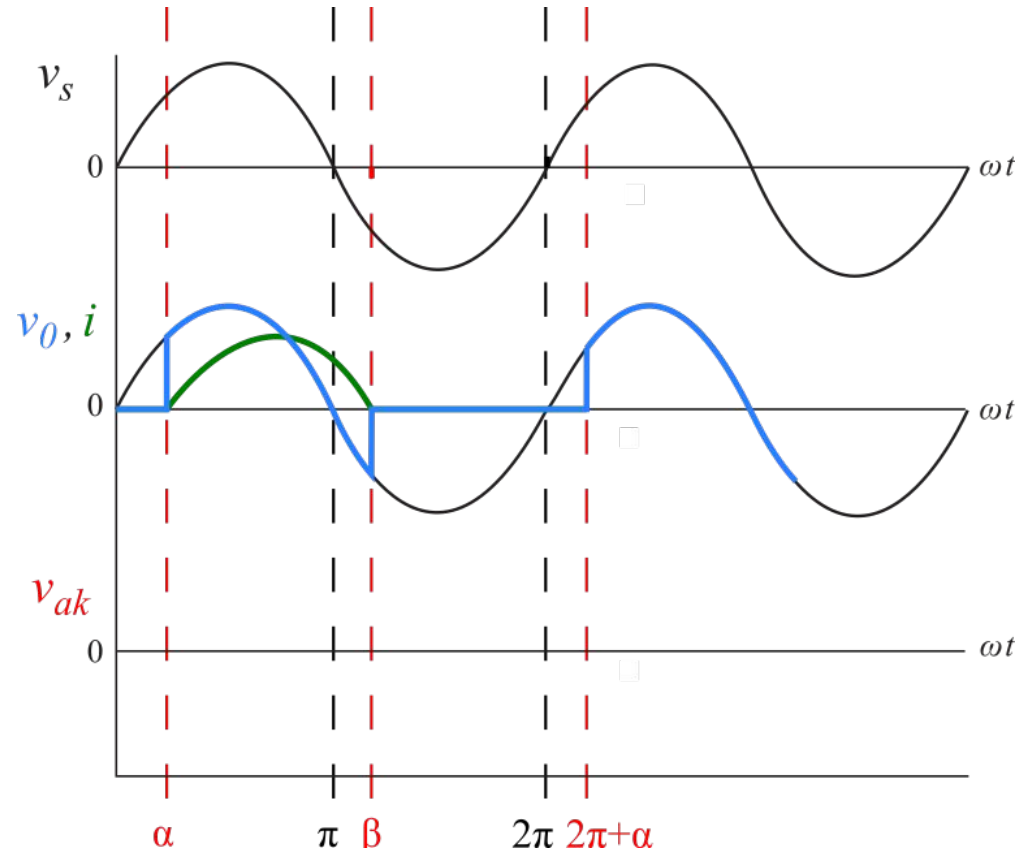
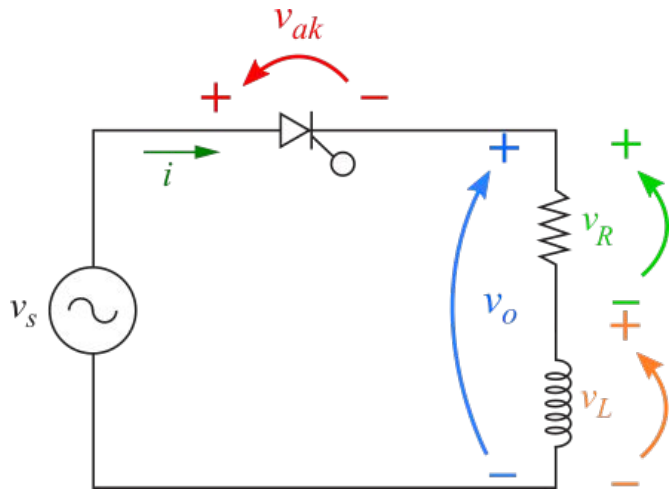
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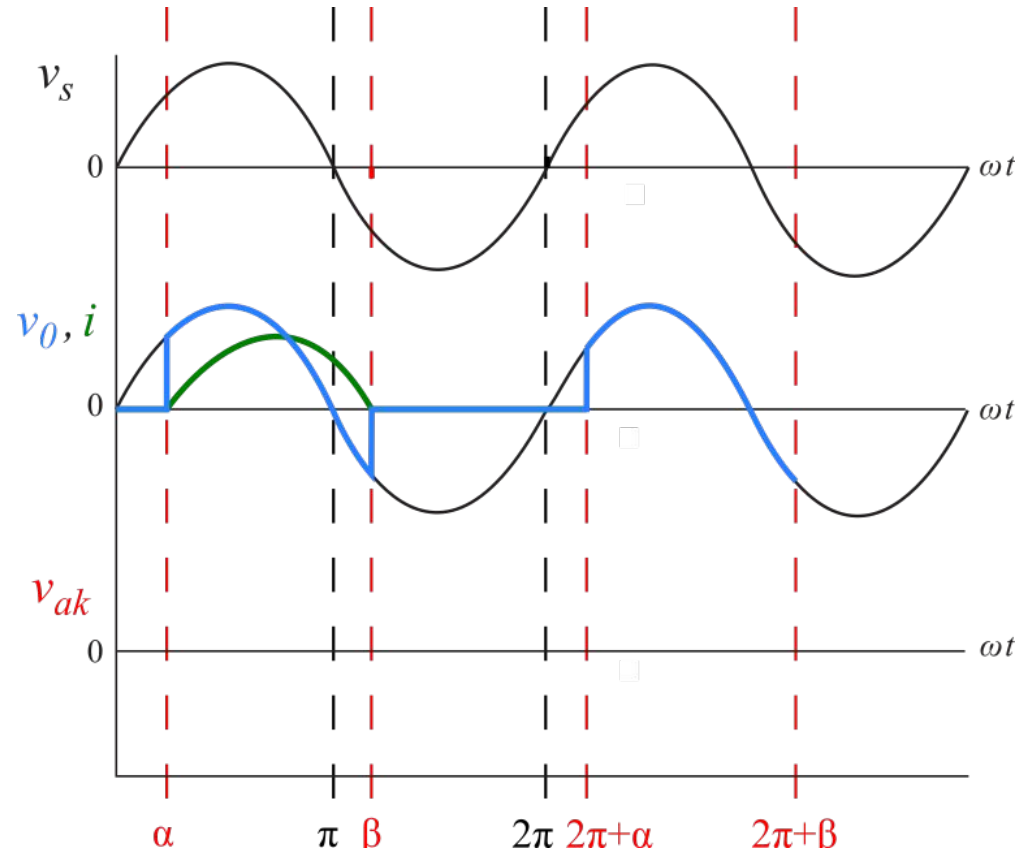
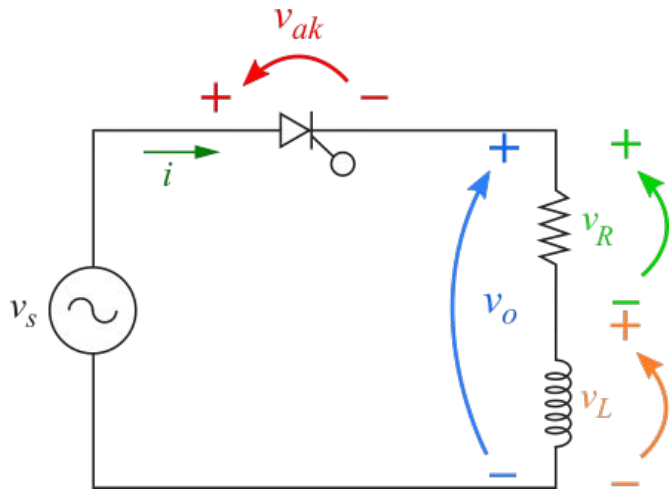
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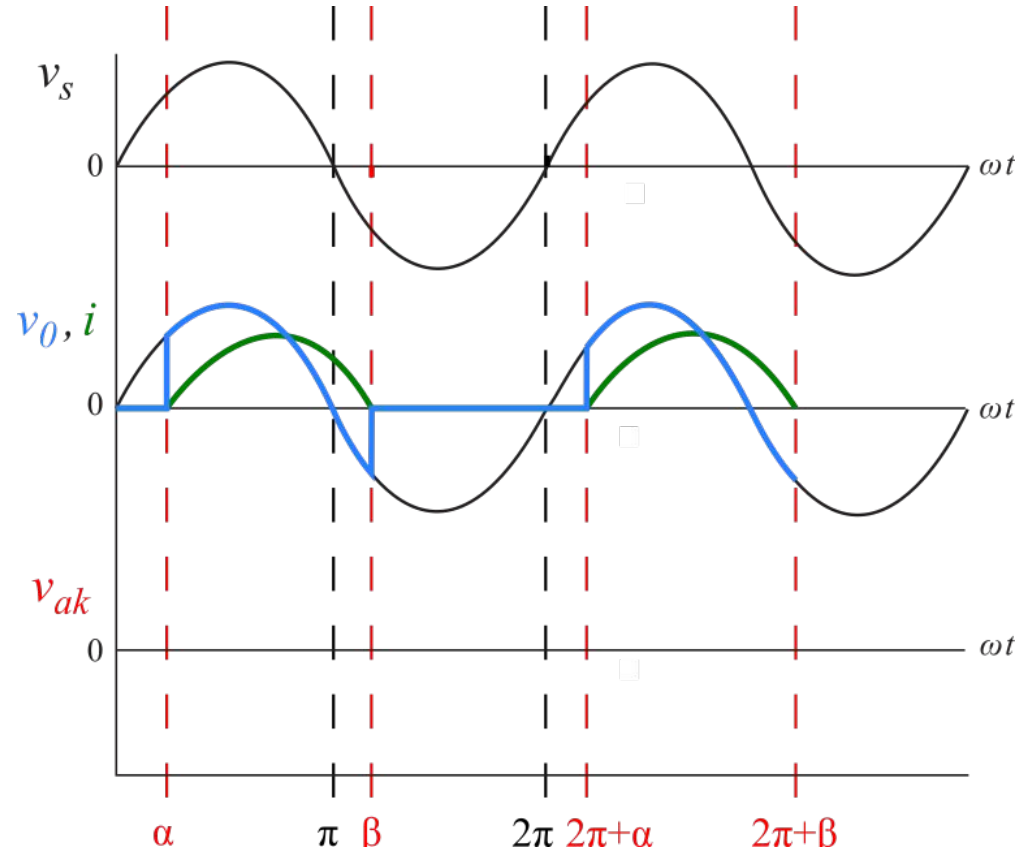
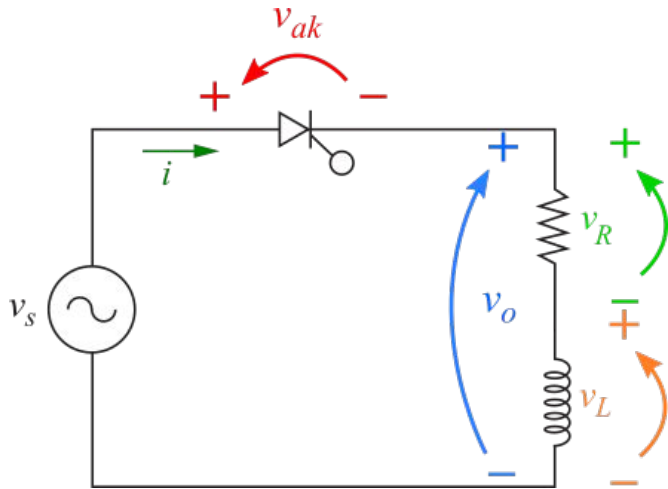
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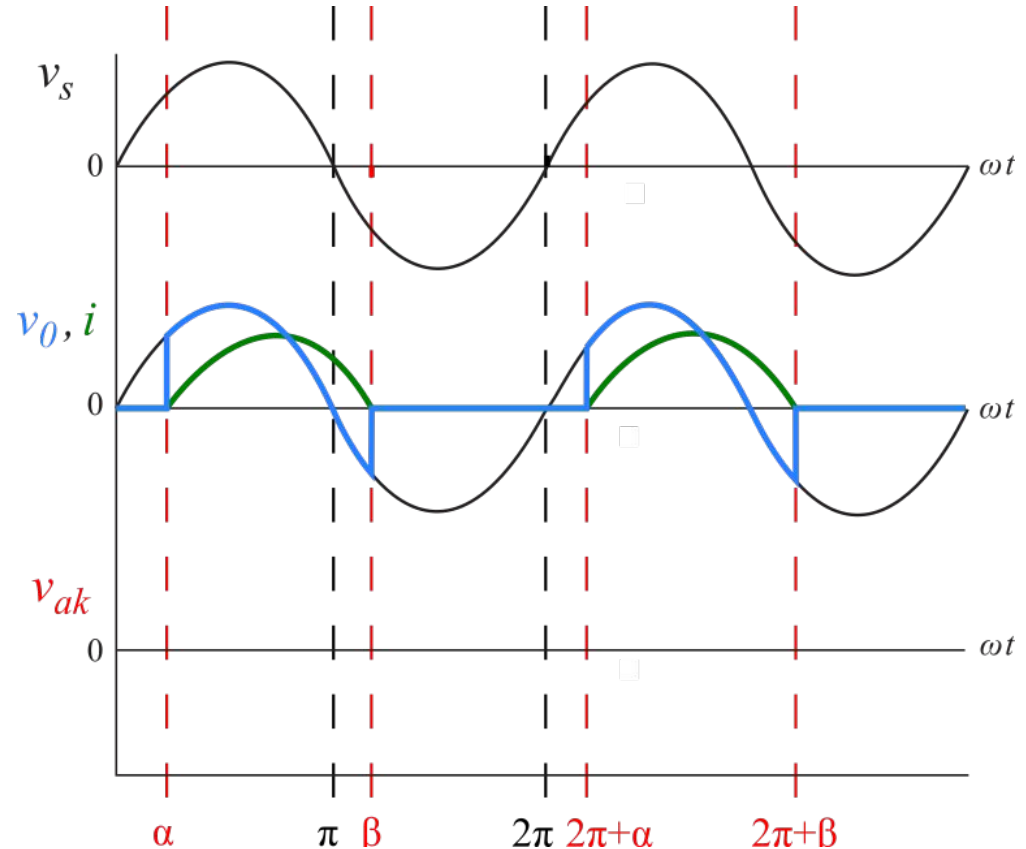
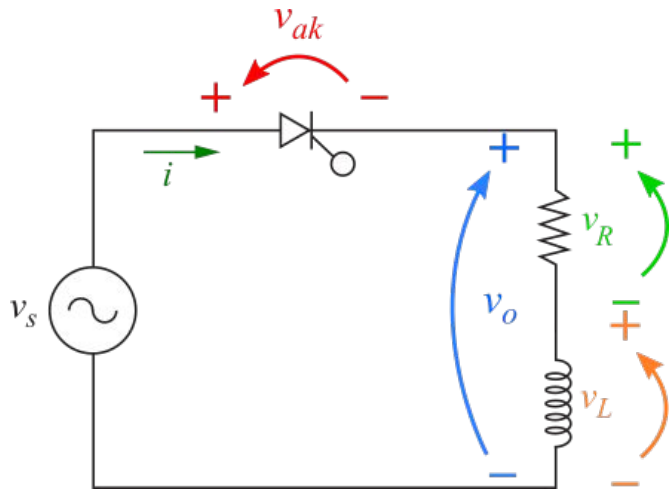
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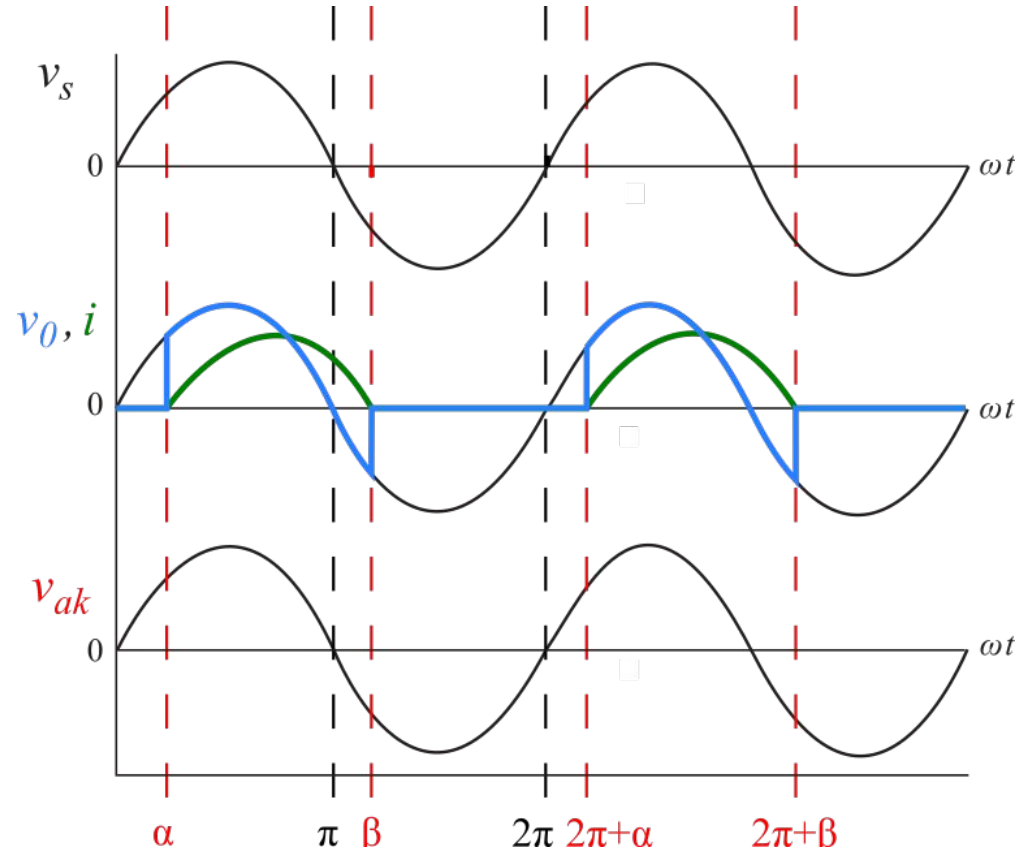
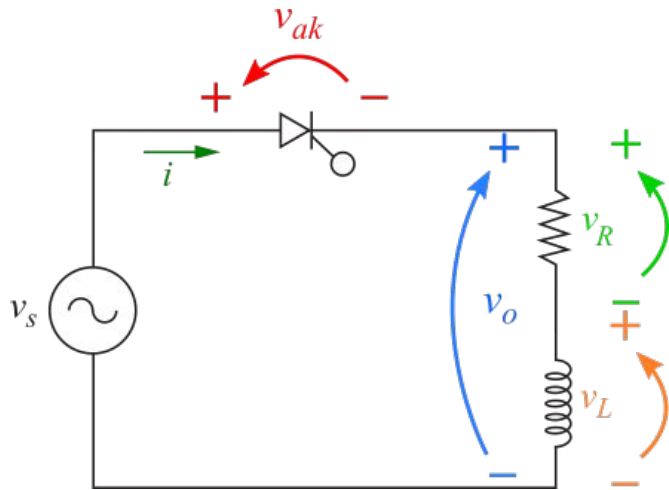
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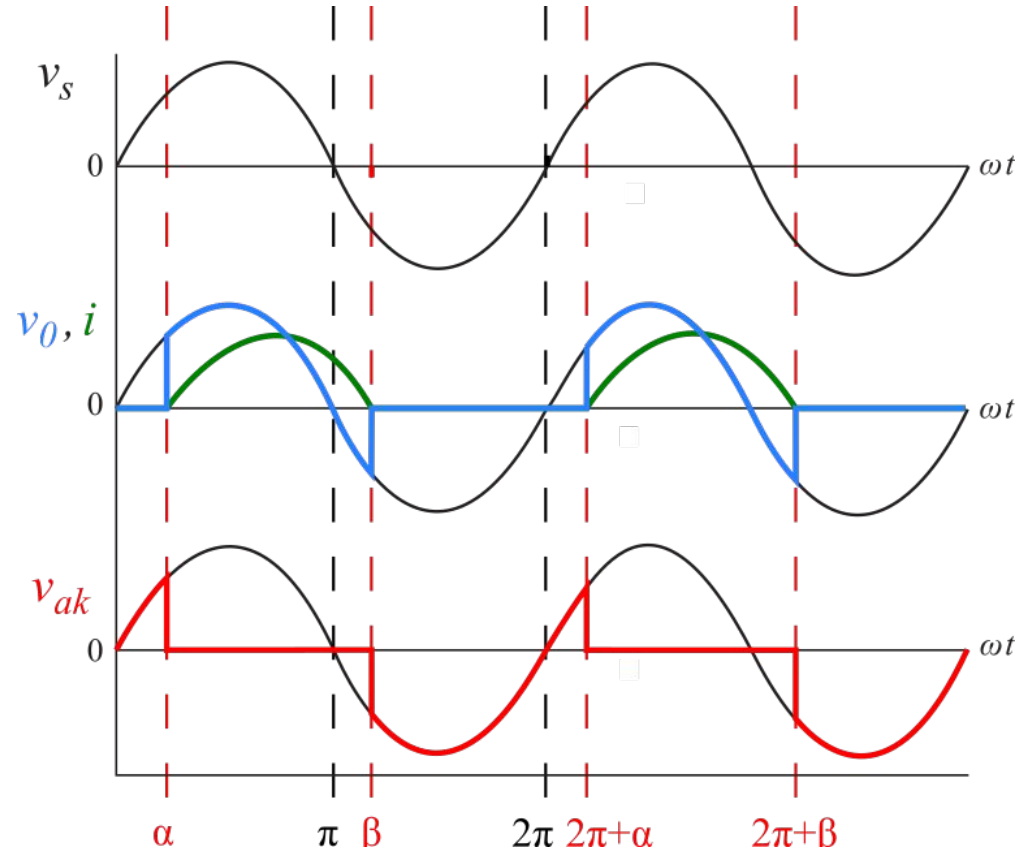
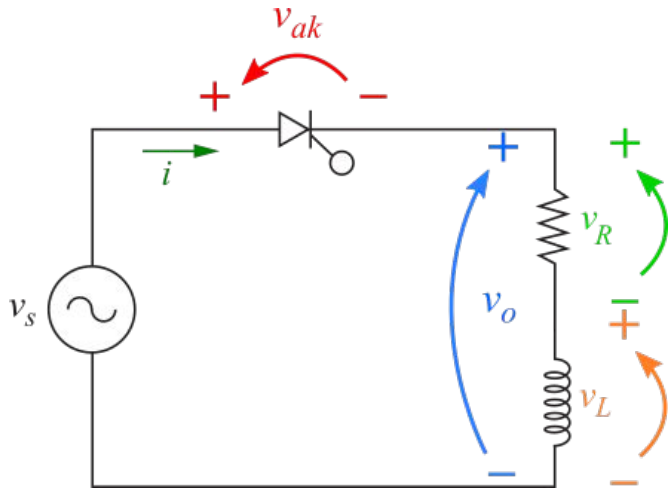
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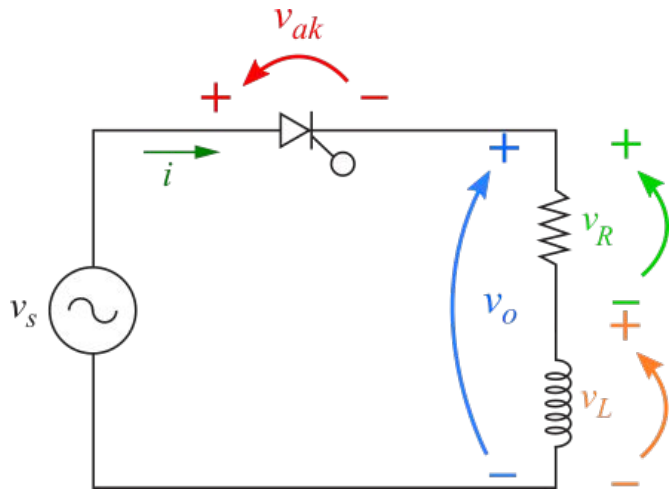
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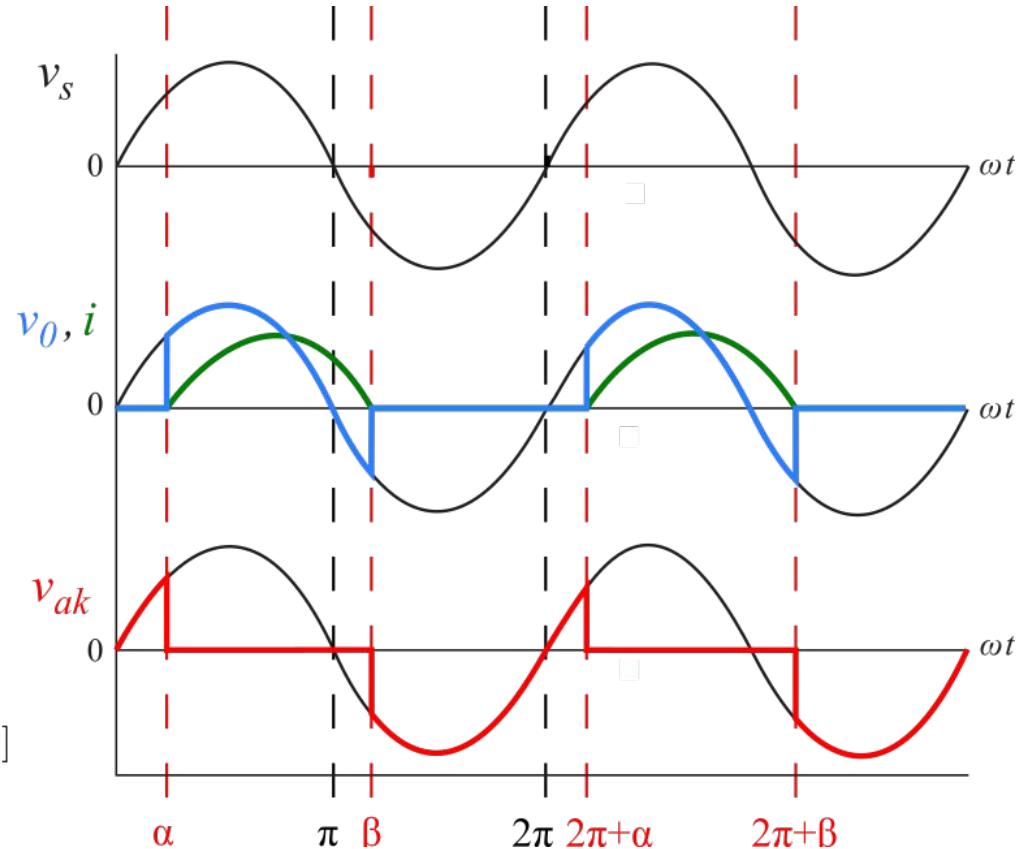


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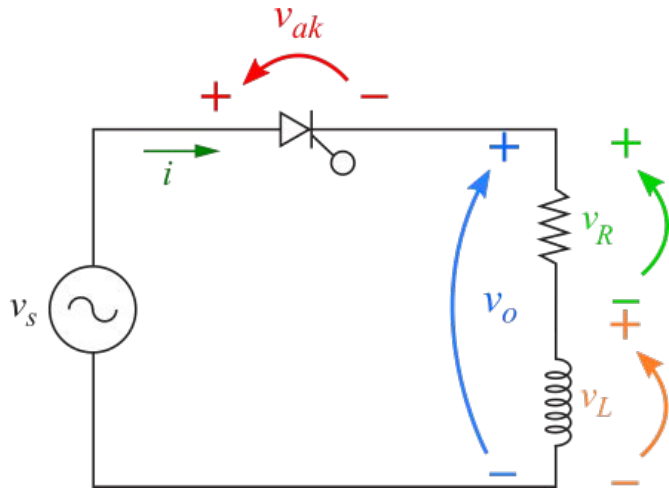


$$V_0 = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin(\omega t) d\omega t = \frac{V_m}{2\pi} [\cos(\alpha) - \cos(\beta)]$$

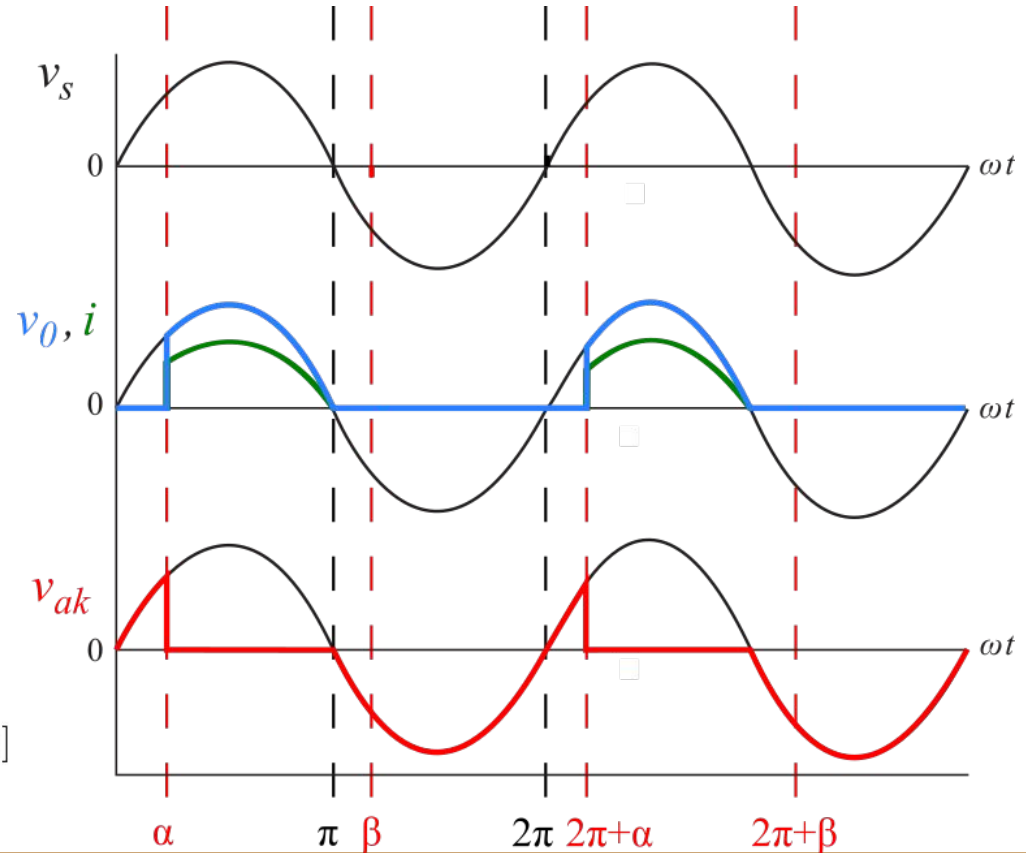


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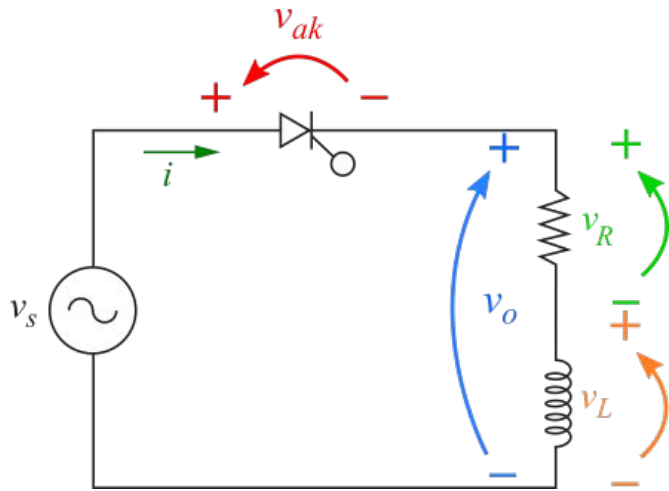


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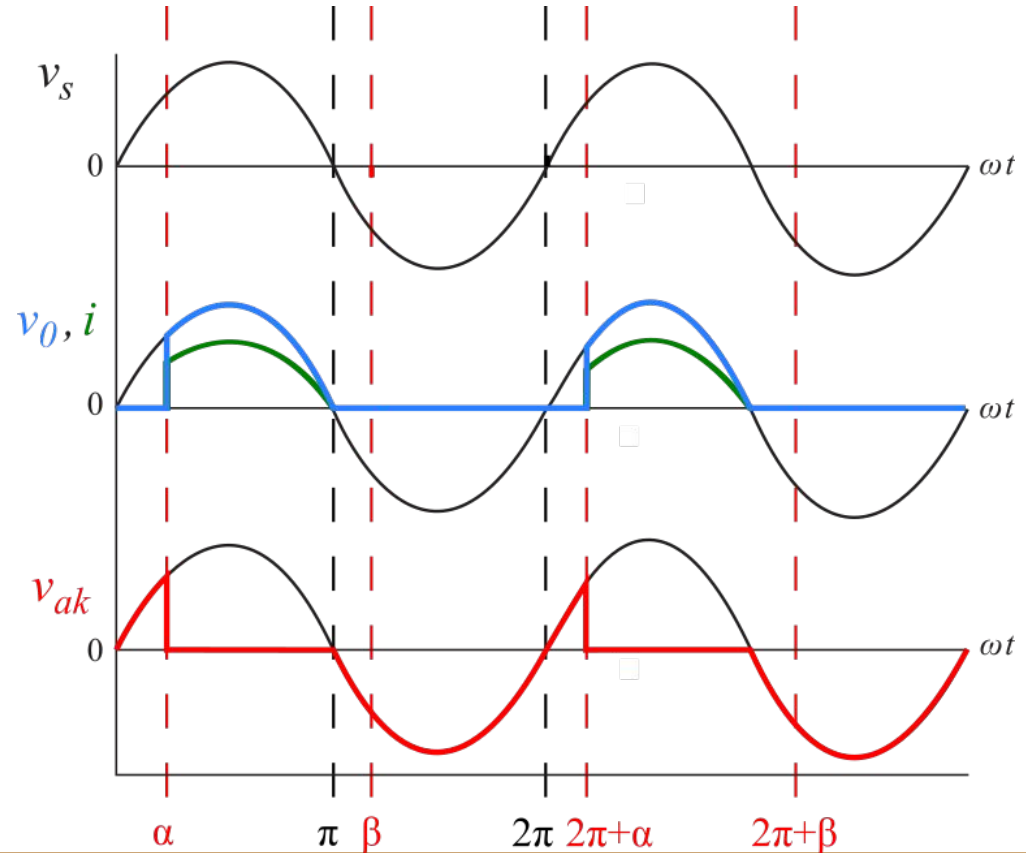


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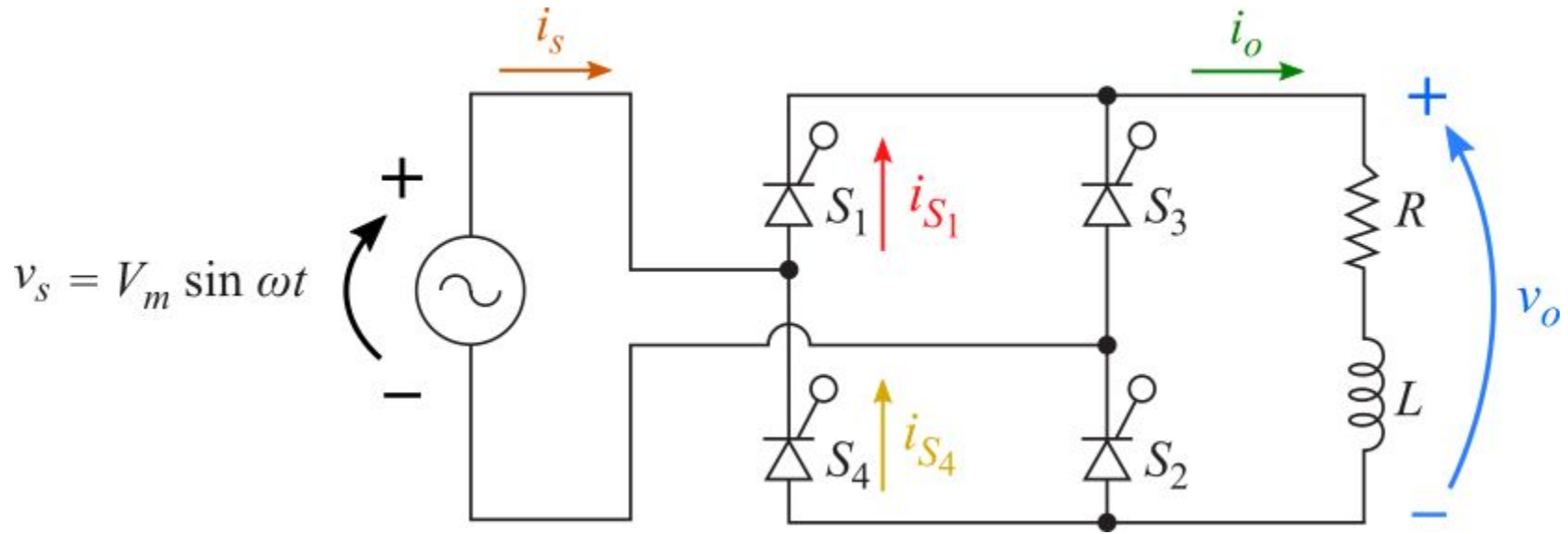


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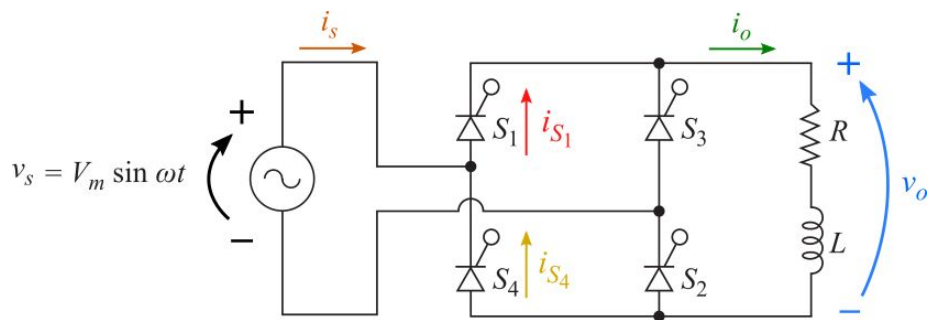
Rectificador controlado de onda completa

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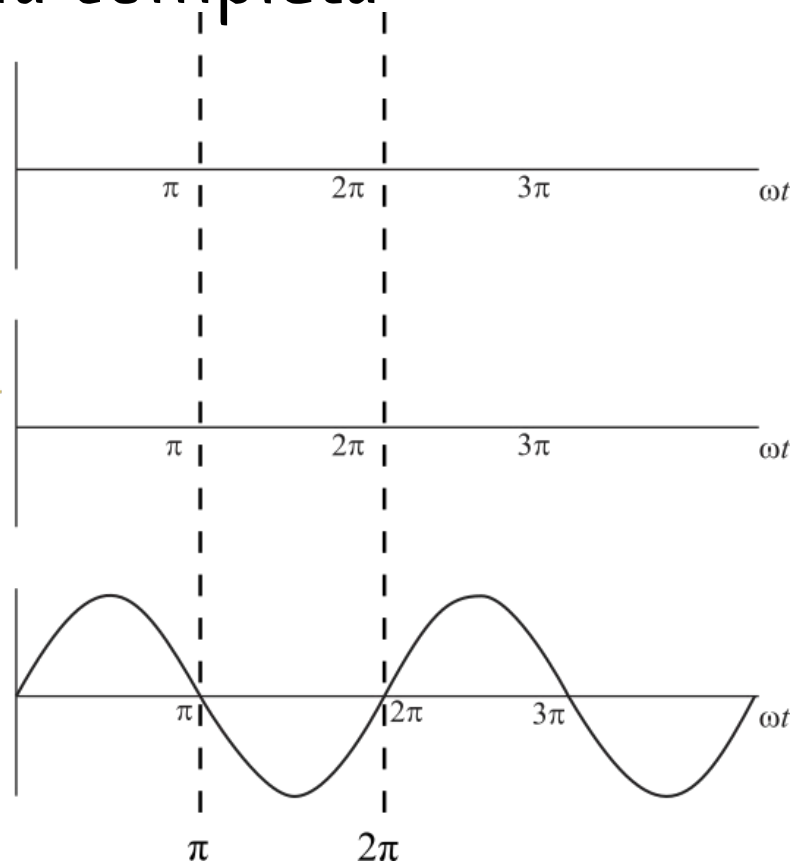
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v_o, i_o

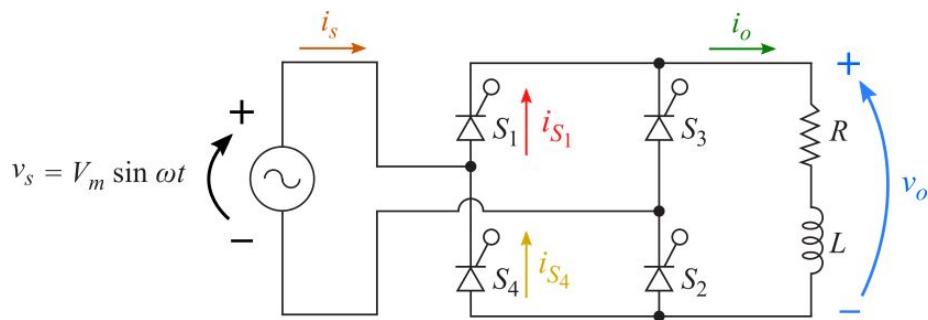
i_{S1}, i_{S4}

v_s, i_s

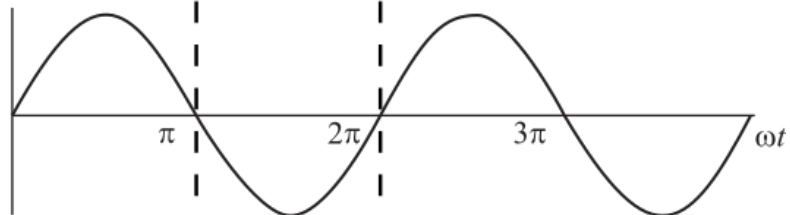


Rectificador controlado de onda completa

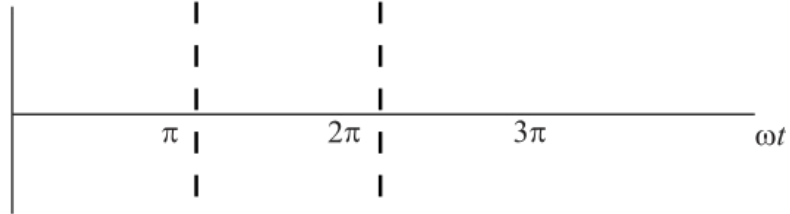
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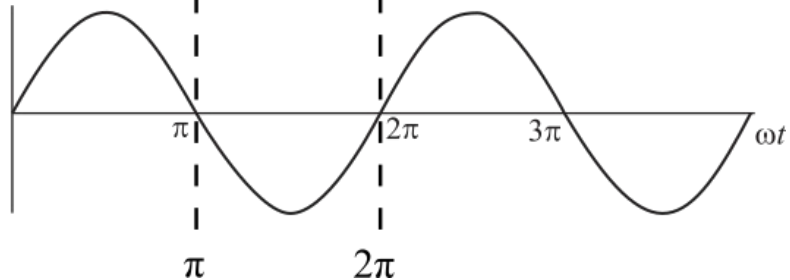
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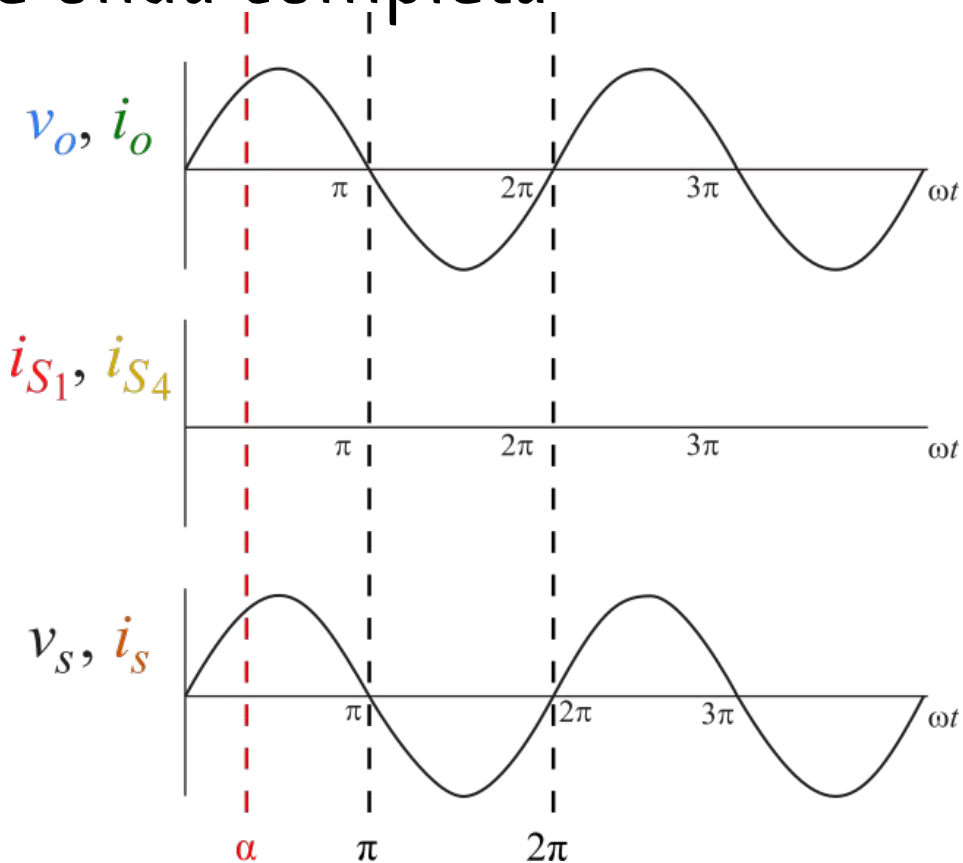
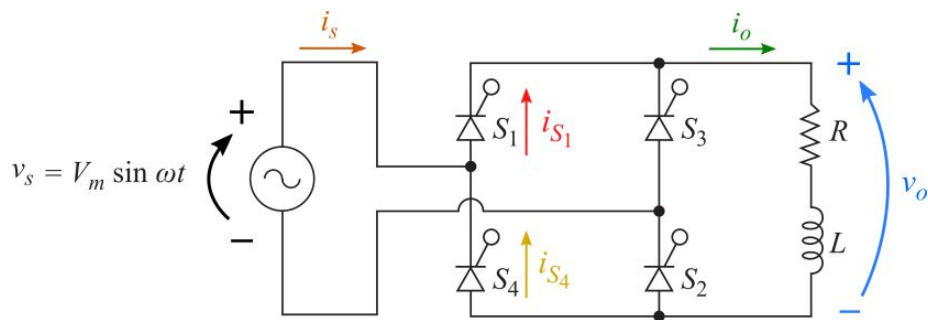


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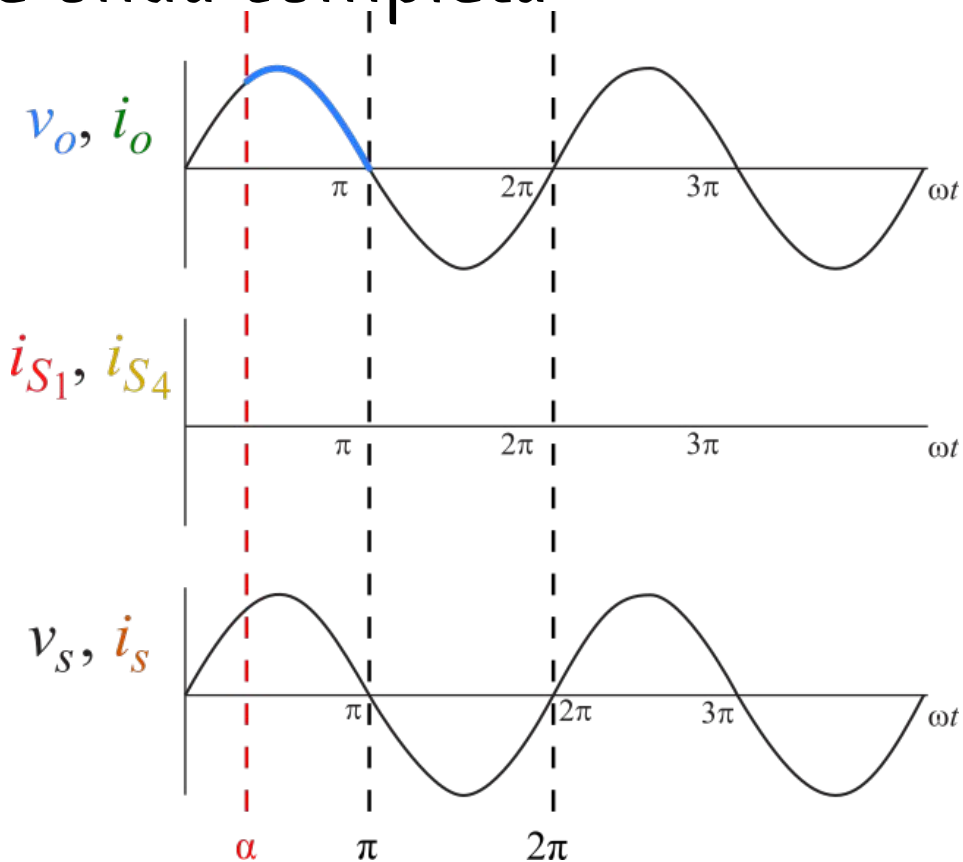
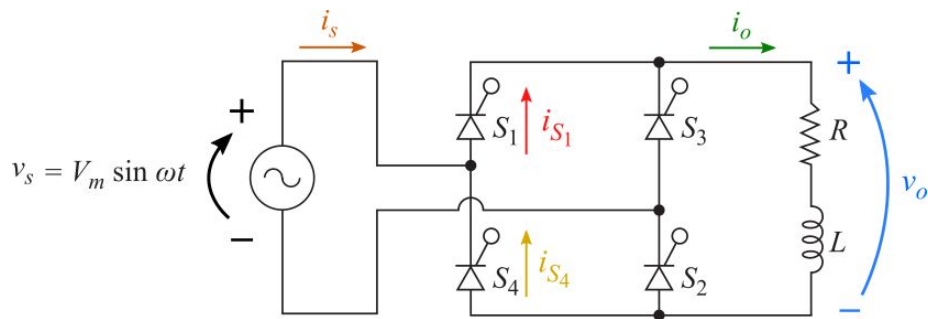
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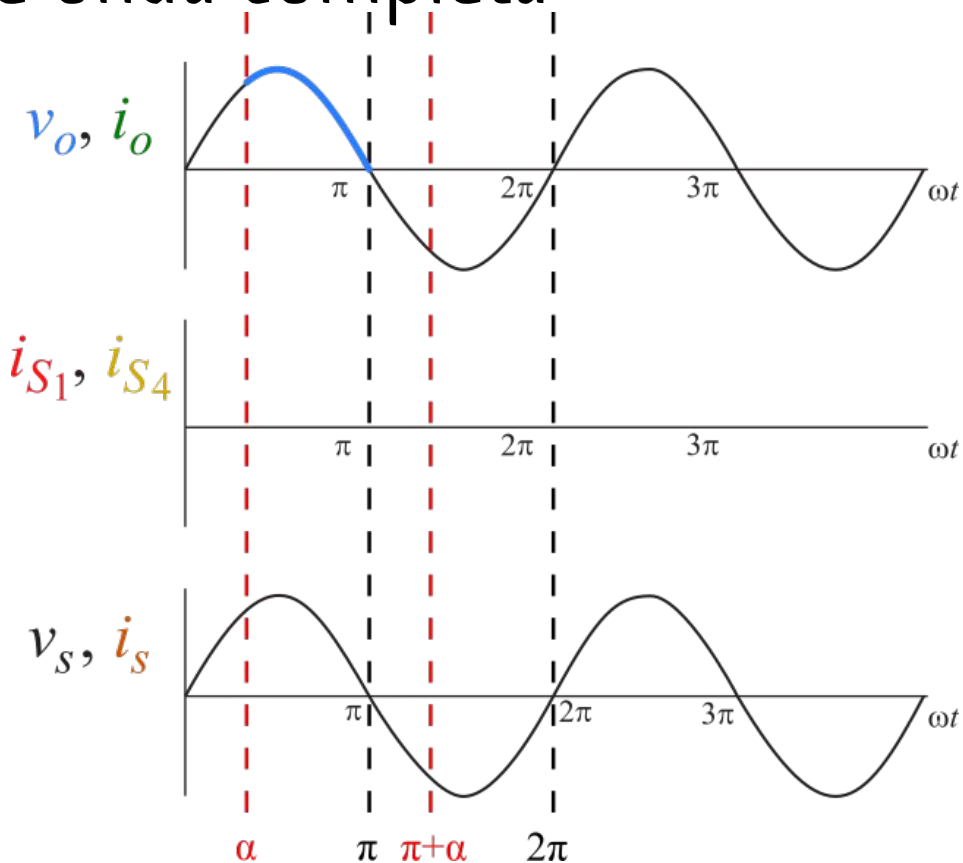
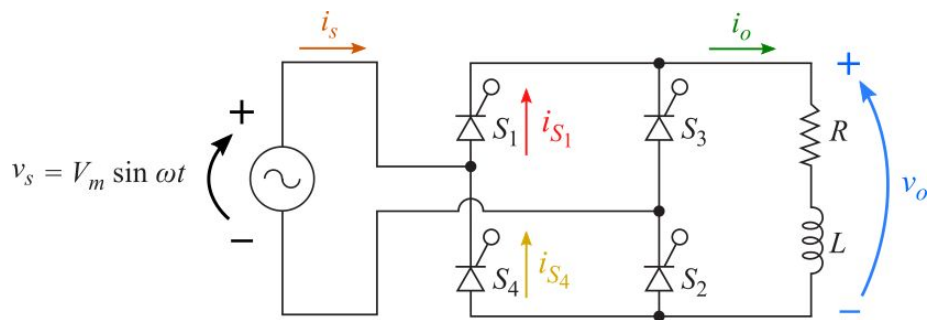
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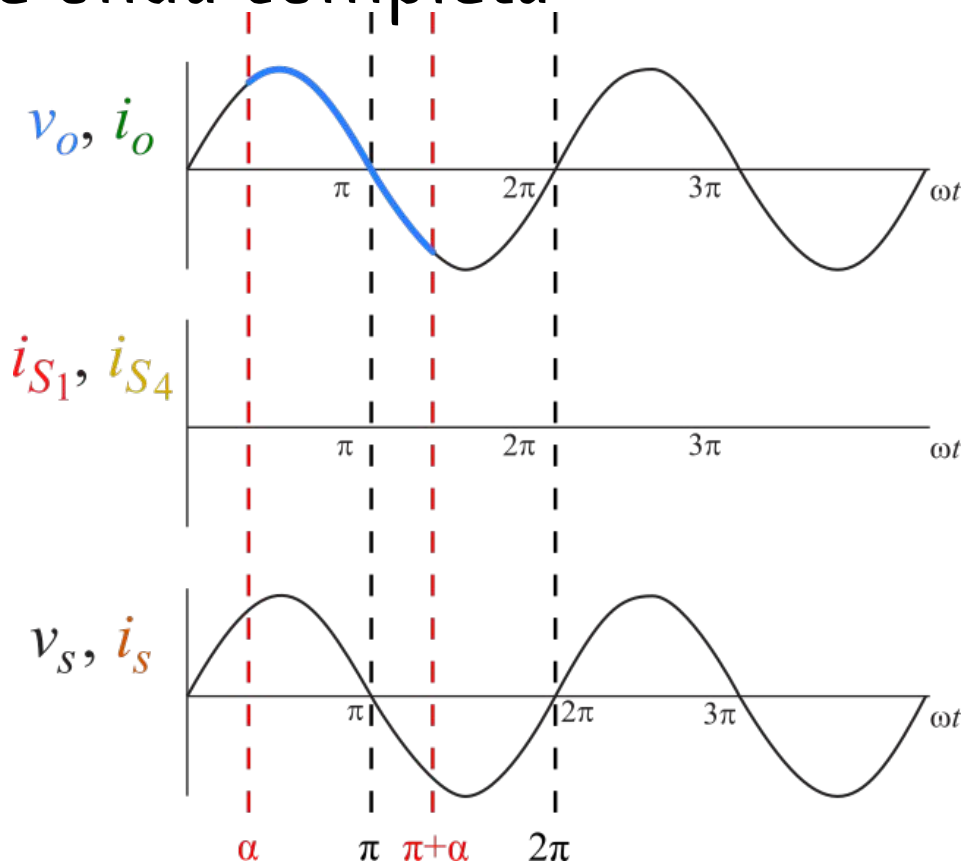
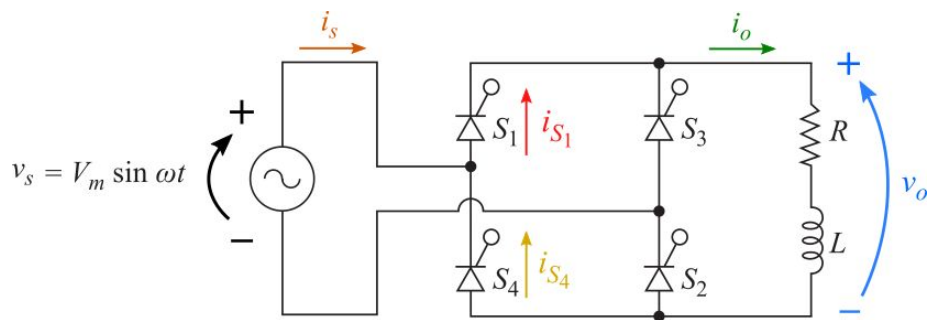
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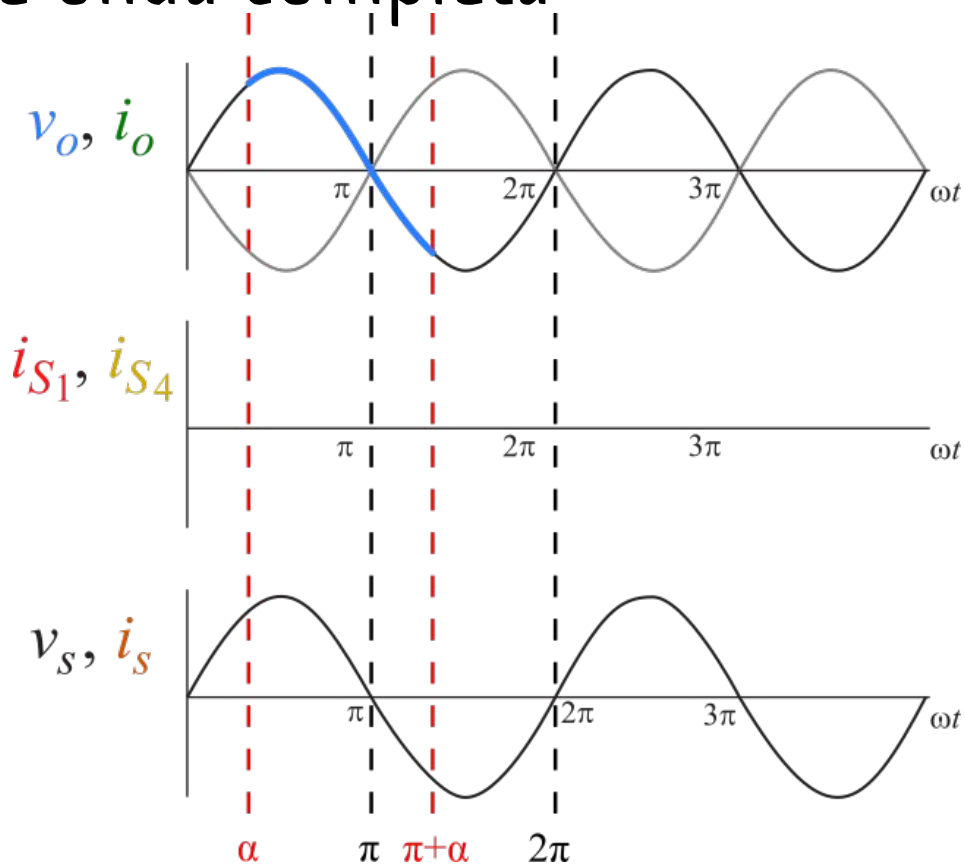
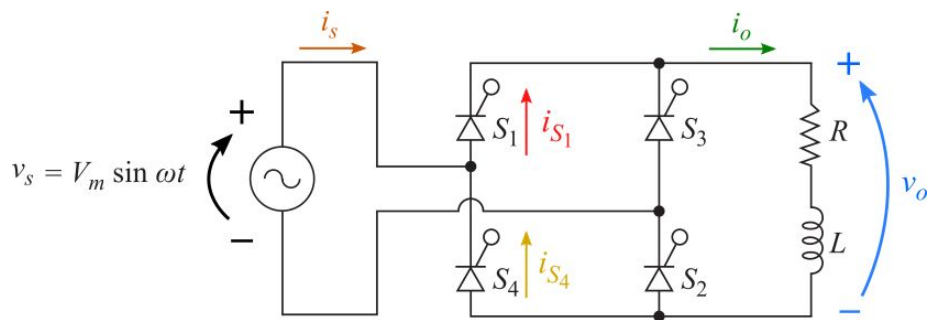
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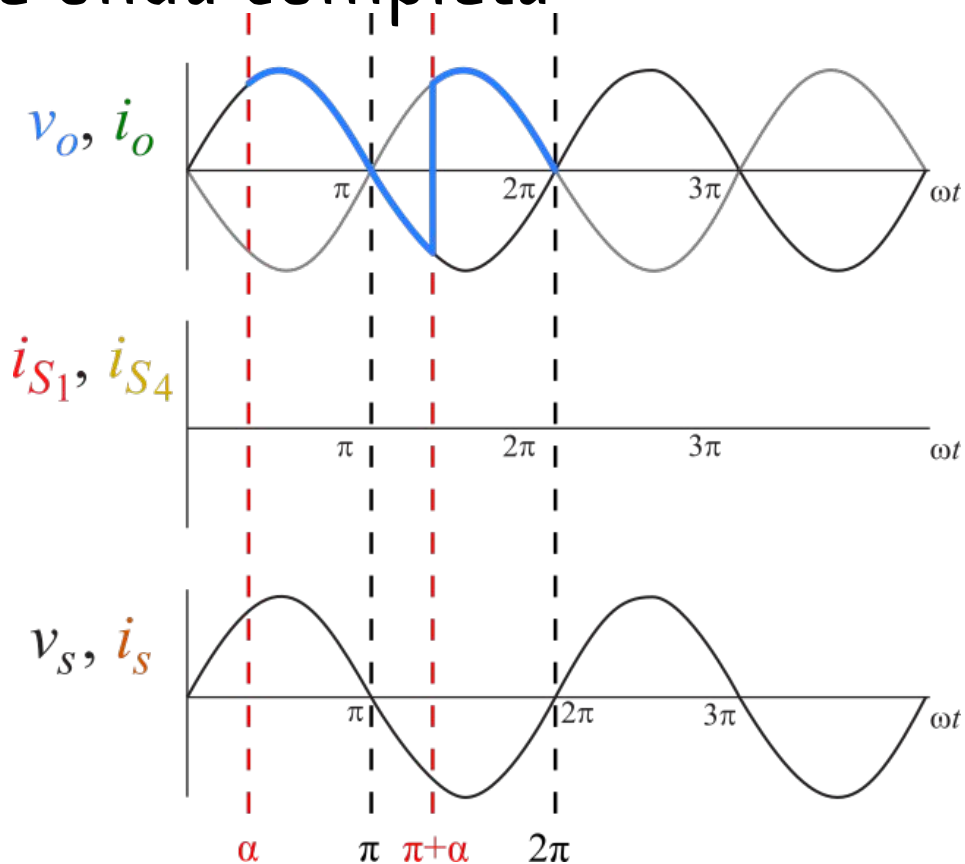
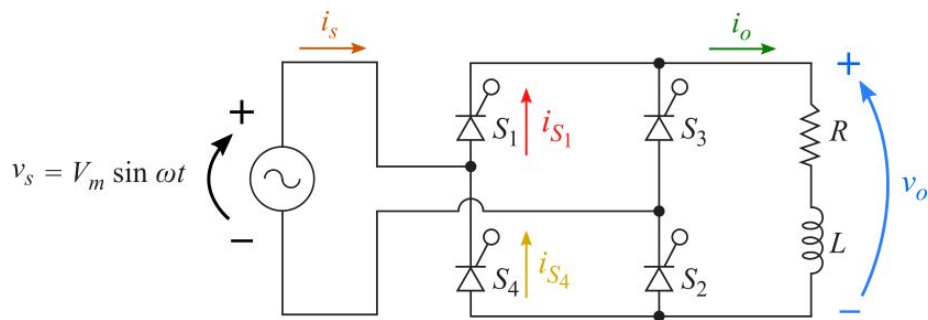
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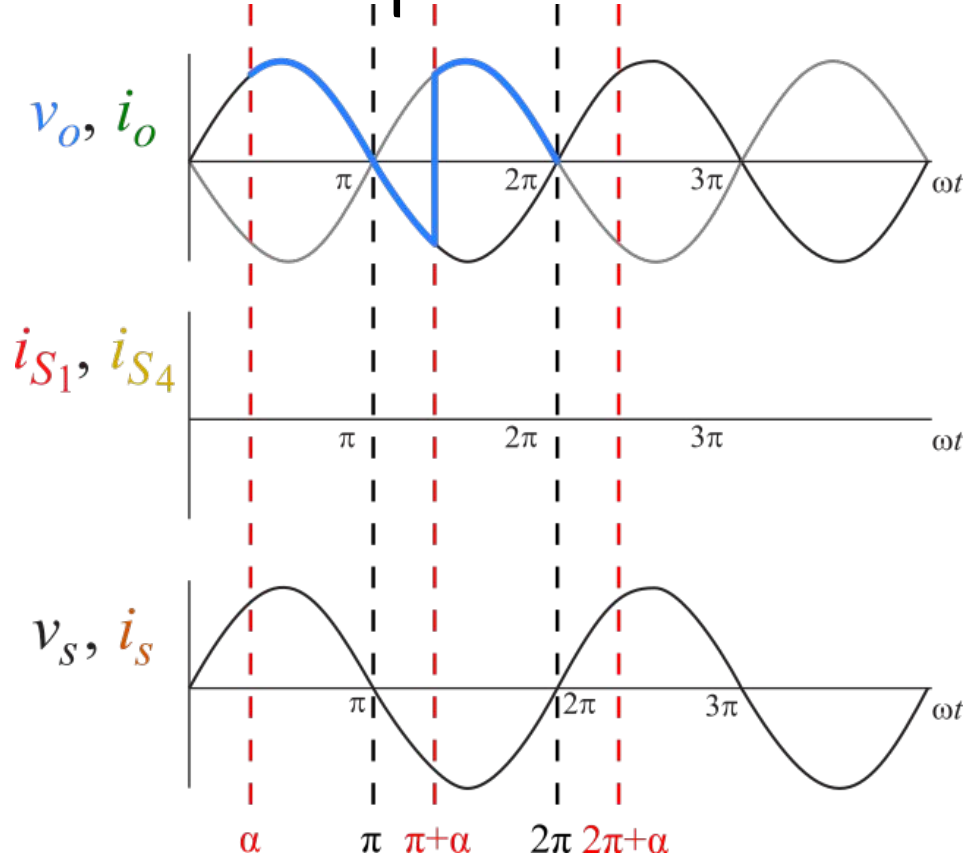
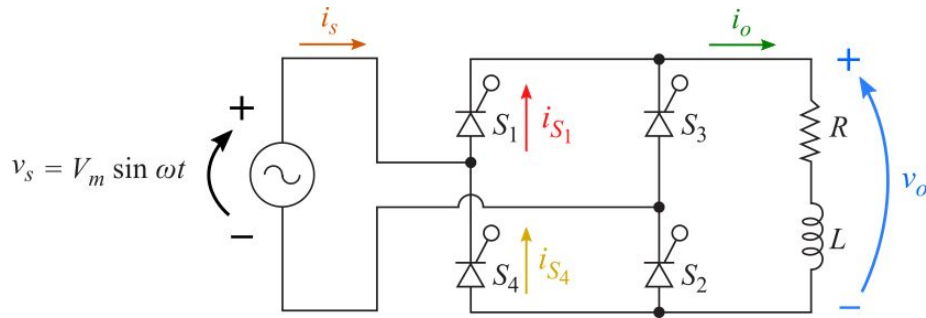
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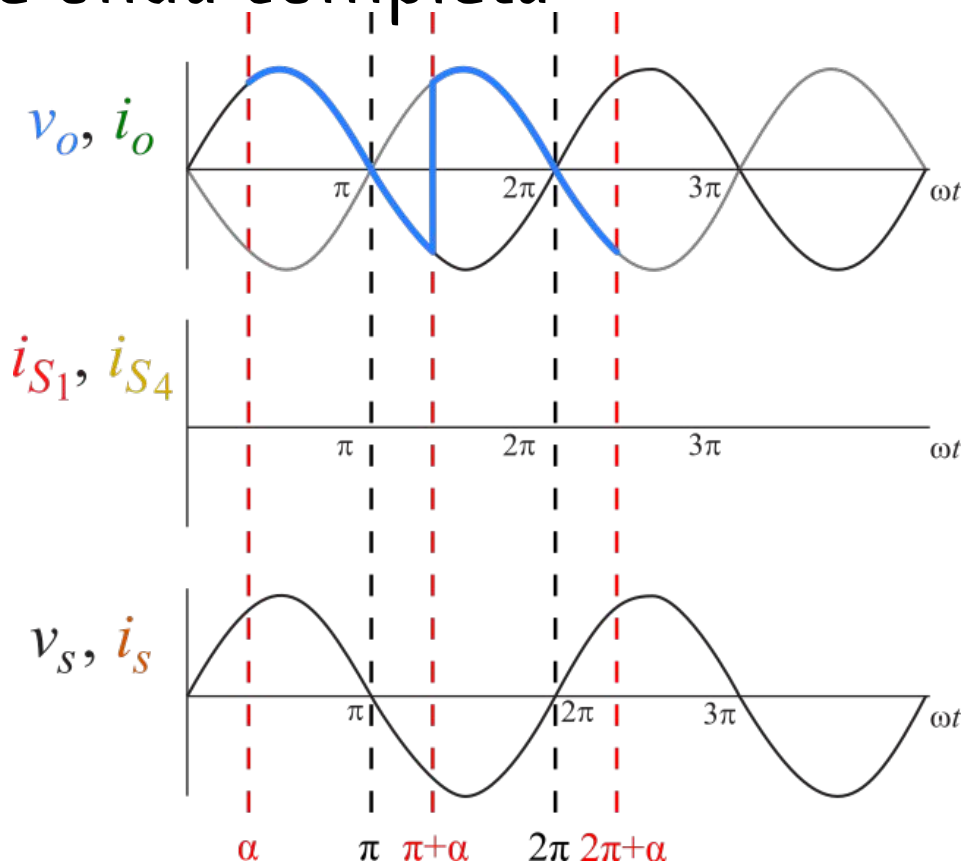
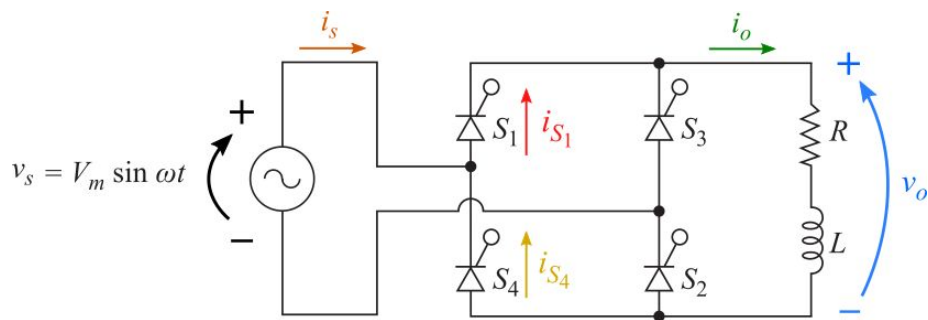
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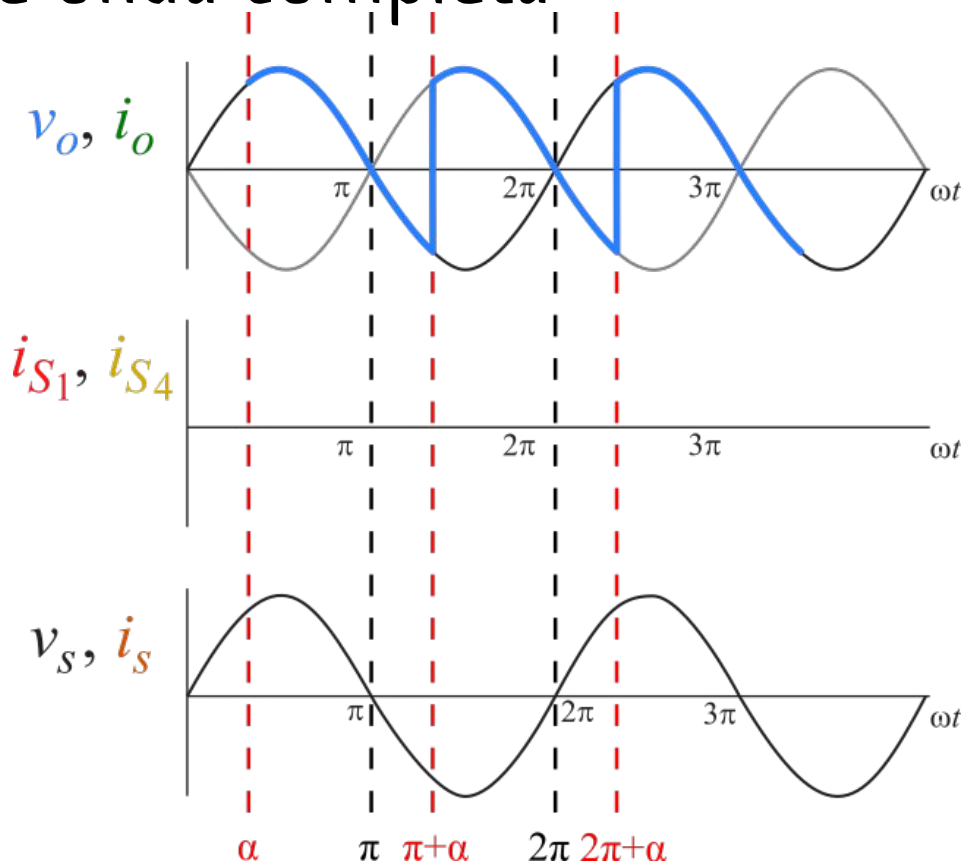
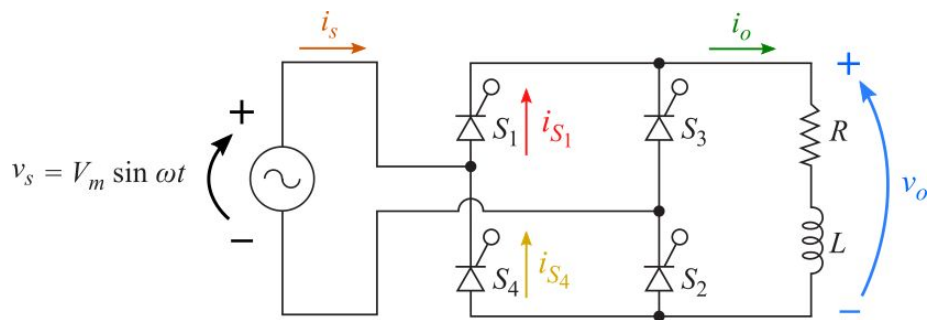
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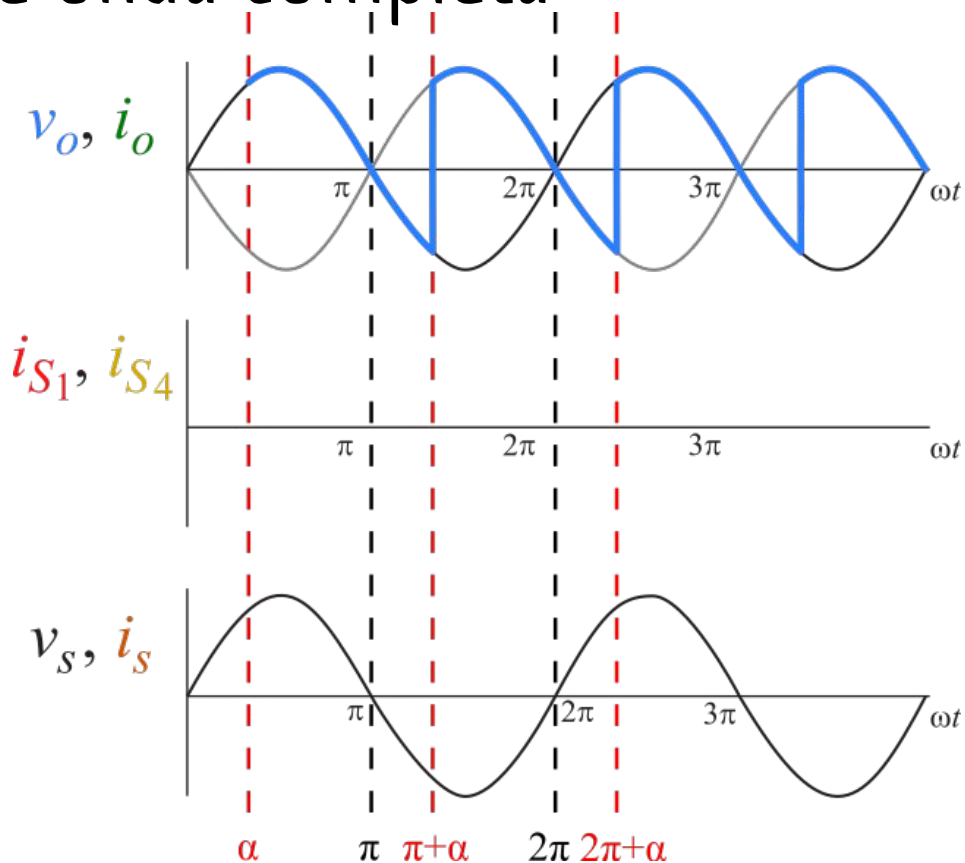
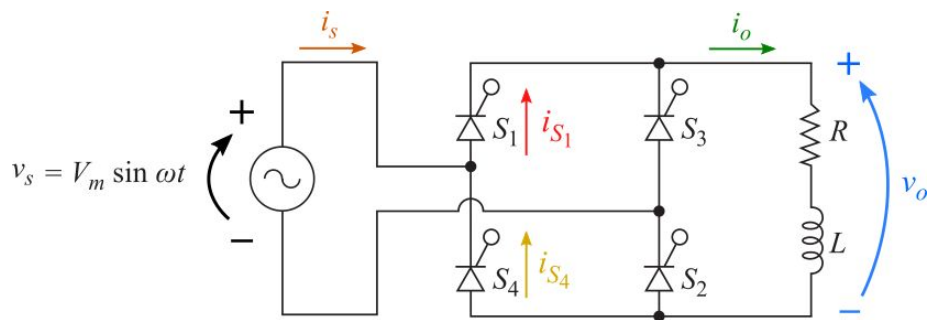
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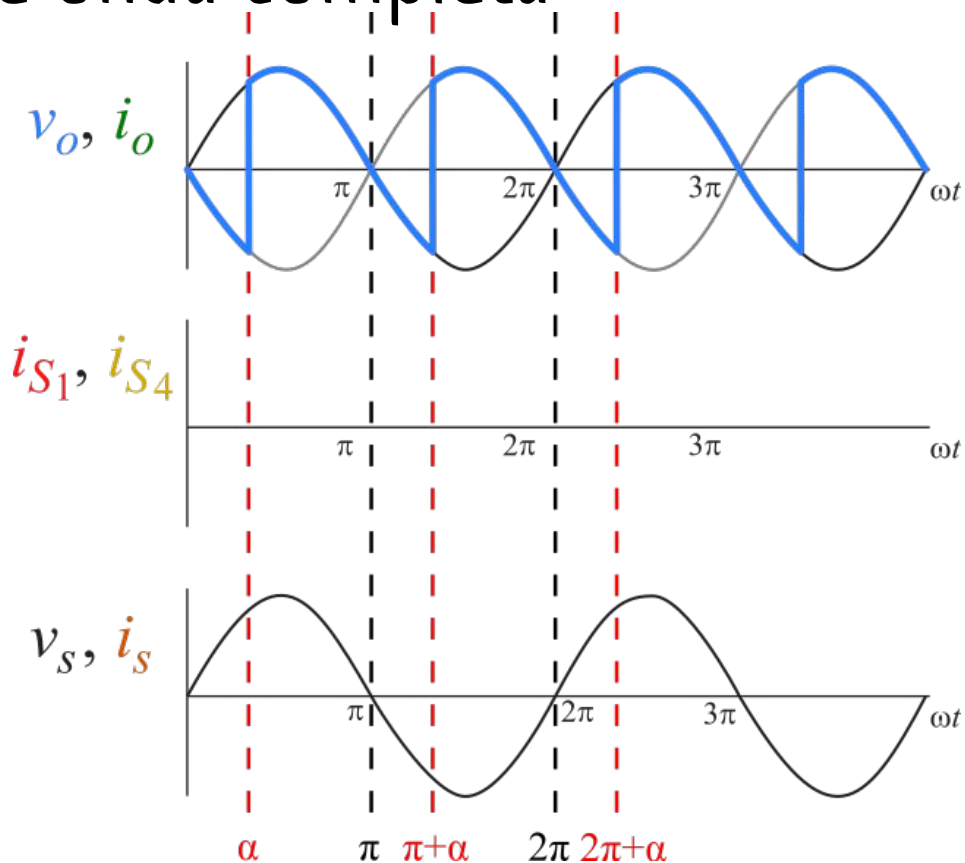
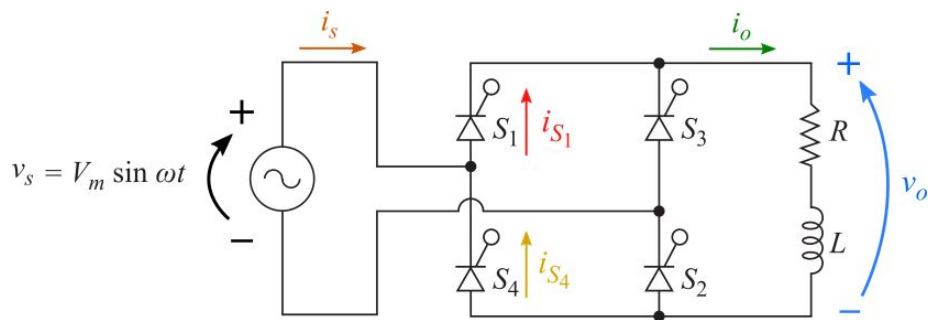
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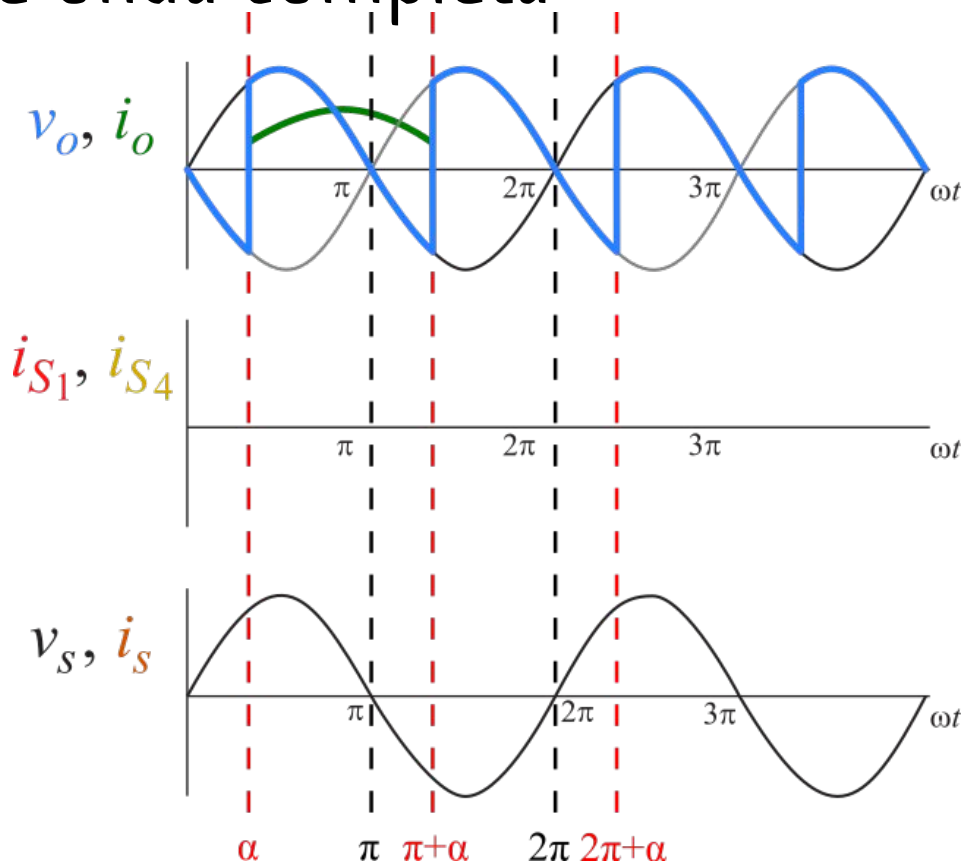
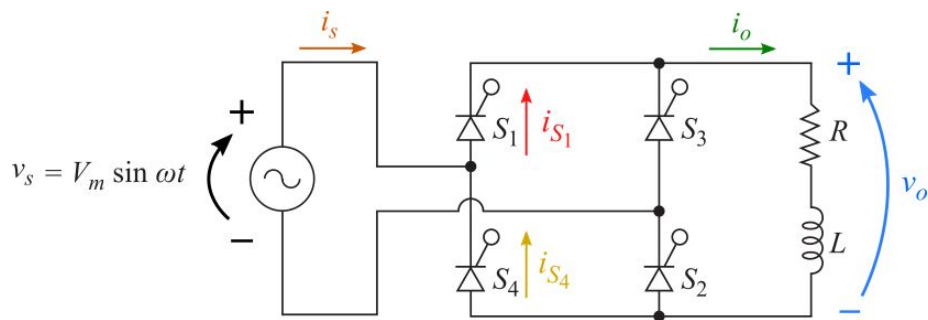
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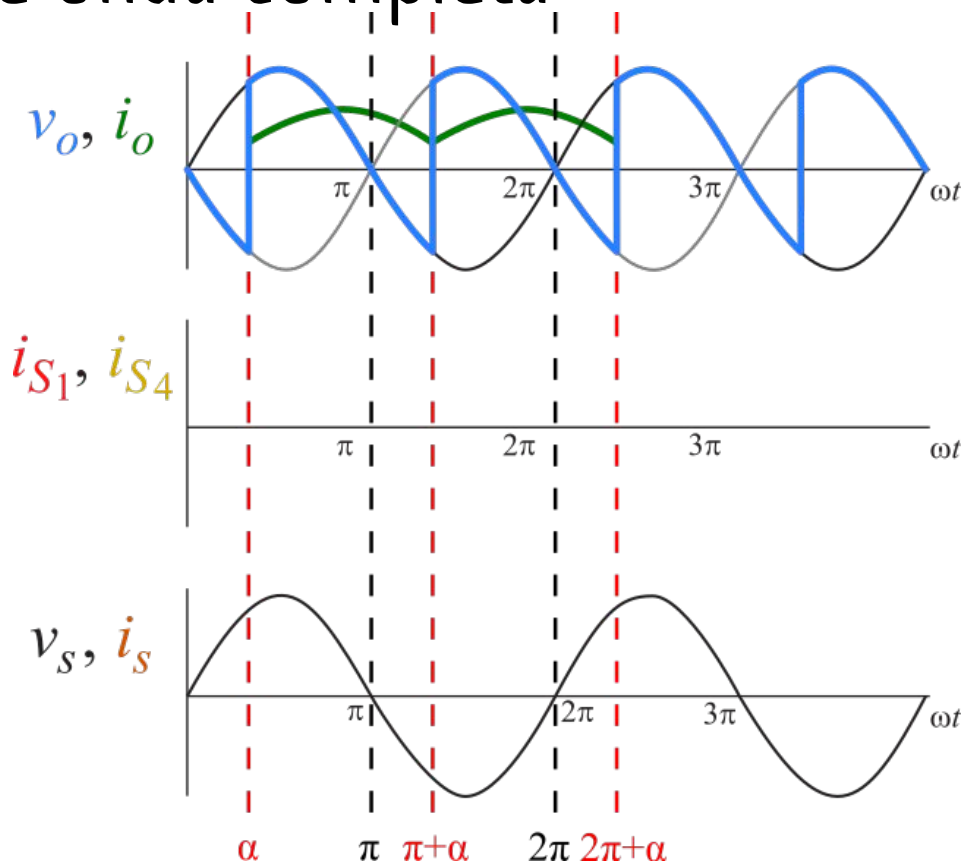
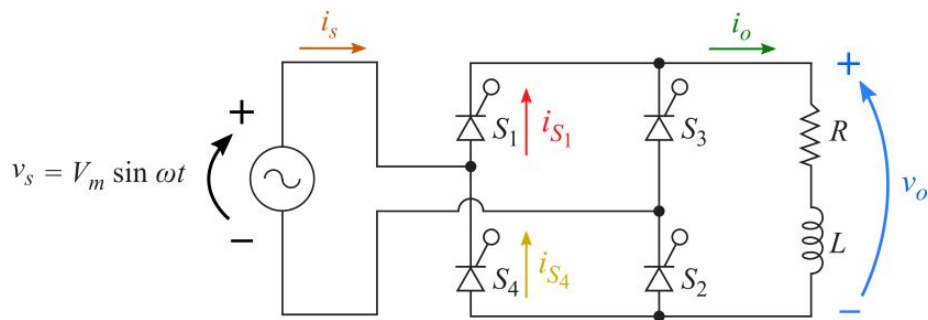
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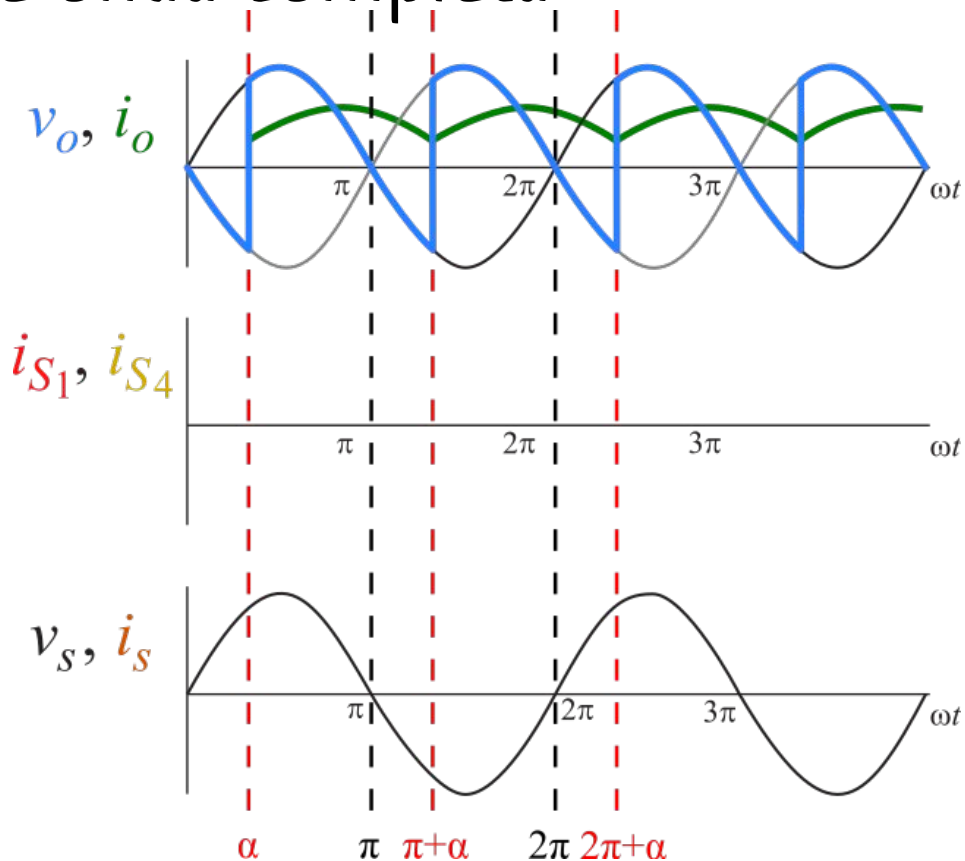
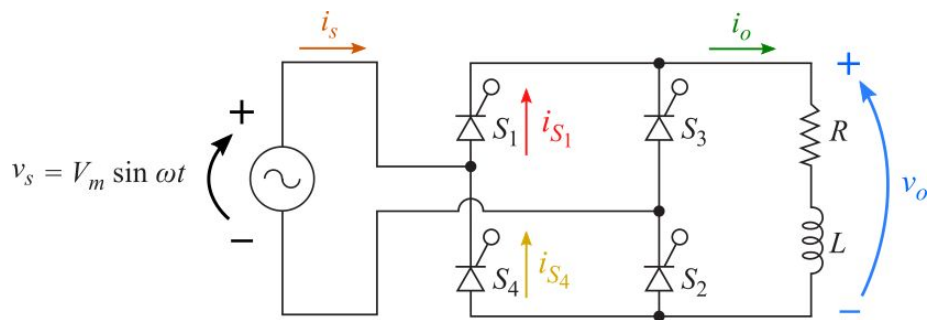
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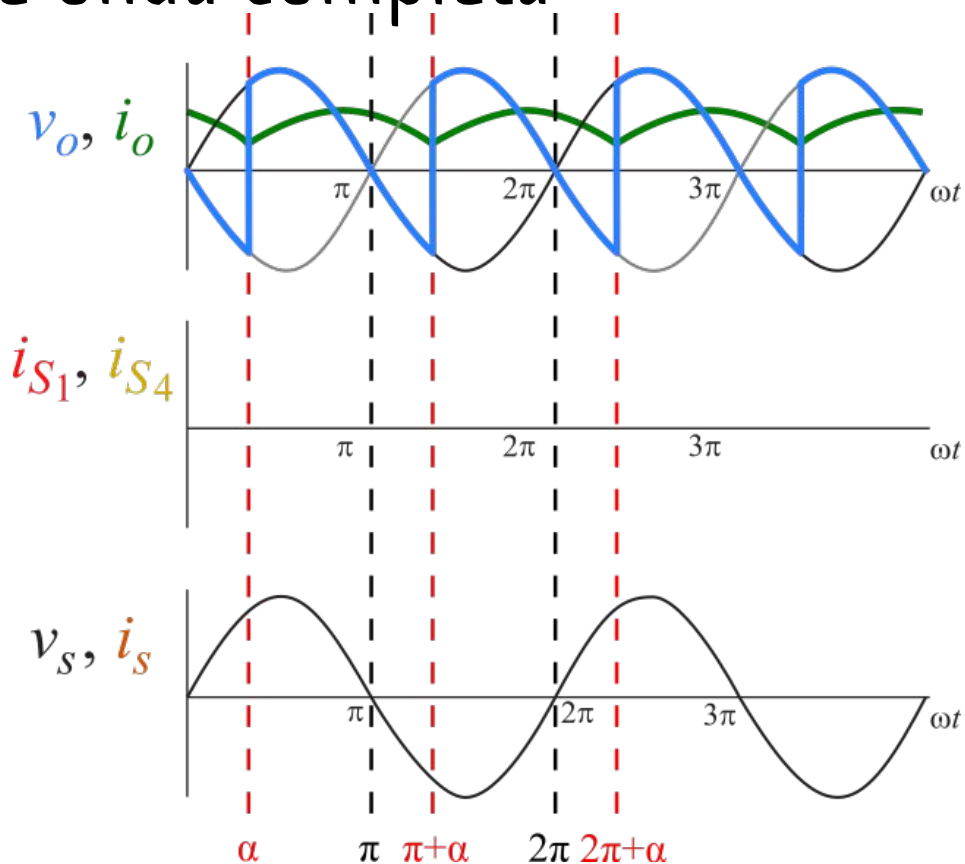
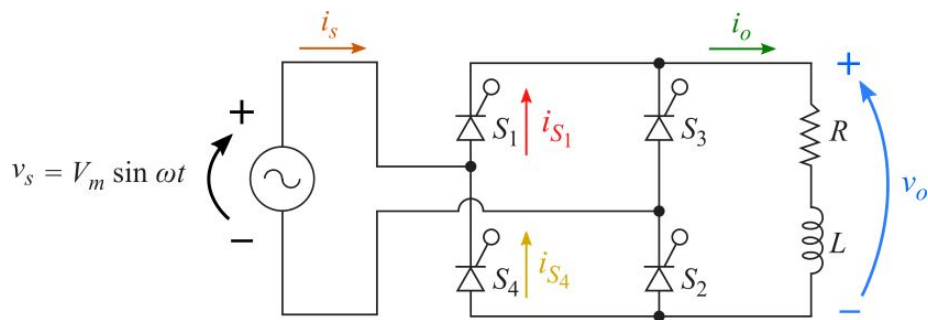
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- Carga R-L



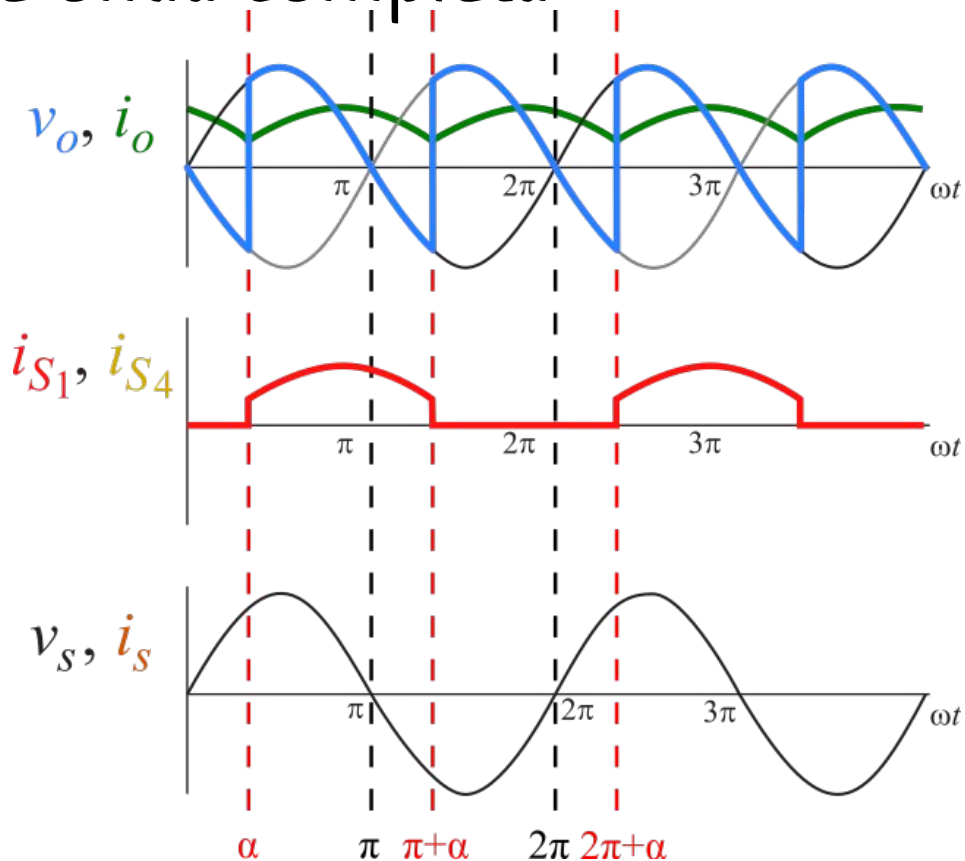
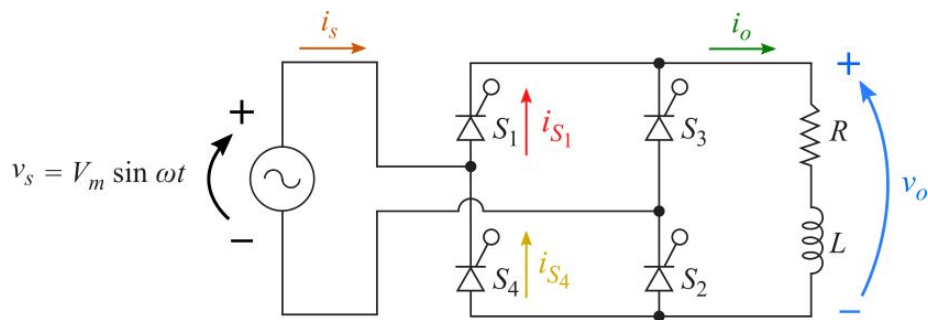
Rectificador controlado de onda completa

- Carga R-L



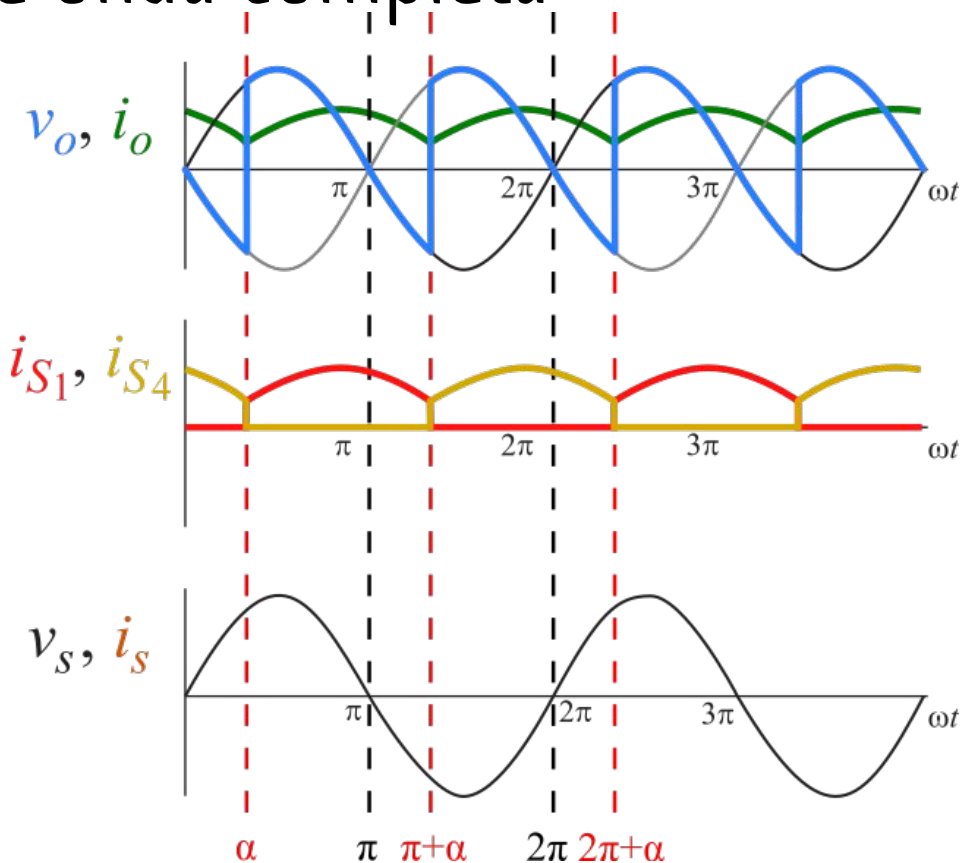
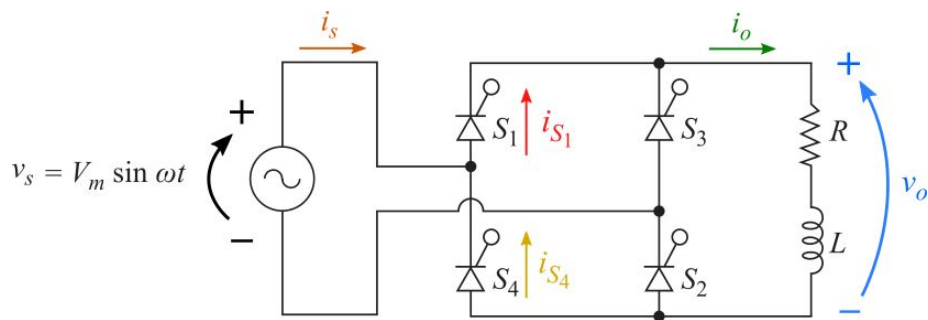
Rectificador controlado de onda completa

- Carga R-L



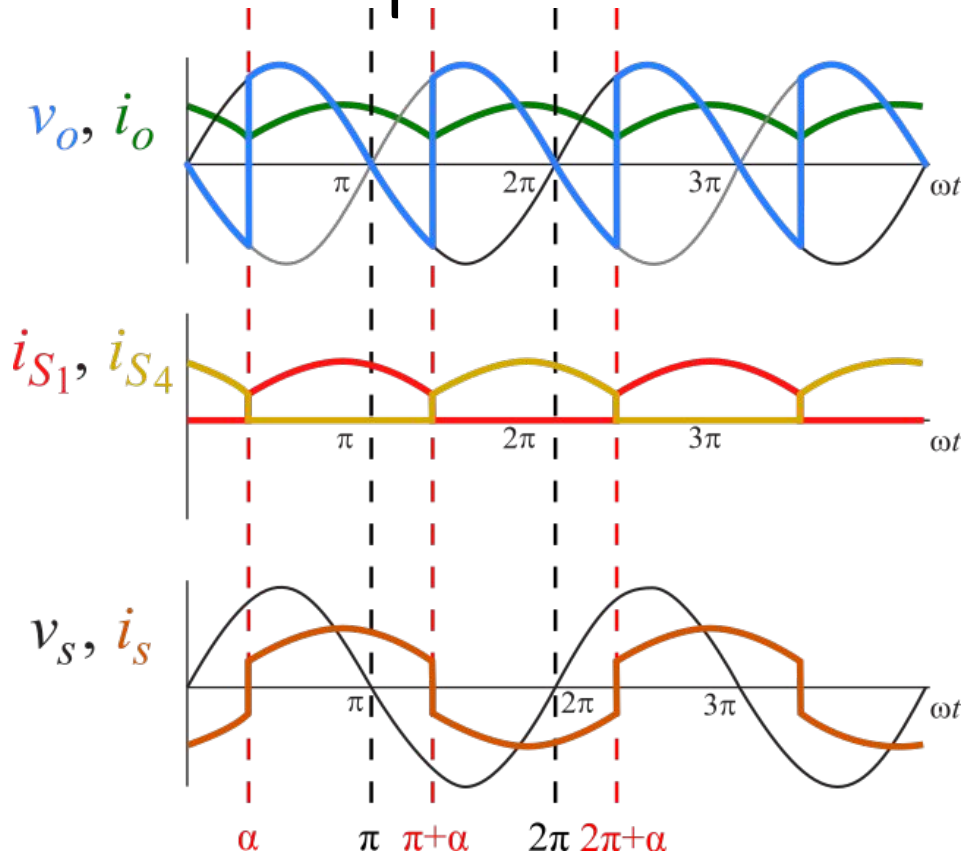
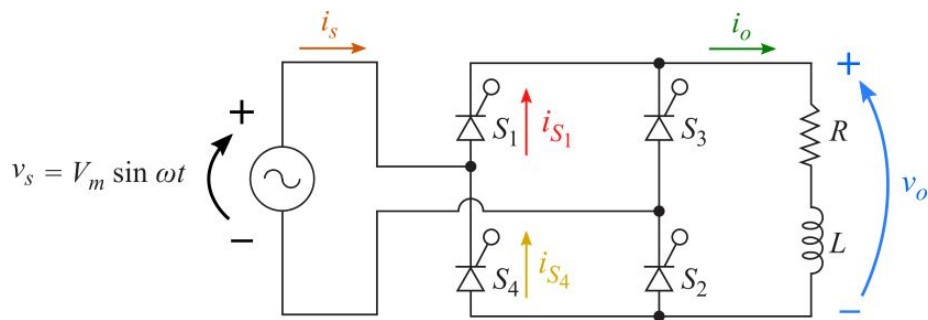
Rectificador controlado de onda completa

- Carga R-L



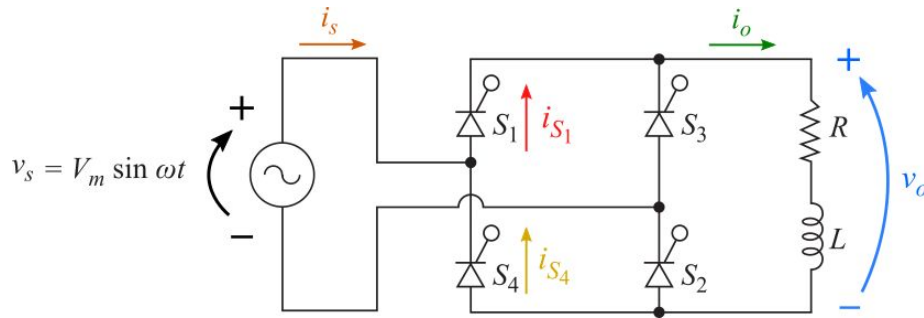
Rectificador controlado de onda completa

- Carga R-L

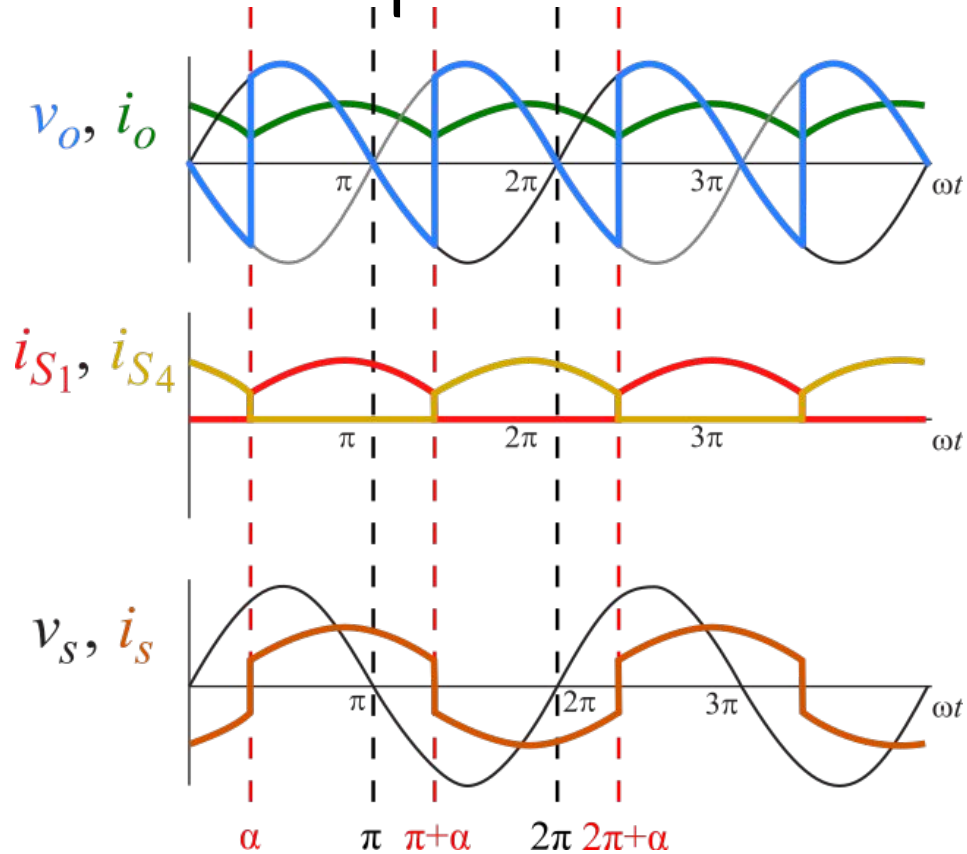


Rectificador controlado de onda completa

- Carga R-L

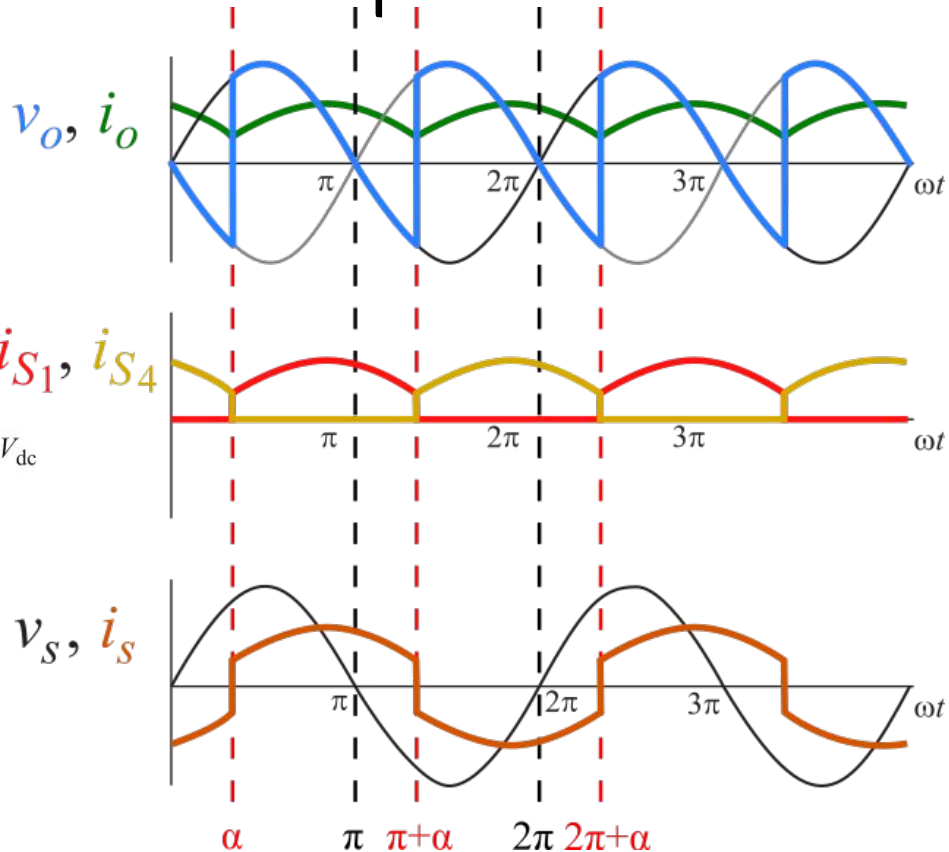
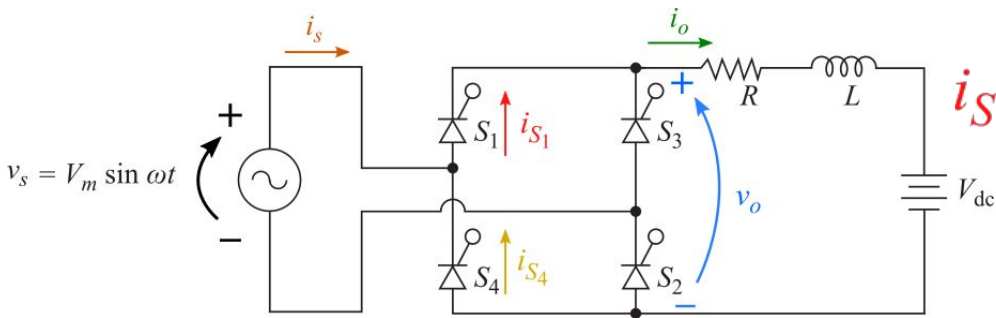


$$V_0 = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} V_m \sin(\omega t) d\omega t = \frac{2V_m}{\pi} \cos(\alpha)$$



Rectificador controlado de onda completa

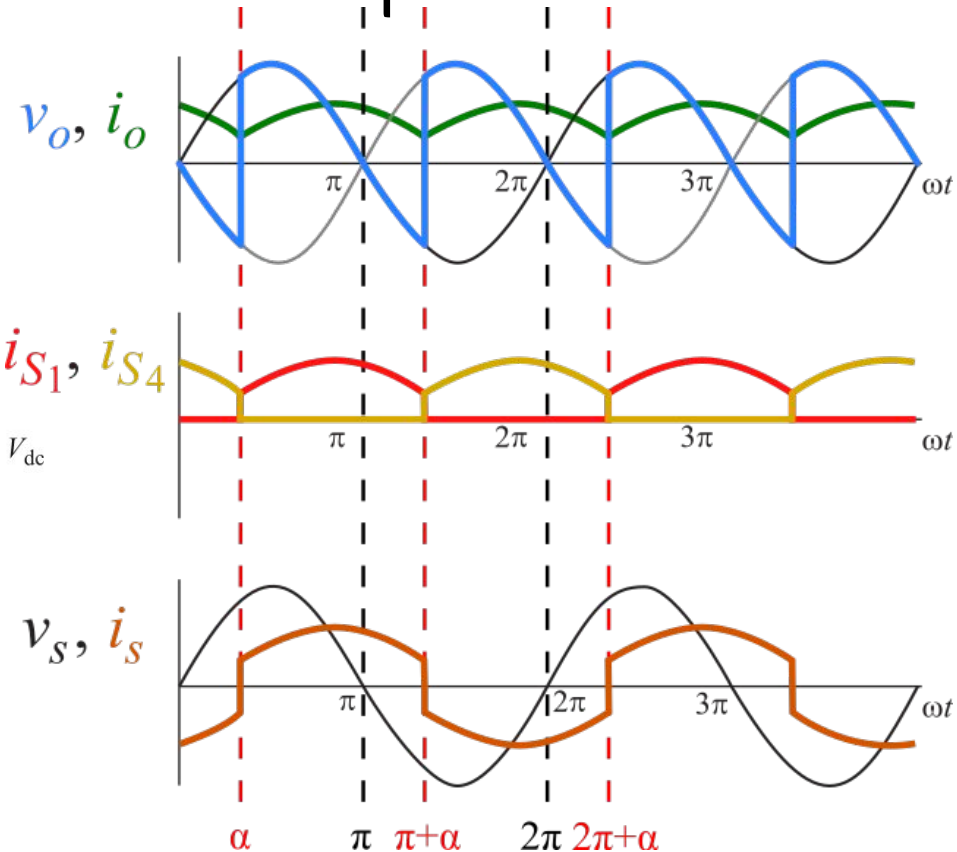
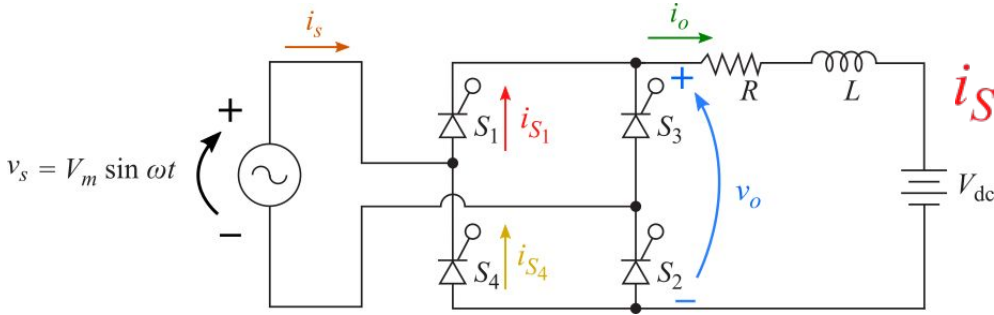
- Carga R-L-E



Rectificador controlado de onda completa

- Carga R-L-E

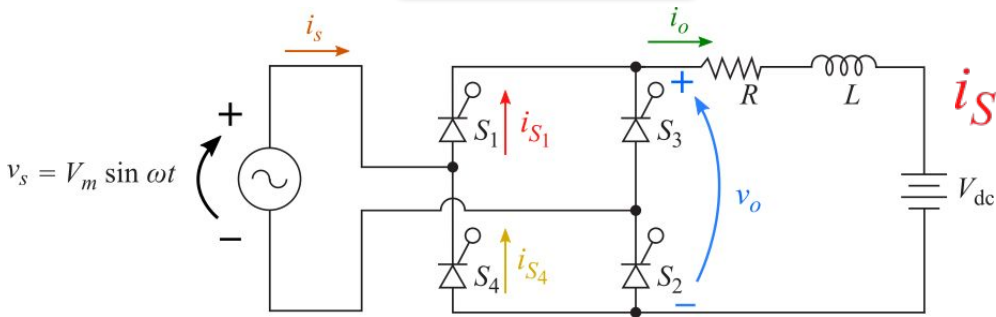
$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



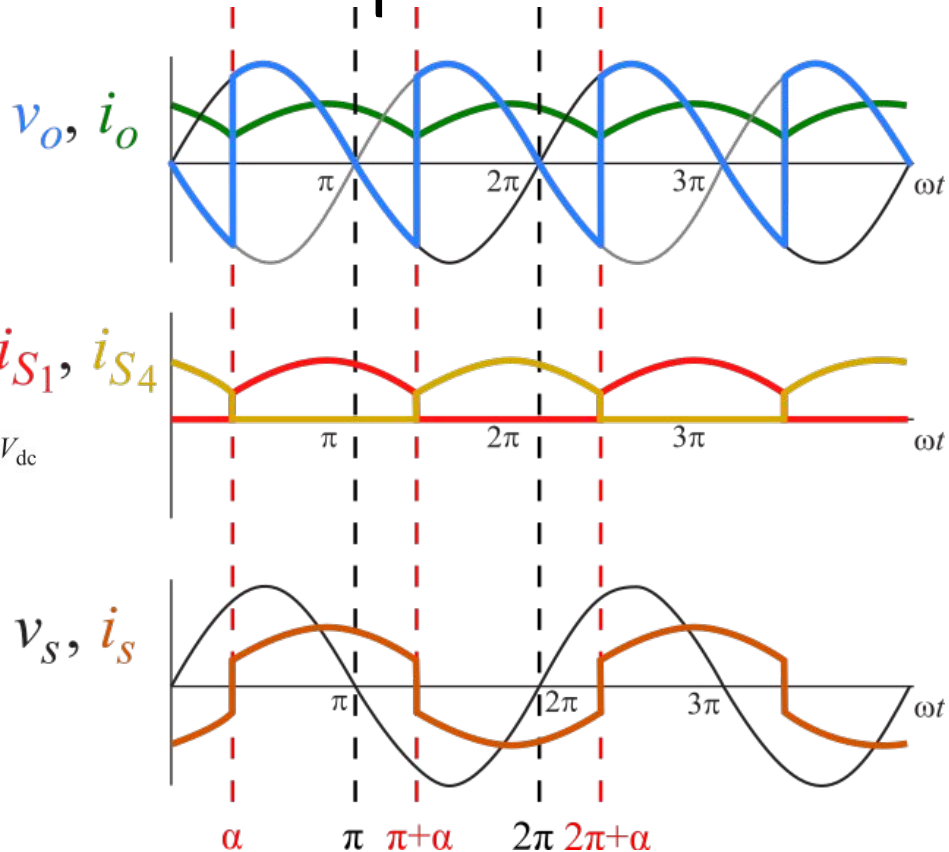
Rectificador controlado de onda completa

- Carga R-L-E

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$

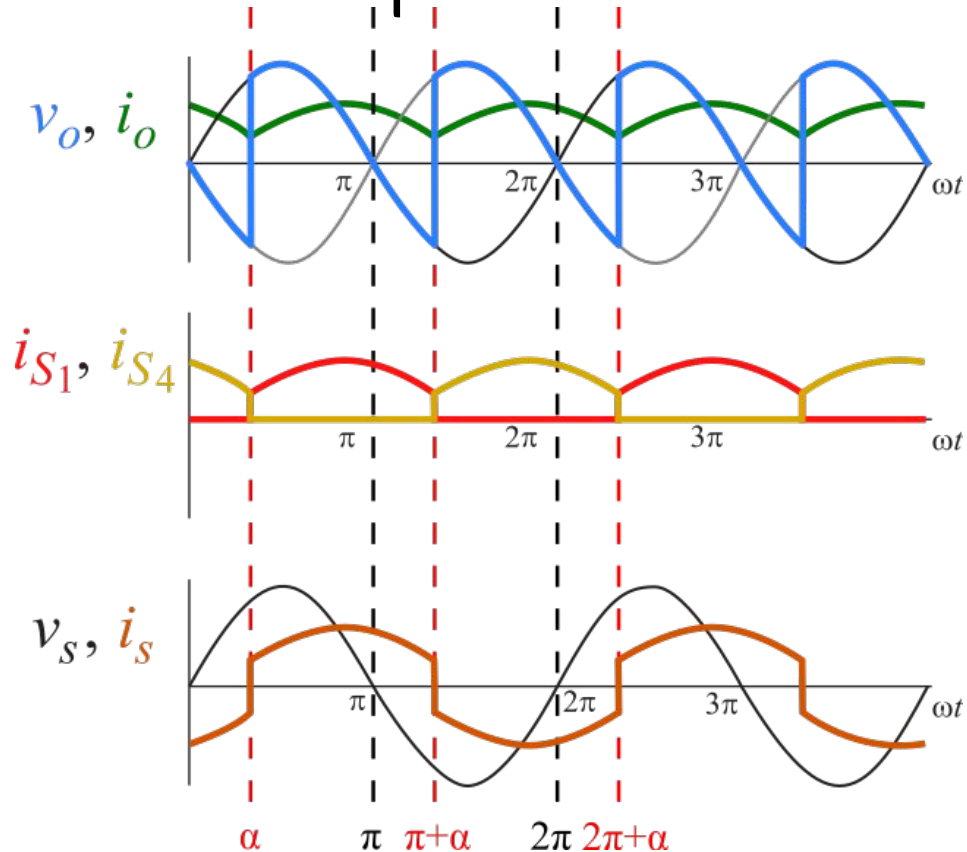
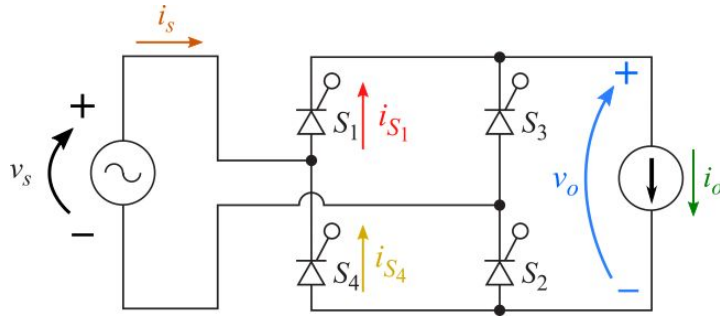


$$I_0 = \frac{V_0 - V_{cc}}{R} = \frac{\frac{2V_m}{\pi} \cos(\alpha) - V_{cc}}{R}$$



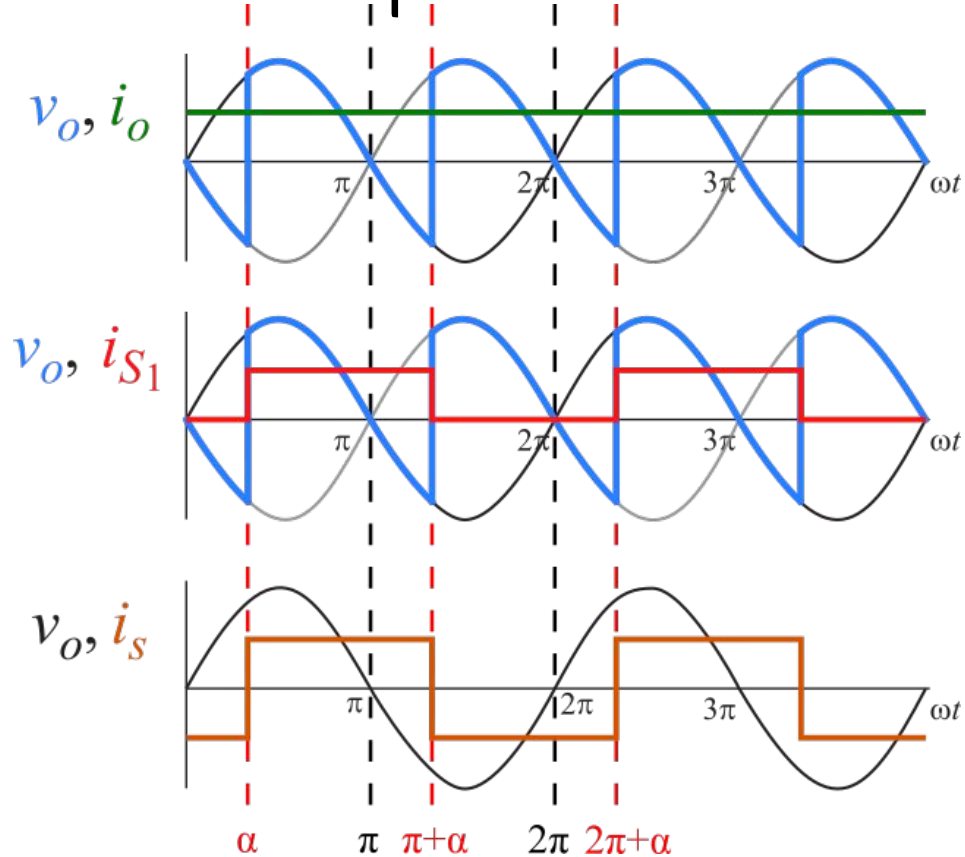
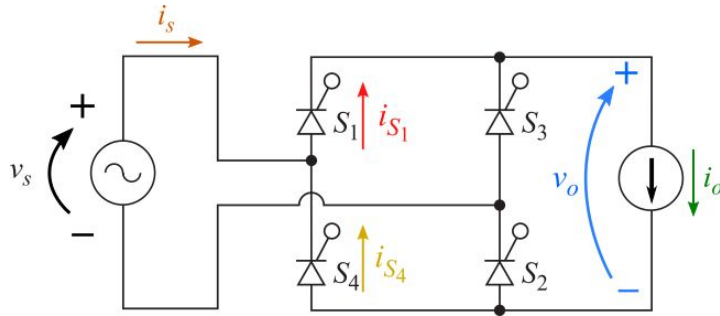
Rectificador controlado de onda completa

- Corriente constante



Rectificador controlado de onda completa

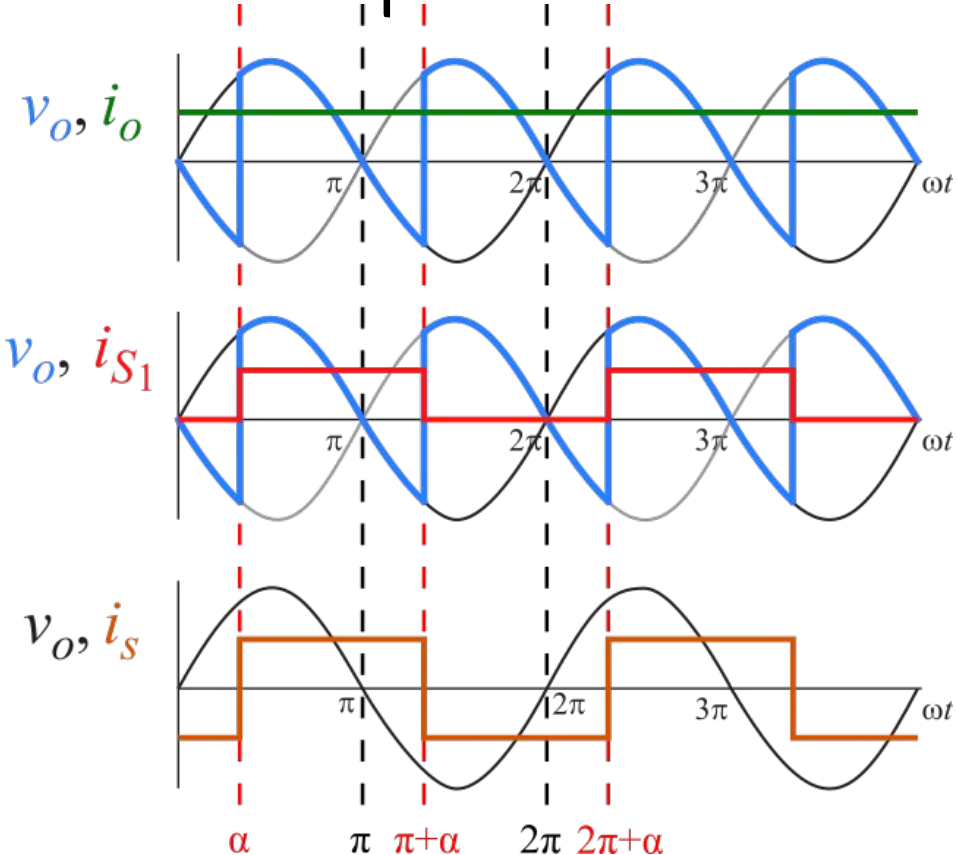
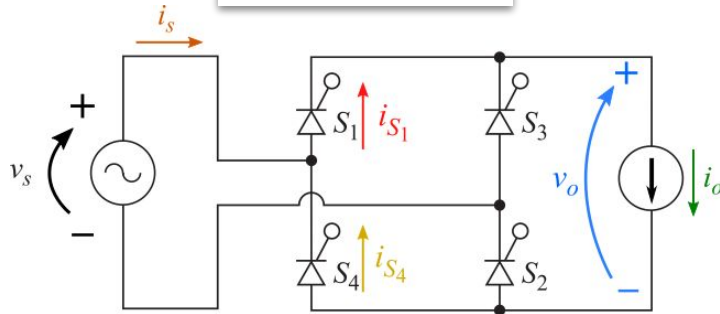
- Corriente constante



Rectificador controlado de onda completa

- Corriente constante

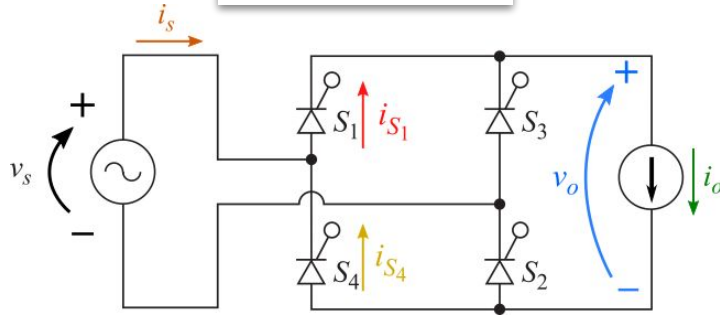
$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



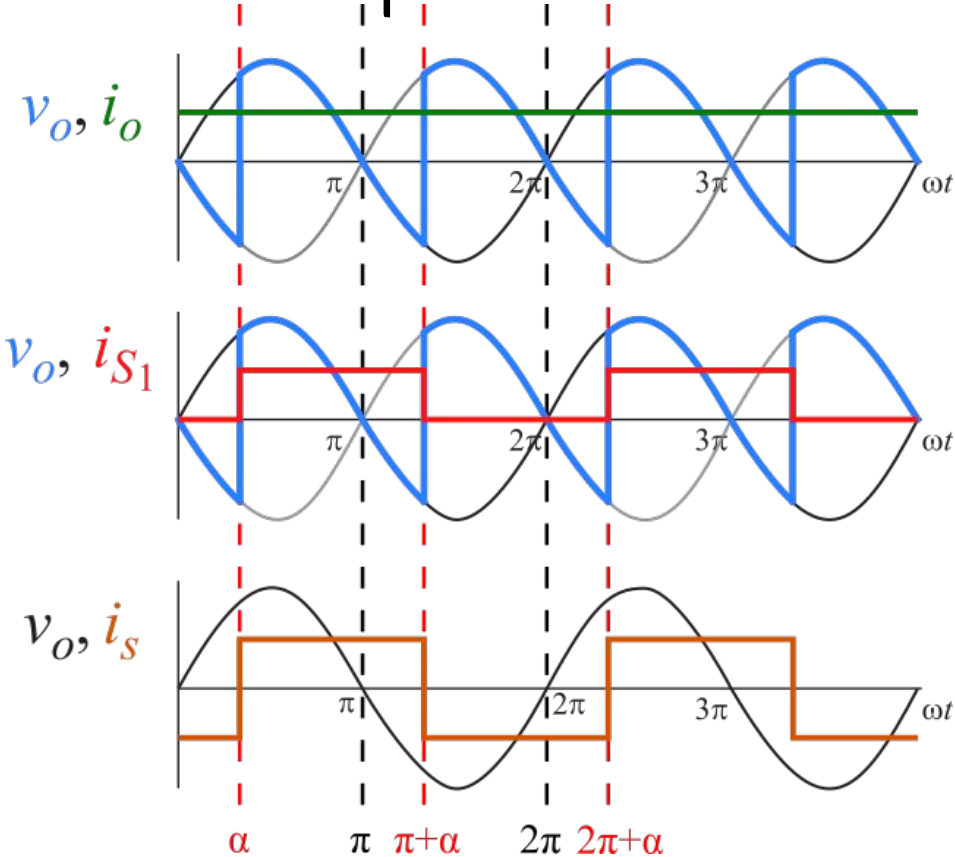
Rectificador controlado de onda completa

- Corriente constante

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



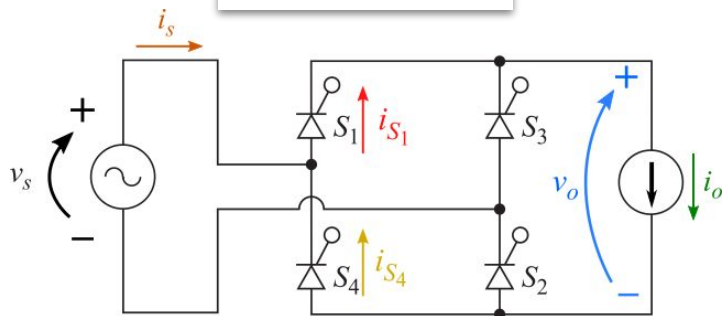
$$0^\circ < \alpha < 90^\circ \longrightarrow V_0 > 0,$$



Rectificador controlado de onda completa

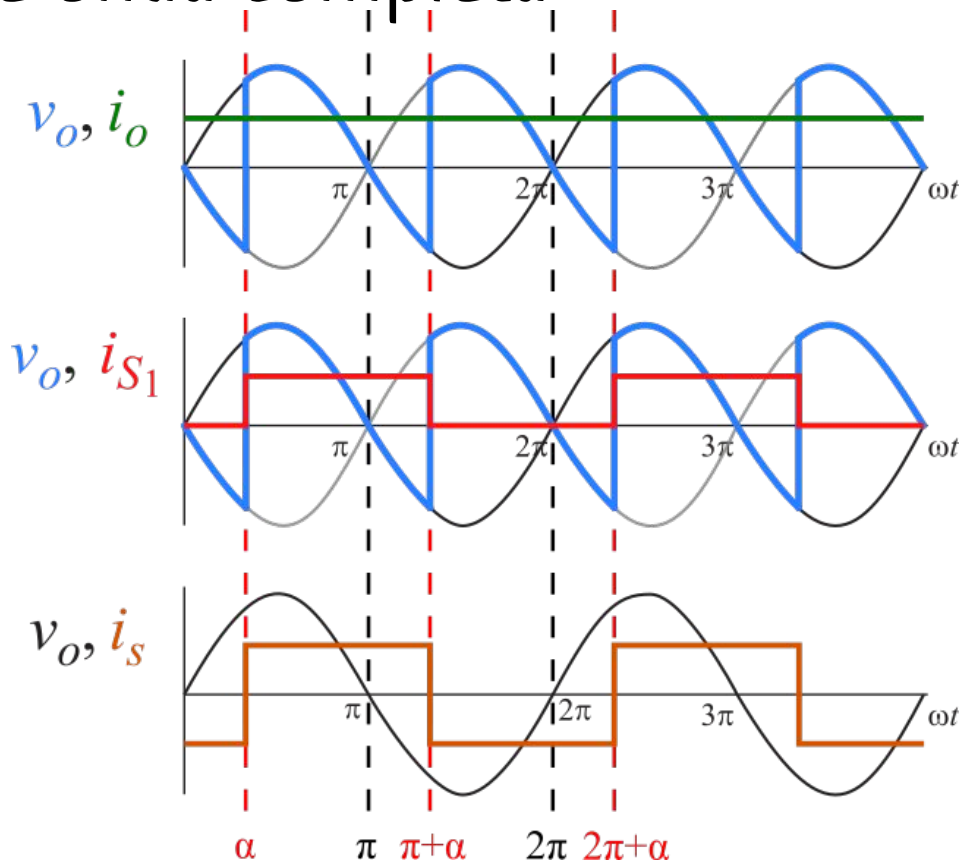
- Corriente constante

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



$$0^\circ < \alpha < 90^\circ \longrightarrow V_0 > 0,$$

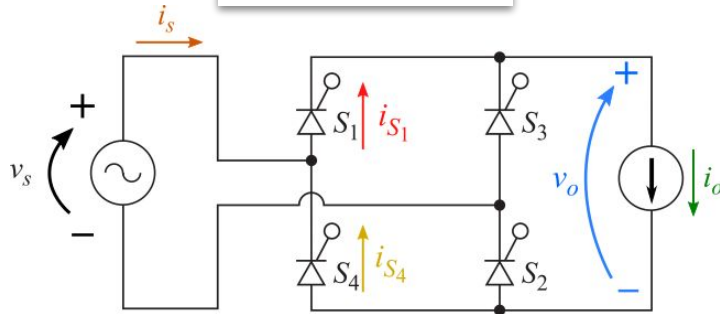
$$90^\circ < \alpha < 180^\circ \longrightarrow V_0 < 0,$$



Rectificador controlado de onda completa

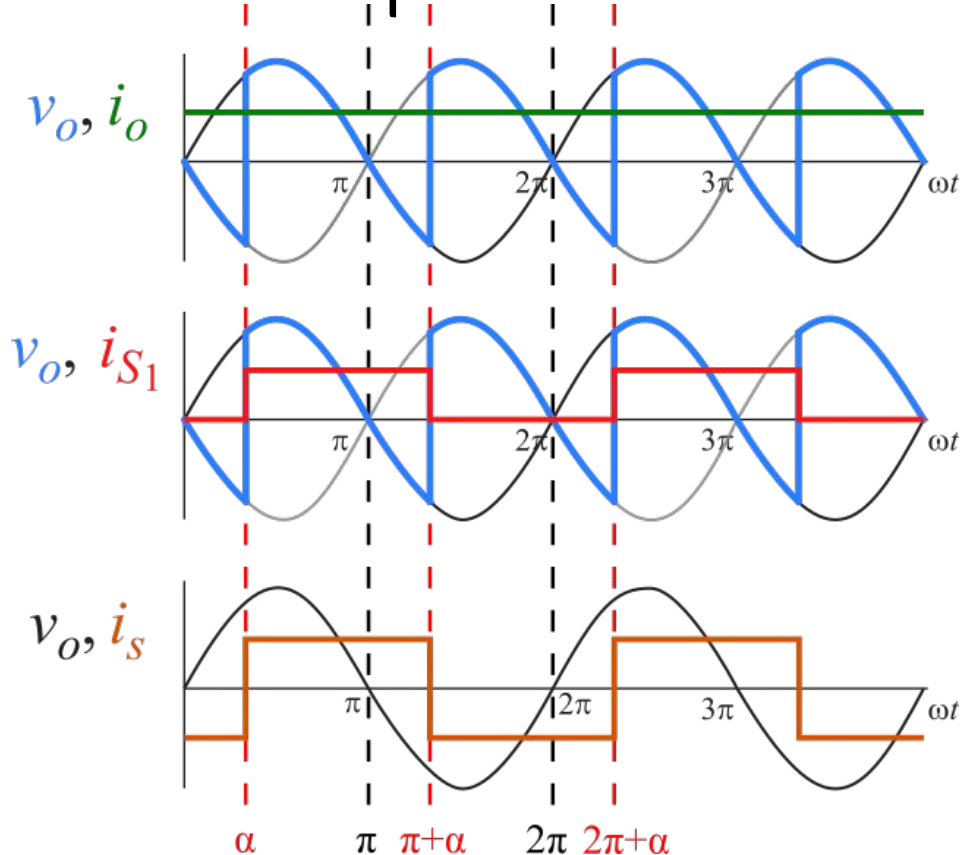
- Corriente constante

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



$0^\circ < \alpha < 90^\circ \rightarrow V_0 > 0$, $\rightarrow$ **rectificador**;

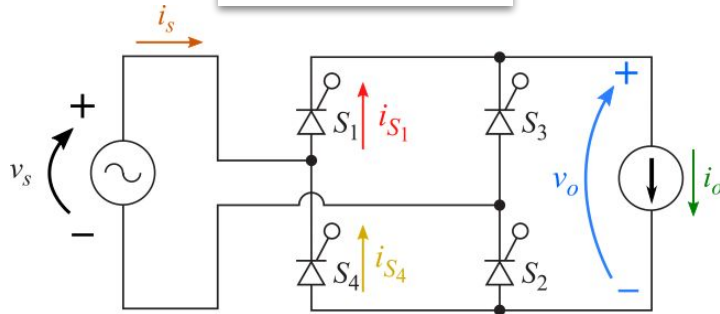
$90^\circ < \alpha < 180^\circ \rightarrow V_0 < 0$,



Rectificador controlado de onda completa

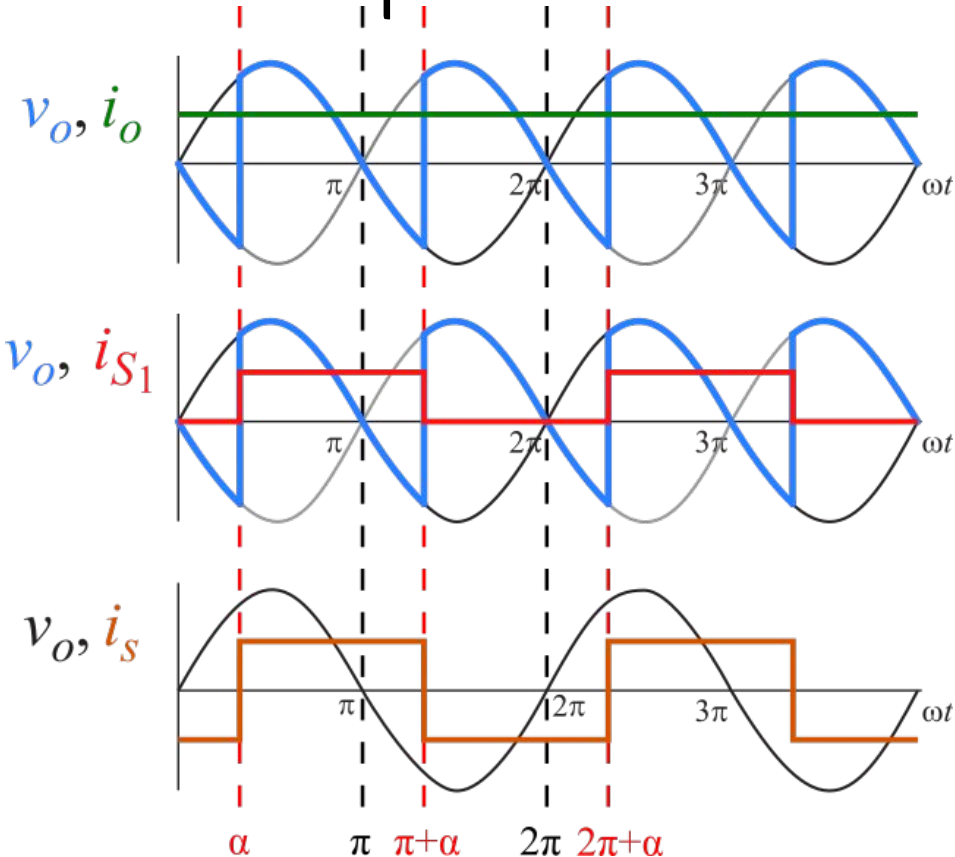
- Corriente constante

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



$0^\circ < \alpha < 90^\circ \rightarrow V_0 > 0$, $\rightarrow$ **rectificador**;

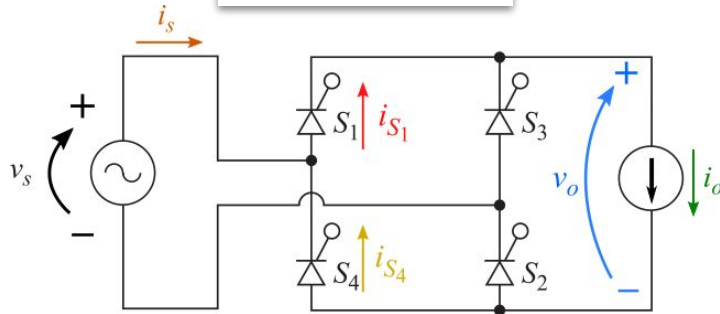
$90^\circ < \alpha < 180^\circ \rightarrow V_0 < 0$, $\rightarrow$ **inversor**;



Rectificador controlado de onda completa

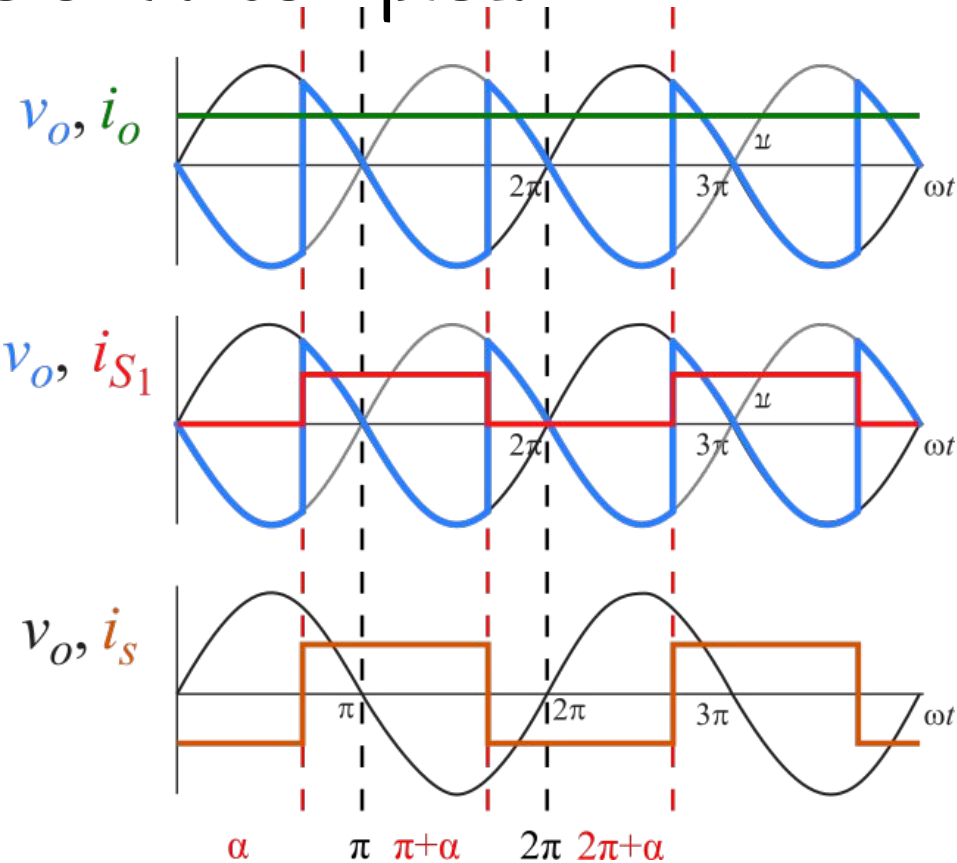
- Corriente constante

$$V_0 = \frac{2V_m}{\pi} \cos(\alpha)$$



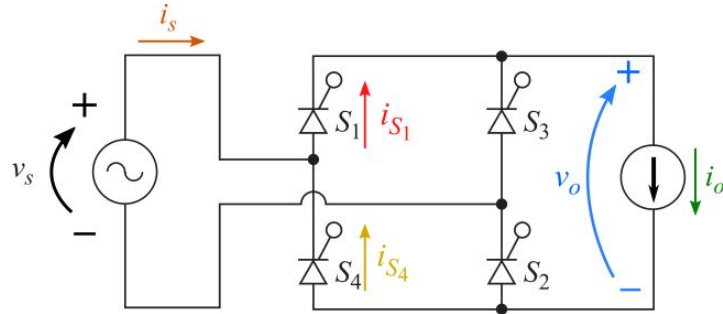
$0^\circ < \alpha < 90^\circ \rightarrow V_0 > 0$,  **rectificador**;

$90^\circ < \alpha < 180^\circ \rightarrow V_0 < 0$,  **inversor**;

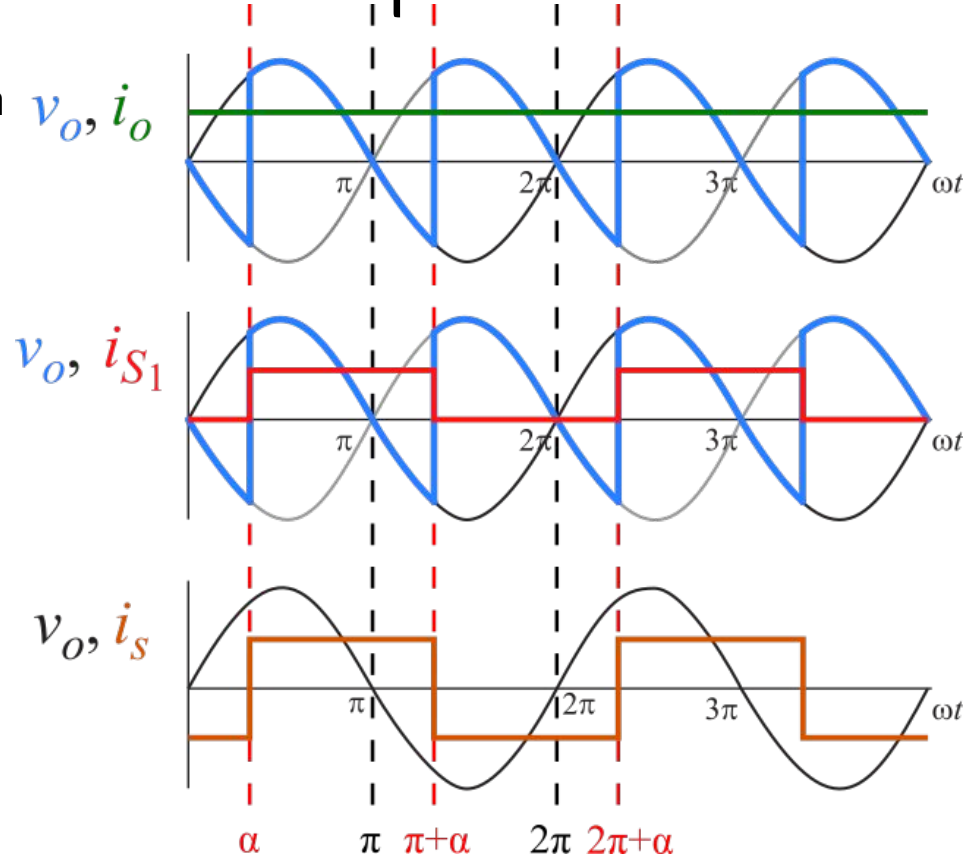


Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o

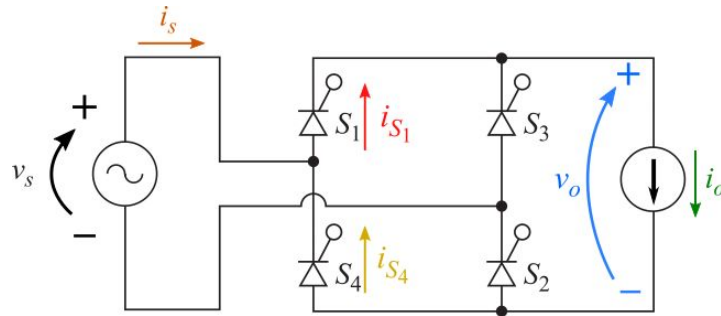


$$I_{ca,n} = \frac{4}{n\pi} I_0, \text{ para } n \text{ impar}$$

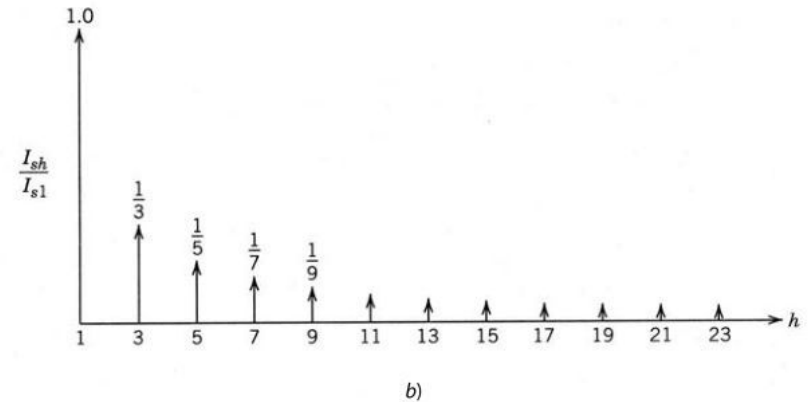
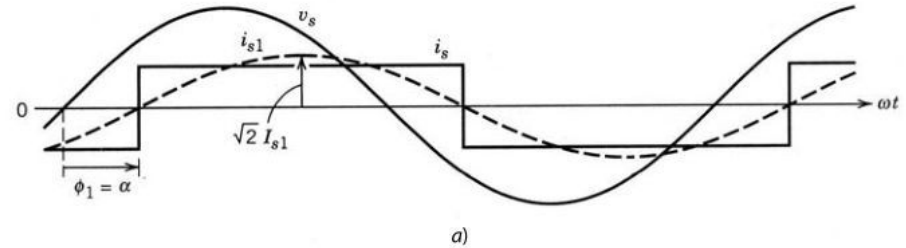


Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia

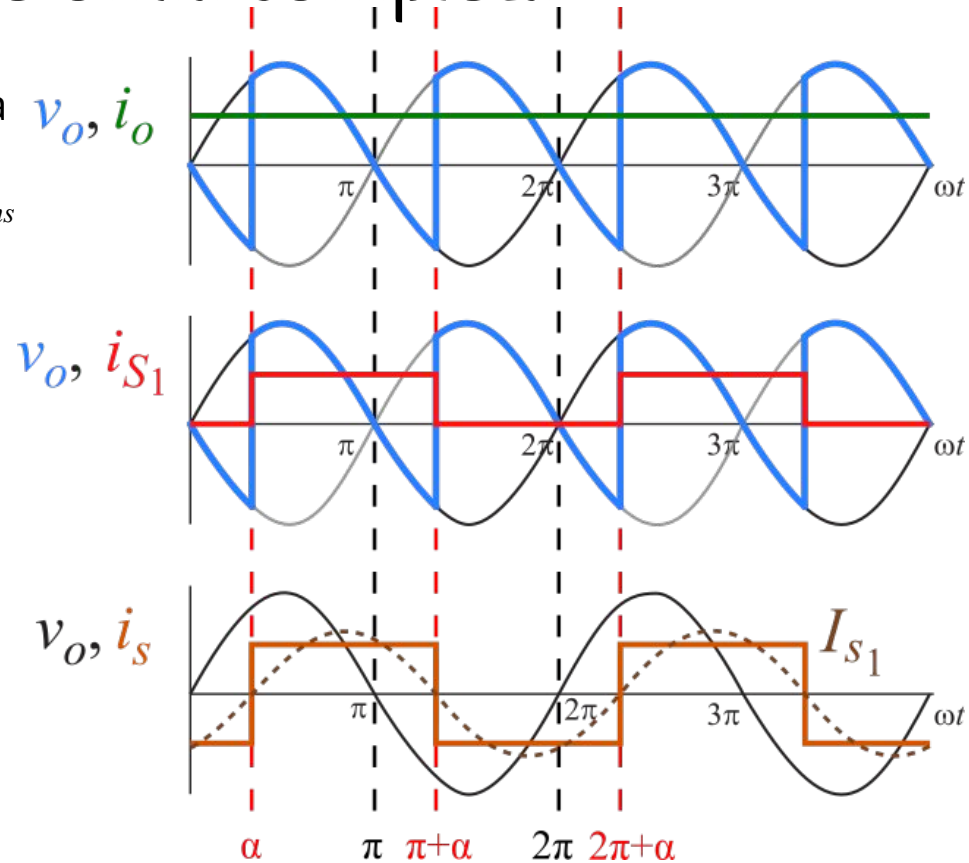


$$I_{ca,n} = \frac{4}{n\pi} I_0, \text{ para } n \text{ impar}$$



Rectificador controlado de onda completa

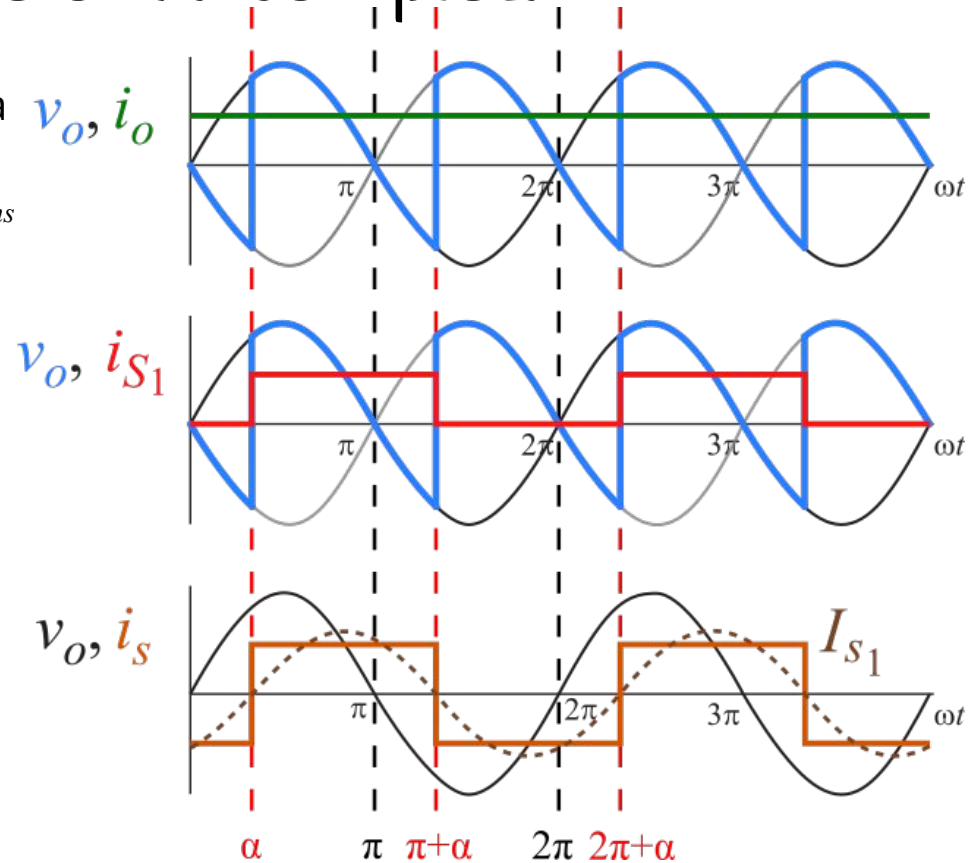
- Corriente de línea y Factor de Potencia v_o, i_o
 → Factor de forma: Relación $I_{l,rms}$ a I_{rms}



Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o
- Factor de forma: Relación $I_{l,rms}$ a I_{rms}

$$FF = \frac{4I_0}{\sqrt{2\pi}I_0} \approx 0,9$$



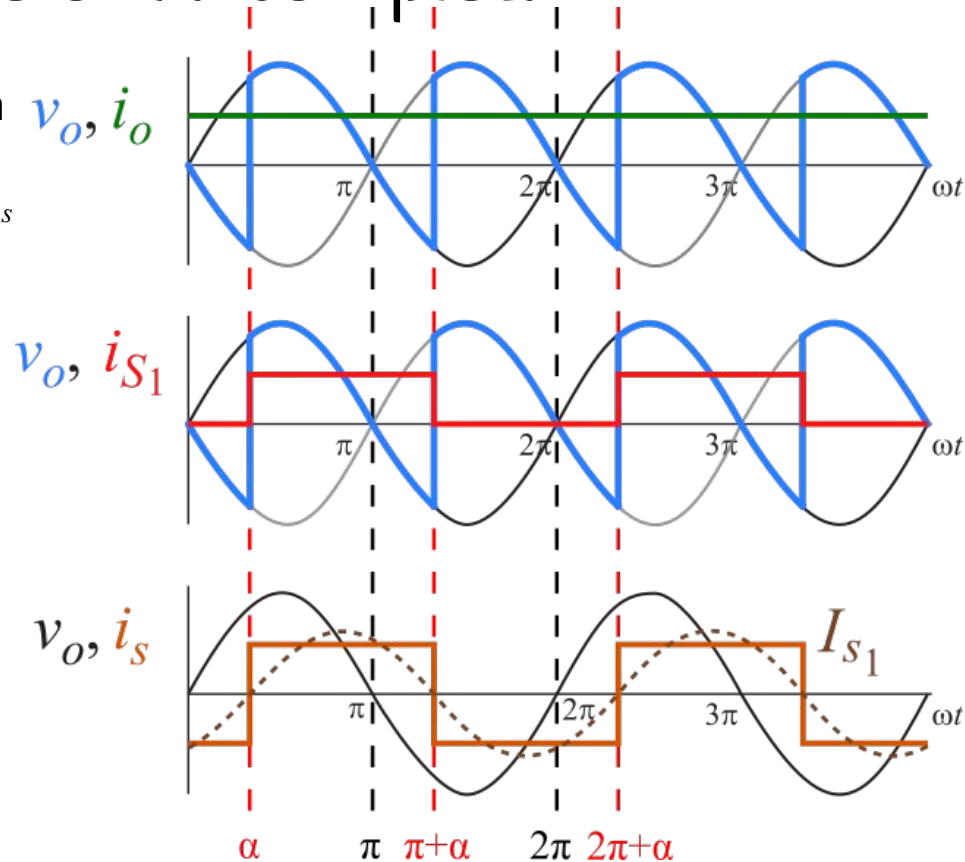
Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o

→ Factor de forma: Relación $I_{l,rms}$ a I_{rms}

$$FF = \frac{4I_0}{\sqrt{2\pi}I_0} \approx 0,9$$

→ Factor de desplazamiento



Rectificador controlado de onda completa

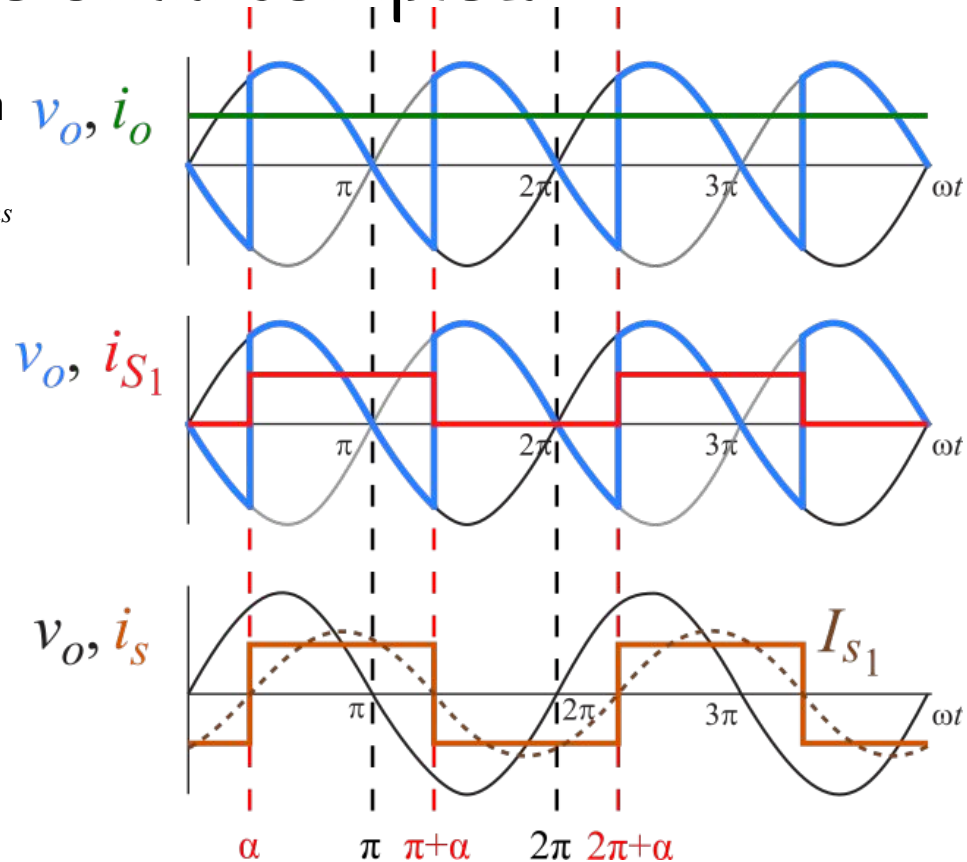
- Corriente de línea y Factor de Potencia v_o, i_o

→ Factor de forma: Relación $I_{l,rms}$ a I_{rms}

$$FF = \frac{4I_0}{\sqrt{2\pi}I_0} \approx 0,9$$

→ Factor de desplazamiento

$$FD = \cos(\varphi) = \cos(\alpha)$$



Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o

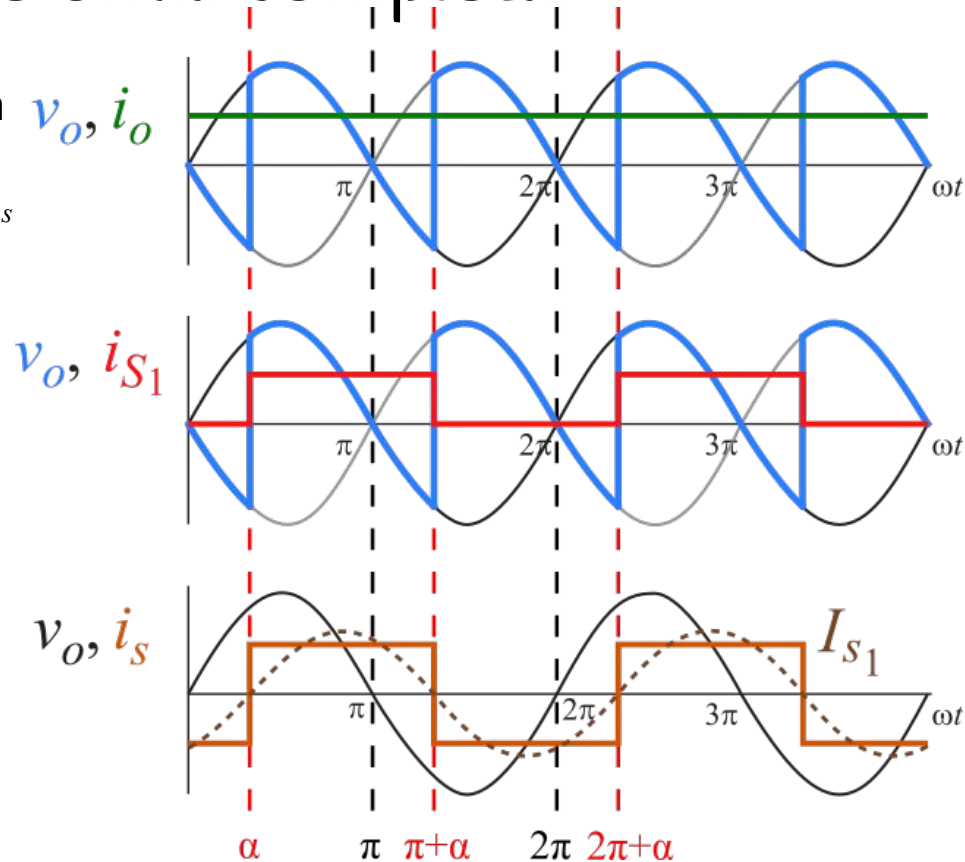
→ Factor de forma: Relación $I_{l,rms}$ a I_{rms}

$$FF = \frac{4I_0}{\sqrt{2\pi}I_0} \approx 0,9$$

→ Factor de desplazamiento

$$FD = \cos(\varphi) = \cos(\alpha)$$

→ Factor de potencia



Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o

→ Factor de forma: Relación $I_{l,rms}$ a I_{rms}

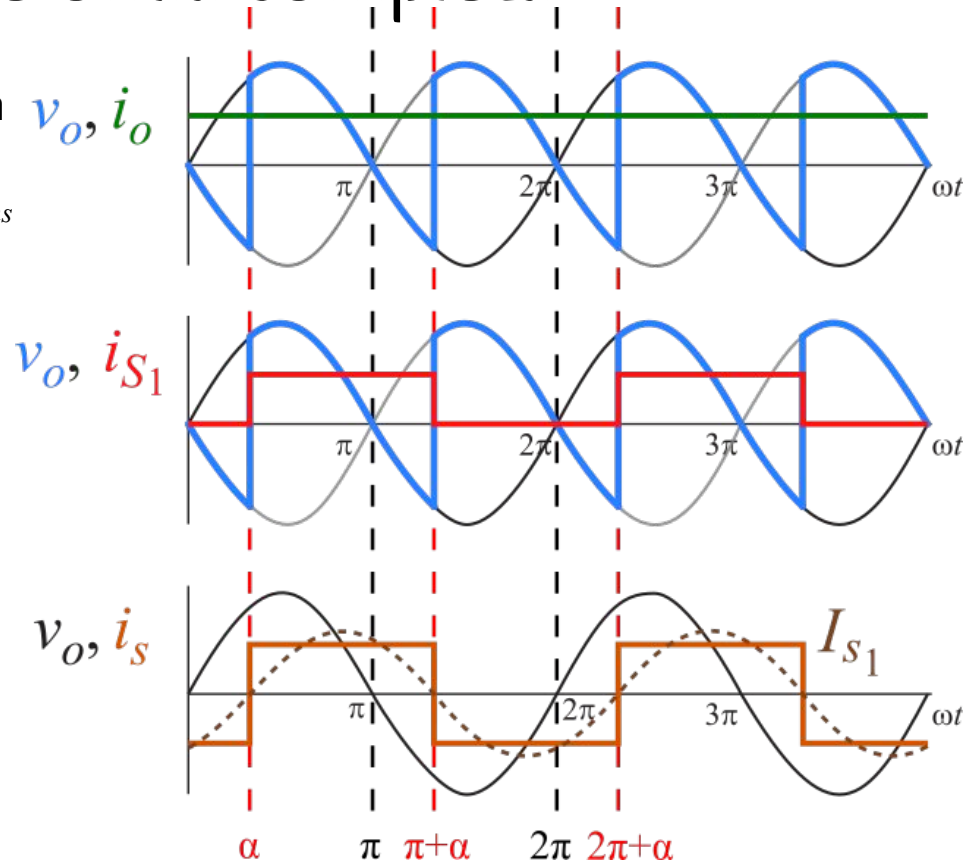
$$FF = \frac{4I_0}{\sqrt{2\pi}I_0} \approx 0,9$$

→ Factor de desplazamiento

$$FD = \cos(\varphi) = \cos(\alpha)$$

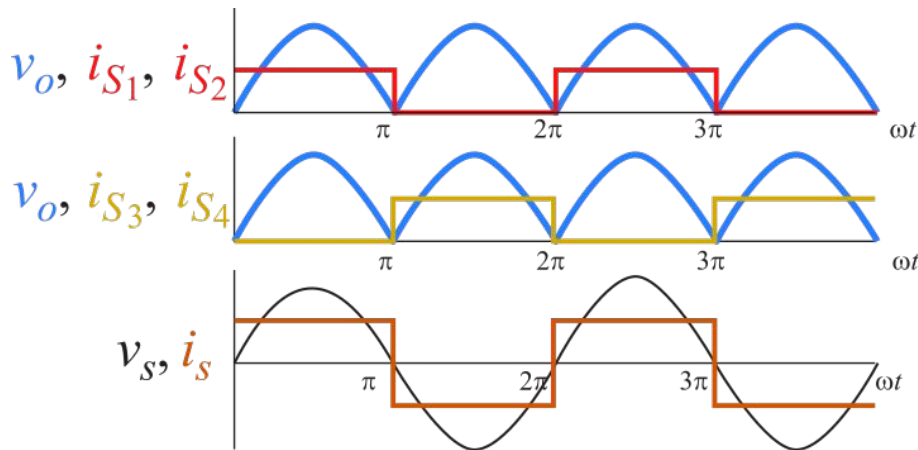
→ Factor de potencia

$$FP = 0,9 \cos(\alpha)$$

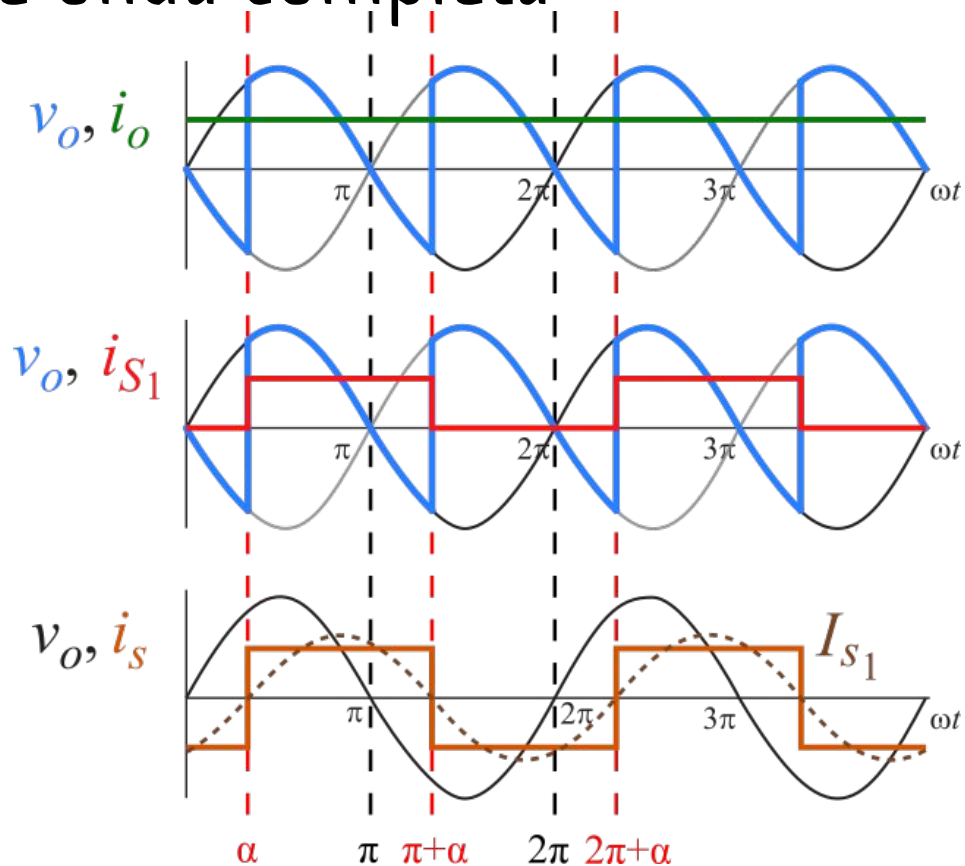


Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o

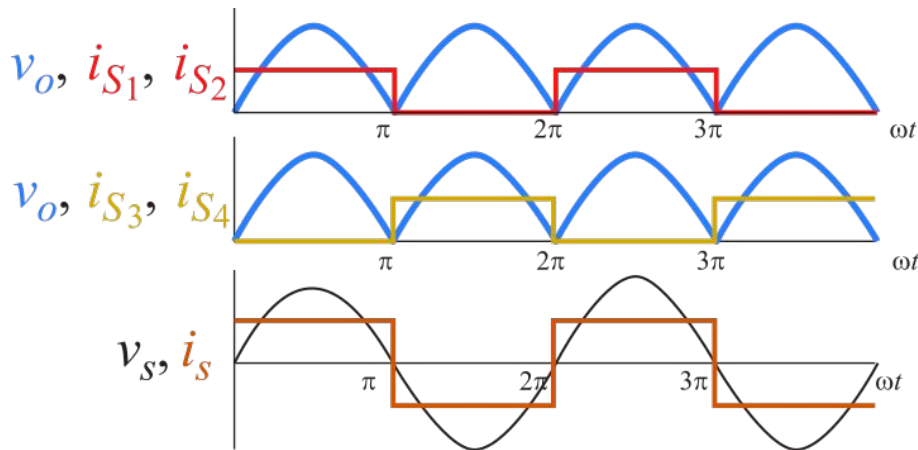


$$FP = 0,9 \cos(\alpha)$$

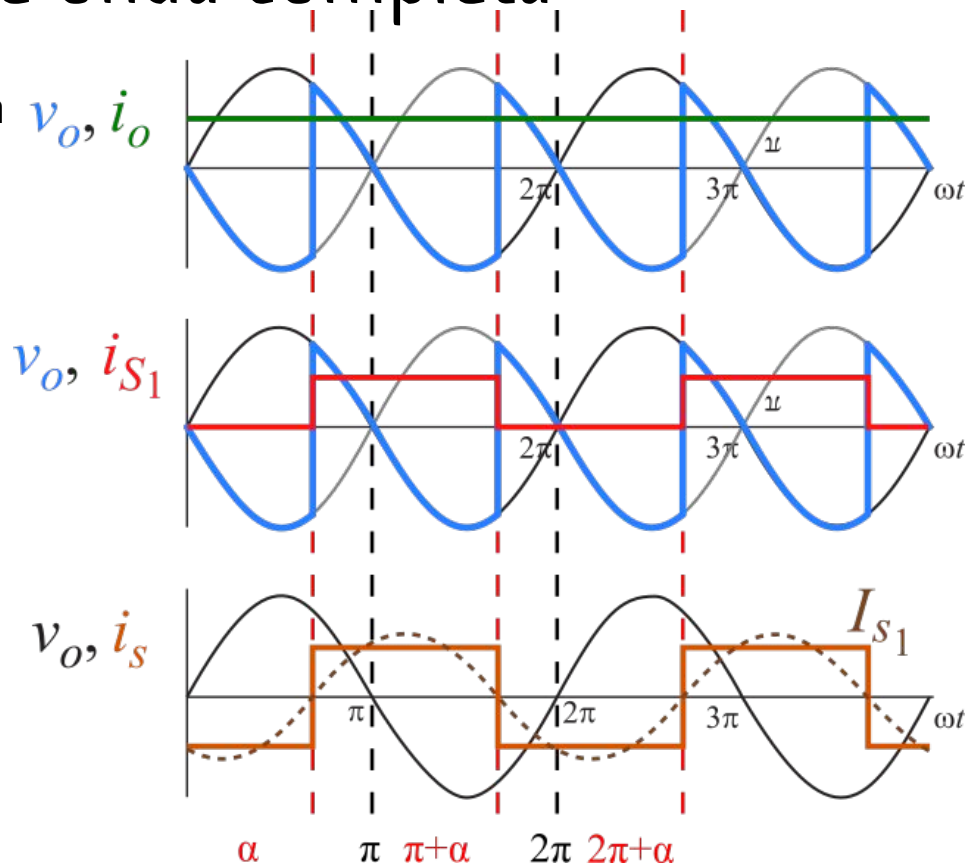


Rectificador controlado de onda completa

- Corriente de línea y Factor de Potencia v_o, i_o



$$FP = 0,9 \cos(\alpha)$$



Bibliografía

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